

The classification of regular Lagrangian fibrations

over compact surfaces

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Preface

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Summary

"Everything is a Lagrangian submanifold"

Alan Weinstein

The main goal of this thesis is to classify Lagrangian fibrations over compact surfaces. The base space of such a fibration comes equipped with an *integral affine structure* which is an invariant of the Lagrangian fibration. We classify the Lagrangian fibrations over a compact surface B by first classifying all the possible integral affine structures on B and then classifying all of the Lagrangian fibrations corresponding to a fixed integral affine structure. Remarkably, if B is simply connected the Lagrangian fibrations over B are determined entirely by this integral affine structure.

In Chapter 1, we develop the basic notions of affine geometry which belongs to a more general class of geometries called (G, X) -structures which were studied by Thurston, Ehresmann and many others [Thu80], [Ehr36]. A common theme when classifying the geometric structures on a manifold M , is that the fundamental group plays an important role in which geometric structures M can be equipped with. To this end, we develop the notion of representation theory for affine manifolds. We also describe how an affine manifold can be equipped with a canonical flat, torsion-free connection and how geodesic completeness of this connection relates to the affine structure. Finally, we introduce the notion of an integral affine manifold and give some properties and examples of these objects.

In Chapter 3, we describe how the condition of the fundamental group being nilpotent relates to completeness of the affine structure. A common theme in the study of affine manifolds is that the fundamental group determines many properties of the affine structure. It is conjectured by Auslander in [Aus64] that the fundamental group of a complete affine manifold should be solvable. This was shown to be true up to dimension six in [AMS20]. We intertwine the ideas of this conjecture with another open problem called the *Markus conjecture*. This states that for affine manifolds, having parallel volume is equivalent to being complete. The main result of this chapter is that the Markus conjecture holds if the fundamental group is nilpotent as first shown in [FGH81].

Chapter 2 introduces Lagrangian fibrations and we describe some of their properties and give some examples. We show how the base space of a regular Lagrangian torus fibration inherits the structure of an integral affine manifold. The fibration allows us to realise the integral affine structure on the base space via so called action-angle coordinates. The action coordinates will determine the integral affine structure on the base and the angle coordinates will give coordinates on the fibres of the fibration. We will look at the obstructing to our manifold admitting *global* action-angle coordinates which was studied in detail by Duistermaat in [Dui80].

In Chapter 4, we show that the only compact surfaces that can admit an integral affine structure are the Klein bottle and the torus. Using the notion of completeness for affine structures and exploiting that the fundamental group of the torus is nilpotent, we can reduce the problem of classifying the integral affine structures on these surfaces to a problem in linear algebra and

group theory. Classifying the injective homomorphisms from the fundamental group of these surfaces to the integral affine transformation group of $\mathbb{R}^2$ such that the quotient is a smooth manifold, is shown to be equivalent to classifying the integral affine structures on these surfaces.

In Chapter 5, we classify the Lagrangian fibrations over each of these integral affine surfaces. The integral affine structure determines the local nature of such a Lagrangian fibration through the so called *symplectic reference fibration*. The symplectic reference fibration can be glued together in various ways to produce all Lagrangian fibrations over the base space that induce the given integral affine structure. Exploiting completeness of the integral affine structures on these surfaces, we can classify these Lagrangian fibrations topologically using techniques from group cohomology and sheaf theory. We explicitly calculate the invariants for these Lagrangian fibrations and show how we can construct them from their invariants using Luttinger surgery.

List of symbols

$C^\infty(M)$	Smooth functions on M .
dev	The developing map.
hol	The affine holonomy representation.
$\text{Aff}(\mathbb{R}^n)$	The group of affine transformations of $\mathbb{R}^n$.
$\text{Aff}_{\mathbb{Z}}(\mathbb{R}^n)$	The group of integral affine transformations of $\mathbb{R}^n$.
∇	A connection on the tangent bundle of a manifold.
$\pi_1(M)$	The fundamental group of M .

Contents

Preface	i
Summary	iii
Introduction	1
1 Preliminaries	5
1.1 Geometric Structures on Manifolds	5
1.1.1 Basic definition and properties	5
1.1.2 The developing map and holonomy representation	6
1.2 Affine Manifolds	10
1.2.1 Affine manifolds and connections	11
1.2.2 Completeness of affine structures	13
1.2.3 Parallel Volume	15
1.2.4 Affine Structures on the Torus	16
1.3 Group (Co)homology	17
1.4 Representation Theory and Affine Manifolds	20
1.4.1 Affine representations	20
1.4.2 Radiant affine manifolds	22
1.5 Integral Affine Manifolds	26
2 Lagrangian Fibrations and Integral Affine Manifolds	29
2.1 Symplectic Geometry	29
2.1.1 The Cotangent Bundle	30
2.1.2 Lagrangian Submanifolds	31
2.2 Lagrangian Fibrations	32
2.2.1 Definition of Lagrangian Fibration	32
2.3 Integral Affine Structure on the Base	34
2.3.1 Torus Bundles	34
2.3.2 (Global) Action-Angle Coordinates	40
2.3.3 Completely integrable Hamiltonian systems	42
3 Nilpotent Holonomy and Completeness	45
3.1 Nilpotent Holonomy	45
3.1.1 Nilpotent Groups	45
3.1.2 Unipotence and the Fitting submodule	46
3.2 Unipotent Representations and Completeness	50
3.2.1 Expansions in the linear holonomy	50
3.2.2 Algebraic groups and representation theory of Lie algebras	50

3.2.3	Consequences of expansions in linear holonomy	54
3.3	Parallel volume	56
4	Classification of Integral Affine Closed Surfaces	61
4.1	Classification of integral affine circles	61
4.2	Some basic results	62
4.3	Classification of Integral Affine structures on the torus	63
4.3.1	Completeness and freeness of the action	63
4.3.2	Finding the possible generators	65
4.3.3	When are two structures equivalent?	67
4.4	Classification of Integral Affine structures on the Klein Bottle	71
4.4.1	Completeness of integral affine structures on K^2	71
4.4.2	Determining the linear holonomy	72
4.4.3	Translational component and freeness of the action	74
4.4.4	Fundamental domains for the action of the holonomy group on $\mathbb{R}^2$. . .	80
5	Classification of Lagrangian Fibrations over Closed Surfaces	81
5.1	The Classification Problem	81
5.2	Latticed one-forms	81
5.2.1	Calculating Cohomological Invariant of Shifts	83
5.3	Luttinger Surgery	86
5.4	Chern Class and Lagrangian Class	88
5.4.1	The Sheaf Theory Approach	90
5.4.2	The Chern Class	91
5.4.3	The Lagrangian Class	92
5.5	Computing the Chern Class	95
5.5.1	Local Systems	95
5.5.2	Cohomology of Local Systems	95
5.5.3	Relation to group cohomology	97
5.5.4	Classifying Lagrangian fibrations over S^1	100
5.6	Computing $H^2(\Lambda)$ for the torus	100
5.7	Computing $H^2(\Lambda)$ for the Klein bottle	102
5.7.1	Spectral Sequences	102
5.7.2	Calculating $H^1(\mathbb{Z}, H^1(\mathbb{Z}, A))$	105
5.8	Constructing the fibrations from their invariants.	108
5.8.1	The construction	109
	Conclusion	111

Introduction

Affine manifolds

If one forgets to bring their ruler to Euclidean space, they may stumble upon the notion of an *affine geometry*. Although a ruler has been lost, an affine geometry still maintains the notion of straight lines and constant speed. An *affine manifold* is a manifold that is locally modelled on an affine geometry, by which we mean the charts take values in $\mathbb{R}^n$ viewed as an affine space and the transition maps are automorphisms of this affine space. The fundamental object on such manifolds turns out to be a flat, torsion free connection ∇ . This connection allows one to make sense of differentiation on an affine manifold and hence make sense of the notion of a parameterized curve having zero acceleration. Curves on an affine manifold satisfying this property will be called *geodesics*. Equivalently, a curve on an affine manifold is a geodesic if it is a straight line in every affine chart.

Of fundamental importance in the theory of affine manifolds is the case when every geodesic can be continued indefinitely. Affine manifolds satisfying this property are said to be *complete*. In [AM55] Auslander and Markus showed that completeness of an affine manifold M is equivalent to M being the quotient of a discrete group of affine transformations acting freely and properly on $\mathbb{R}^n$. There are many open questions surrounding completeness of affine manifolds. For example, it was conjectured by Auslander in [Aus64] that the fundamental group of such an affine manifold cannot be too "big" - namely that it should be solvable up to finite index. If this conjecture holds, then it implies that a free group should not be able to act on $\mathbb{R}^n$ by affine transformations such that the quotient is a smooth, compact affine manifold.

The motivation behind this conjecture comes from the fact that affine geometry shares many of the same properties as the more familiar Euclidean geometry. A group which acts isometrically on $\mathbb{R}^n$ such that the quotient is a smooth, compact manifold was shown to be abelian up to finite index by Bieberbach. He classified these groups called *Euclidean crystallographic groups* in [Bie10] and [Bie12]. Unexpectedly, Margulis showed in [Mar83] that a free group could act freely and properly on $\mathbb{R}^n$ by affine transformations, however, the quotient was not a *compact* manifold. Even though these affine manifolds failed to be compact, they still garnered a lot of interest and were dubbed *Margulis spacetimes*, owing to the fact that they inherit a unique parallel Lorentzian metric.

Another open conjecture in affine geometry is the *Markus conjecture* which was posed by Markus in [Mar63]. It states that if a compact affine manifold possesses a volume form which is *parallel* with respect to the affine connection, then the affine manifold is complete. In Chapter 3, we give the positive answer to this conjecture in the case the fundamental group is nilpotent which was first proven in [FGH81]. This conjecture is open in full generality, however it has been solved in some other special cases. For example, in [Car89] they showed the conjecture holds in the case that the affine manifold is a compact, flat Lorentzian manifold.

Integral affine manifolds and Lagrangian fibrations

If an affine manifold (or its orientable double cover) admits a parallel volume form, then the linear component of all its transition maps must live in the group of invertible matrices of determinant ± 1 . An interesting subgroup of this is the group $GL(\mathbb{Z}^n)$, consisting of invertible *integer* valued matrices whose inverse is integer valued. Affine manifolds where the linear component of their transition maps lie in this group are called *integral affine manifolds*. As well as having a characterization in terms of admitting a flat, torsion-free connection whose holonomy lies in $GL(\mathbb{Z}^n)$, an integral affine structure on a manifold M can also be shown to be equivalent to a full-rank lattice in the cotangent bundle of M that is locally spanned by closed one-forms. It is well known that the cotangent bundle of any manifold admits a *symplectic structure* and this observation connects the study of integral affine manifolds to symplectic geometry.

A *symplectic manifold* is a manifold that admits a closed, non-degenerate two-form called the symplectic form. By necessity, such a manifold is always even dimensional. There is an important class of submanifolds of symplectic manifolds called *Lagrangian submanifolds*. These are submanifolds which are half the dimension of the ambient manifold and such that the symplectic form restricts to zero along them. If B is any manifold, then its cotangent bundle T^*B is symplectic and the fibres of the projection map $\pi : T^*B \rightarrow B$ are Lagrangian submanifolds of the cotangent bundle. This leads into the more general notion of a *Lagrangian fibration*. If M is a symplectic manifold then a Lagrangian fibration is a surjection $\pi : M \rightarrow B$ where B is any smooth manifold, such that the fibres of the projection are Lagrangian submanifolds.

Of particular interest is when π is a surjective submersion and the fibres are compact and connected. In this case, the fibres are tori and the base space B inherits the structure of an integral affine manifold. The integral affine structure on the base turns out to be an invariant of the Lagrangian fibration. Moreover, the lattice in the cotangent bundle of B which induces this integral affine structure gives rise to a canonical Lagrangian torus fibration over B . This fibration is a local model for all Lagrangian torus fibrations over B which induce the given integral affine structure. The obstruction to a generic Lagrangian torus fibration being globally equivalent to this local model was studied in detail by Duistermaat in [Dui80]. He defined a characteristic class called the *Chern class* and its vanishing gives rise to a global section of the Lagrangian fibration which can be used to identify it with the local model.

This identification is only topological in the sense that the symplectic structure of the fibration is not preserved. However, if the global section determines a Lagrangian submanifold of the total space of the fibration then the fibration identifies both topologically and symplectically with the local model as shown also in [Dui80]. This was expanded upon further by Zung in [Zun01]. He defined another characteristic class called the *Lagrangian class* which maps onto the Chern class through a certain group homomorphism. If this characteristic class vanishes then the Lagrangian fibration will be both topologically and symplectically equivalent to the local model.

Outline

In this section we will briefly present the main structure of the thesis which can hopefully be used as a guide to the reader. The primary goal of this thesis is to classify the Lagrangian fibrations over closed surfaces. Our path to achieving this classification will require us to use tools from many different fields of mathematics such as symplectic geometry, algebraic topology and group theory.

In Chapter 1 we develop some results from the general theory of (integral) affine manifolds, the representation theory of groups and group cohomology. We describe what is meant by *completeness* of an affine manifold and give two equivalent perspectives on this property. The first view point introduces the notion of the *developing map* which is a local diffeomorphism from the universal cover of the affine manifold into $\mathbb{R}^n$. An affine manifold is then called complete if this developing map is a global diffeomorphism. The other equivalent definition of completeness is obtained by noting that an affine manifold inherits a canonical flat, torsion-free connection. Geodesic completeness of this connection is then equivalent to completeness of the affine structure. We also introduce integral affine manifolds which can be realised as a full-rank lattice in the cotangent bundle of the manifold that is locally spanned by closed one-forms. These will become key objects in the classification of Lagrangian fibrations as the base space always inherits such a structure.

In Chapter 2 we explore these objects in more detail. It is well known that the cotangent bundle of any smooth manifold inherits the structure of a *symplectic manifold*. A symplectic manifold is a manifold equipped with a closed, non-degenerate two form. The fact that an integral affine structure is induced from a lattice in this cotangent bundle provides an important link between integral affine geometry and symplectic geometry. The main result of this section is that any Lagrangian fibration over a fixed base B gives B the structure of an integral affine manifold. There is an action of the cotangent bundle of B on any Lagrangian fibration over B and the fixed-point set of this action corresponds to a lattice which induces the integral affine structure.

In Chapter 3 we start developing the tools required to classify the integral affine structures on closed surfaces. The main result of this section is that if the fundamental group of a compact affine manifold is nilpotent, completeness of the affine structure is equivalent to the existence of a volume form which is parallel with respect to the canonical affine connection. This is a partial resolution to a conjecture of Markus in [Mar63] which states that this should hold true even when the fundamental group is not nilpotent.

Chapter 4 contains the classification of the integral affine structures over closed surfaces. This is one of the main invariants of a Lagrangian fibration over these surfaces. Using the tools built up in Chapter 1 and Chapter 3 we are able to reduce the problem of classifying these structures to a relatively simple computation involving linear algebra and group theory. We also make use of a powerful theorem proved by Milnor in [Mil58] which states that a closed, orientable affine surface must have genus less than 2.

In the last chapter, we complete the classification of the Lagrangian fibrations over the Klein bottle and the torus. In order to do this, we find two more invariants of the fibration. These are the *Chern class* and the *Lagrangian class*. The Chern class determines the topological nature while the Lagrangian class is a refinement of this and determines the symplectic structure of the fibration. These invariants are defined in terms of sheaf cohomology which can make explicit calculation of them difficult in practice. However, we show that these cohomology group can be computed using group cohomology which we developed in Chapter 1. We calculate these explicitly for both the torus and the Klein bottle.

CHAPTER 0. INTRODUCTION

Finally, we show how to explicitly construct a Lagrangian fibration over the torus or Klein bottle from their invariants. This is done using a symplectic version of Dehn surgery called Luttinger surgery and first described in [Lut95]. This completes the main content of this thesis and gives an explicit way to construct any given Lagrangian torus fibration over a closed surface.

Preliminaries

The goal of this chapter is to provide some of the background needed to begin our classification of Lagrangian fibrations over compact surfaces. We will start with some of the theory of affine manifolds as these will prove to be a crucial structure that the base space of such a fibration inherits. These belong to a wider class of geometries called locally homogeneous geometric structures. We will develop the theory of affine manifolds in the broader context of these geometries.

This will lead us to consider representations of the fundamental group of such manifolds through the *holonomy representation*. We will spend sometime to introduce results from the representation theory of affine manifolds which we will later use in our classification results. After this, we specialize to the case of integral affine manifolds as these will turn out to be an invariant of the base space of a Lagrangian fibration.

1.1 Geometric Structures on Manifolds

1.1.1 Basic definition and properties

One of the main objects of study throughout this thesis are affine manifolds. These belong to a class of geometries called *locally homogeneous geometric structures*. The idea behind these structures is that we can describe our geometry through charts into some model space X and that we can stitch these charts together through elements of some Lie group G which acts transitively on the model space.

We will require that our group G be a Lie group acting transitively on X .

Definition 1.1. Let $U \subset X$ be an open set of our model space X . We say that a smooth map $f : U \rightarrow X$ is *locally- G* if for each component $U_i \subset U$ there is a $g_i \in G$ such that $f|_{U_i} = g_i|_{U_i}$.

We also require the map f to satisfy the following unique extension property. If $U \subset X$ and $f : U \rightarrow X$ is a locally- G map then there is a unique $g \in G$ restricting to f .

Definition 1.2. A (G, X) -atlas is a pair $(\mathcal{U}, \Phi)$ consisting of an open covering for M and a collection of charts such that the transition maps are locally- G on their respective overlaps.

Definition 1.3. A map $f : M \rightarrow N$ between two (G, X) -manifolds is a (G, X) -map if for each pair of charts (ϕ, φ) then $\phi \circ f \circ \varphi^{-1}$ is locally- G on the relevant intersection.

Example 1.4. A *hyperbolic surface* Σ is a (G, X) -structure where X is the hyperbolic plane $\mathbb{H}^2$ and $G = \text{PSL}(2, \mathbb{R})$.

Example 1.5. A *real projective structure* on M is a (G, X) -structure where $X = \mathbb{RP}^n$ and $G = \text{PGL}(n+1, \mathbb{R}) = \frac{\text{GL}(n+1, \mathbb{R})}{\mathbb{R}^*}$.

Example 1.6. A *flat Riemannian structure* or *Euclidean structure* on M is a (G, X) -structure where $G = O(n) \ltimes \mathbb{R}^n$ and $X = \mathbb{R}^n$.

If M is a (G, X) -manifold and $\pi : \tilde{M} \rightarrow M$ is a regular covering, it is always possible to pull back the (G, X) -structure on M to $\tilde{M}$.

Lemma 1.7. Let M be equipped with a (G, X) -structure and suppose that $\pi : \tilde{M} \rightarrow M$ is a regular covering map. Then the (G, X) -structure on M can be pulled back to $\tilde{M}$.

Proof. Since π is a covering map for each $x \in M$ there exists an open set U_x of x such that $\pi^{-1}|_{U_x}$ is a disjoint union $\bigsqcup_i V_i$ and $\pi|_{V_i}$ is a homeomorphism. If necessary, we shrink the open set U_x so that we may assume it lies entirely inside a (G, X) -chart (W, φ) . Now, doing this for each $x \in M$, we define charts on $\tilde{M}$ by $(V_i, \varphi \circ \pi|_{V_i})$. Since π is surjective and restricts to a homeomorphism on each V_i , it is clear that this defines an atlas on $\tilde{M}$. It remains to check that our transition maps lie in G .

Suppose $V_i \cap V_j \neq \emptyset$. Then $\pi(V_i) \cap \pi(V_j) \neq \emptyset$ but we know that they lie inside (G, X) -charts (W_i, φ_i) and (W_j, φ_j) respectively with W_i possibly equal to W_j . If they are distinct charts then since M has a (G, X) -structure we know that their transition maps lie in G . But the transition maps for (V_i, φ_i) , (V_j, φ_j) on $\tilde{M}$ are

$$\begin{aligned} (\varphi_i \circ \pi|_{V_i}) \circ (\varphi_j \circ \pi|_{V_j})^{-1} &= (\varphi_i \circ \pi|_{V_i}) \circ (\pi|_{V_j}^{-1} \circ \varphi_j^{-1}) \\ &= \varphi_i \circ \varphi_j^{-1} \end{aligned}$$

which is the transition map between W_i and W_j and hence in G as M has a (G, X) -structure. $\square$

1.1.2 The developing map and holonomy representation

For (G, X) -manifolds, there is an inherently nice way to "globalize" the charts via a (G, X) -map called the developing map. The study of this map will be of importance to us throughout this thesis.

Proposition 1.8. Let M be a simply connected (G, X) -manifold. Then there exists a (G, X) -map

$$\text{dev} : M \rightarrow X.$$

Proof. We pick a base point $x_0 \in M$ and a chart (U_0, φ_0) around x_0 . Now for any $x \in M$ we define the map dev as follows. We pick a curve $x(t)$ on M satisfying $x(0) = x_0$ and $x(1) = x$. As the image of our curve is compact, we can cover it by finitely many coordinate charts (U_i, φ_i) with $i \in \{0, \dots, n\}$ such that $x(t) \in U_i$ for $t \in (a_i, b_i)$ with

$$a_0 < 0 < a_1 < b_0 < a_2 < b_1 < a_3 \cdots < a_n < b_{n-1} < 1 < b_n$$

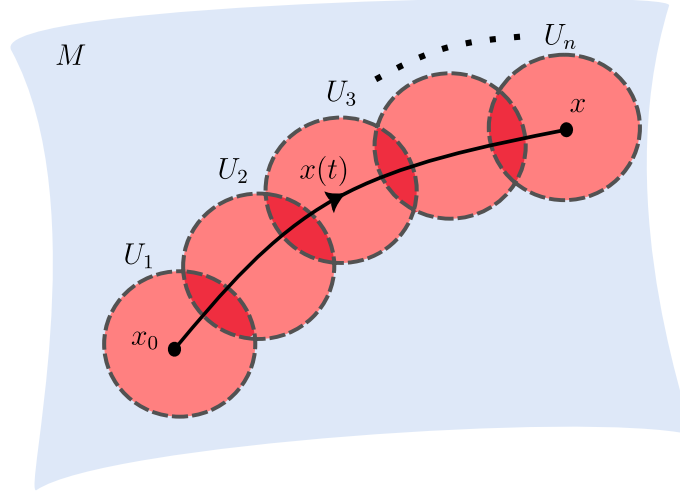


Figure 1.1: Developing along a curve

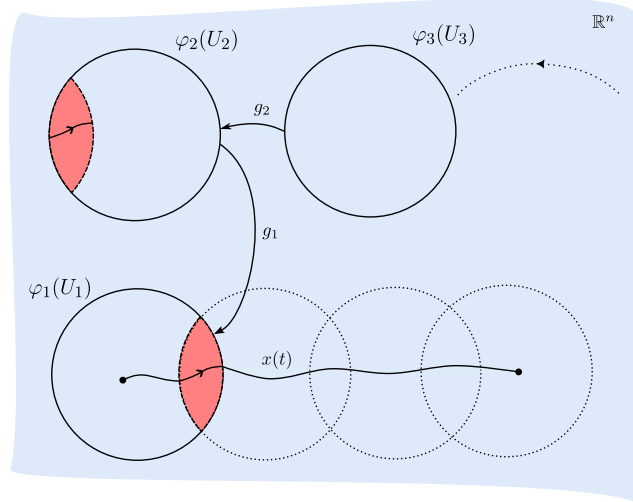


Figure 1.2: We successively glue our charts together using the transition maps

Now we consider where these charts overlap. Since our manifold is equipped with a (G, X) -structure our transition maps differ by an element of G . Let $g_i = \varphi_{i-1} \circ \varphi_i^{-1} \in G$ be the transition map between U_i and U_{i-1} . We then define

$$\text{dev}(x) = g_1 g_2 \dots g_n \varphi_n(x).$$

We must show that this is a well-defined map. We first show that this map does not change if we refine the cover. Suppose we insert a chart (V, ϕ) between U_i and U_{i-1} . Let $h_i, h_{i-1} \in G$ be the unique group elements such that

$$\begin{aligned} \varphi_{i-1} &= h_{i-1} \circ \phi \text{ on } V \cap U_{i-1} \text{ and} \\ \phi &= h_i \varphi_i \text{ on } V \cap U_{i-1}. \end{aligned}$$

Then on $V \cap U_i \cap U_{i-1}$ we have that $\varphi_{i-1} = h_{i-1} \circ h_i \circ \varphi_i$. This gives us that $g_i = \varphi_{i-1} \circ \varphi_i^{-1} = h_{i-1} \circ h_i$ as we require that our transition maps satisfy a unique extension property.

CHAPTER 1. PRELIMINARIES

Now when we develop along our curve with respect to this new covering, we obtain that:

$$\begin{aligned}\text{dev}(x) &= g_1 \dots g_{i-1} h_{i-1} h_i g_{i+1} \dots g_n \varphi_n(x) \\ &= g_1 \dots g_{i-1} g_i g_{i+1} \dots g_n \varphi_n(x)\end{aligned}$$

This defines dev along $x(t)$. We now need to show that it does not depend on our choice of curve. As M is simply connected, all curves with the same start point and end point will be homotopic. By compactness we can split our homotopy into smaller homotopies such that we can break up our curve into regions

$$a_1 = 0 < a_2 < \dots < a_n = 1$$

such that during each small homotopy the segment $x((a_i, a_{i+1}))$ lies entirely inside a coordinate patch. It then follows that dev is independent of our choice of paths completing the proof. $\square$

This shows that we have a well defined developing map on any simply connected (G, X) -manifold. However, this map depended on the choice of base point and initial chart. We immediately get the following corollary from this fact.

Corollary 1.9. Let M be a (G, X) -manifold and let $\tilde{M}$ be its universal cover. Then there exists a pair (dev, hol) satisfying

$$\begin{aligned}\text{dev} : \tilde{M} &\rightarrow X \\ \text{hol} : \pi_1(M) &\rightarrow G\end{aligned}$$

where $\text{hol}(\gamma)$ is the element $g_1 \dots g_n$ of G obtained from developing around a loop $\gamma \in M$. The map dev is a local diffeomorphism and the map hol is a group homomorphism.

The development pair satisfy the following equivariance condition. Let $\gamma \in \pi_1(M)$ act by deck transformations on $\tilde{M}$. Then we have the following.

Lemma 1.10. Let $\gamma \in \pi_1(M)$. Then

$$\text{dev}(\gamma \cdot x) = \text{hol}(\gamma) \text{dev}(x)$$

for $x \in \tilde{M}$.

Proof. Let $\gamma \in \pi_1(M)$ and let $\{(V_i, \phi_i)\}$ be a cover for γ . We pick a lift of γ to the universal cover $\tilde{M}$ which we denote by $\tilde{\gamma}$. We can pull back our cover of γ to a covering of the lift $\tilde{\gamma}$. Denote the transition maps on $V_i \cap V_{i+1}$ by g_i . We now wish to compute the action of γ on a point $x \in \tilde{M}$.

Letting $x_0 \in \tilde{M}$ be the base point for our developing map, we pick a curve $\tilde{\alpha}$ joining x_0 to x . We consider the composition $\alpha \circ \gamma$. This has a unique lift to $\tilde{M}$ by choosing a lift of α such that $\alpha'(0) = \tilde{\gamma}(1)$. Now, concatenating these curves and evaluating the endpoint gives the action i.e. $\text{dev}(\gamma \cdot x) = (\alpha' \cdot \tilde{\gamma})(1)$.

1.1. GEOMETRIC STRUCTURES ON MANIFOLDS

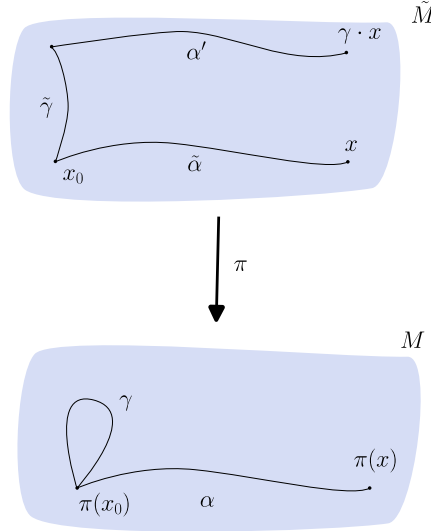


Figure 1.3: The action $\gamma \cdot x$.

Now, we note that starting at $x_0 \in \tilde{M}$ and first developing along $\tilde{\gamma}$ we pick up the group element $\text{hol}(\gamma)$ by definition. We claim that starting at $\tilde{\gamma}(1)$ and developing along α' we pick up exactly $\text{dev}(x)$. This is true because the lifts α' and $\tilde{\alpha}$ both project onto α by the covering map and since the (G, X) -structure on $\tilde{M}$ is defined by pulling back the one on M , it follows that developing along α' yields $\text{dev}(x)$. $\square$

The role of the development pair is to "globalize" the coordinate chart of M . Indeed, the following proposition makes this statement clearer.

Proposition 1.11. Let M be a (G, X) -manifold with development pair (dev, hol) . Denote by Γ the image of $\pi_1(M)$ under the holonomy map. Then there exists a (G, X) -atlas such that the transition maps lie in Γ .

Proof. We pick $m \in M$ and fix a lift $\tilde{m} \in \tilde{M}$ of m . Let $p : \tilde{M} \rightarrow M$ be the universal cover. Then, as dev is a local diffeomorphism, there is an open set $\tilde{U}$ of $\tilde{m}$ such that dev is a diffeomorphism onto some open subset of X . By shrinking $\tilde{U}$ if necessary, we may assume that $V = p(\tilde{U})$ is evenly covered i.e. $p^{-1}(V) = \bigsqcup_{i \in I} V_i$ with p a diffeomorphism on each V_i .

We define coordinate charts on M by $(V_i, \psi_i = \text{dev} \circ p_i^{-1})$ at a point $m \in M$ where $p_i : V_i \rightarrow V$ is a diffeomorphism.

Now suppose that $V_i \cap V_j \neq \emptyset$ and let $\tilde{U}_i = p_i^{-1}(V_i)$ and $\tilde{U}_j = p_j^{-1}(V_j)$. Then the preimage of their intersection will differ by a deck transformation $\gamma \in \pi_1(M)$. Therefore, on the intersection $V_i \cap V_j$

$$\begin{aligned} \text{dev} \circ p_i^{-1} &= \text{dev} \circ \gamma \circ p_j^{-1} \\ \Rightarrow p_i^{-1} \circ p_j &= \gamma \end{aligned}$$

CHAPTER 1. PRELIMINARIES

Now our by definition of our transition maps

$$\begin{aligned}\psi_i \circ \psi_j^{-1} &= \text{dev} \circ p_{i-1} \circ p_j \circ \text{dev}^{-1} \\ &= \text{dev} \circ \gamma \circ \text{dev}^{-1} \\ &= \text{hol}(\gamma) \text{dev} \circ \text{dev}^{-1} \\ &= \text{hol}(\gamma) \in \Gamma\end{aligned}$$

and so we have given our coordinate charts on M in terms of the development pair. $\square$

This shows us that our (G, X) -structure is determined by the development pair. Conversely, the following lemma shows us that once we have fixed our (G, X) -structure, the development pair is determined up to certain factors/conjugations.

Lemma 1.12. Let M be a (G, X) -manifold with development pair (dev, hol) . Suppose that $(\text{dev}', \text{hol}')$ is another development pair for M . Then $\text{dev}' = g \circ \text{dev}$ and $\text{hol}' = g \circ \text{hol} \circ g^{-1}$ for some $g \in G$.

Proof. Our development pair (dev, hol) was determined once we picked a base point $x_0 \in \tilde{M}$ and a chart (U, ϕ) of x_0 . Suppose that we pick a different base point $y_0 \in \tilde{M}$ and initial chart (V, φ) . This determines a new development pair $(\text{dev}', \text{hol}')$ and we wish to determine the relationship between the two pairs.

We pick a curve γ joining $y_0 \mapsto x_0$. For a point $x \in \tilde{M}$, we can now define $\text{dev}'(x)$ by picking a cover $\{V_i\}_{i \in I}$ of the curve γ , developing along γ with respect to this cover, and then developing along any curve connecting x_0 to x . However, by definition of the developing map, if we develop along the curve γ we pick up some group element $g \in G$ by composing the transition maps of the covering. Then, when we develop from x_0 to x , this is by definition exactly the same as the value of our initial developing map $\text{dev}(x)$. Hence we have that $\text{dev}'(x) = g \text{dev}(x)$ for some $g \in G$.

Now to show that $\text{hol}' = g \circ \text{hol} \circ g^{-1}$ we consider

$$\begin{aligned}\text{dev}'(\gamma \cdot \alpha) &= g \circ \text{dev}(\gamma \cdot \alpha) \\ &= g \circ \text{hol}(\gamma) \circ \text{dev}(\alpha) \\ &= (g \circ \text{hol}(\gamma) \circ g^{-1})(g \circ \text{dev}(\alpha)) \\ &= (g \circ \text{hol}(\gamma) \circ g^{-1}) \text{dev}'(\alpha)\end{aligned}$$

and as the development pair satisfied the equivariance property it follows that $\text{hol}' = g \circ \text{hol} \circ g^{-1}$. $\square$

This shows that once we have fixed a development pair, any other development pair is related by $(\text{dev}', \text{hol}') = (g \circ \text{dev}, g \circ \text{hol} \circ g^{-1})$. We should therefore note that if we want to classify (G, X) -structures on a manifold M by classifying the possible development pairs, it is important to consider them up to this relation.

1.2 Affine Manifolds

Of primary interest to us throughout this thesis are *affine structures* on a manifold M . These are (G, X) -structures whereby G is the group of affine transformations of $\mathbb{R}^n$ and the model

space X is simply $\mathbb{R}^n$. There are many interesting properties and open questions surrounding this class of geometric structures which we shall see throughout this thesis. For now we give some of the basic definitions and properties regarding these objects.

Definition 1.13. An affine transformation of $\mathbb{R}^n$ is a map

$$A : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

such that $A(x) = Gx + b$ where $G \in GL(\mathbb{R}^n)$ and $b, x \in \mathbb{R}^n$.

Definition 1.14 (Affine Group). The group

$$\text{Aff}(\mathbb{R}^n) = GL(\mathbb{R}^n) \rtimes \mathbb{R}^n$$

is the affine group of $\mathbb{R}^n$ and consists of all the affine transformations of $\mathbb{R}^n$.

1.2.1 Affine manifolds and connections

In $\mathbb{R}^n$ we can differentiate vector fields $X, Y \in \mathfrak{X}(M)$ by taking the directional derivative $\nabla_X Y$. One can extend the differentiation of vector fields in $\mathbb{R}^n$ to arbitrary manifolds through the notion of a *connection*. One can define a connection for any vector bundle but we will focus only on the case our vector bundle is the tangent bundle of a smooth manifold.

Definition 1.15. Let M be a smooth manifold and denote by TM its tangent bundle. Then a connection on TM is a map

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$$

which satisfies the following properties.

- i) $\nabla_X Y$ is linear over $C^\infty(M)$ in X : for $f_1, f_2 \in C^\infty(M)$ and $X_1, X_2 \in \mathfrak{X}(M)$,

$$\nabla_{f_1 X_1 + f_2 X_2} Y = f_1 \nabla_{X_1} Y + f_2 \nabla_{X_2} Y.$$

- ii) $\nabla_X Y$ is linear over $\mathbb{R}$ in Y : for $a_1, a_2 \in \mathbb{R}$ and $Y_1, Y_2 \in \mathfrak{X}(M)$,

$$\nabla_X (a_1 Y_1 + a_2 Y_2) = a_1 \nabla_X Y_1 + a_2 \nabla_X Y_2.$$

- iii) ∇ satisfies the following product rule: for $f \in C^\infty(M)$,

$$\nabla_X (fY) = f \nabla_X Y + X(f)Y.$$

Remark 1.16. The term connection comes from the fact that a connection provides a way to identify fibres in a vector bundle by transporting a basis along a curve in a way that is covariantly constant with respect to the connection.

Definition 1.17 (Christoffel Symbols). Let $E_1, \dots, E_n$ be a frame on M and let ∇ be a connection in TM . Then we have that

$$\nabla_{E_i} E_j = \sum_k \Gamma_{ij}^k E_k$$

for some smooth functions Γ_{ij}^k . These functions are called the *Christoffel symbols* of ∇ .

Definition 1.18. A vector field $X \in \mathfrak{X}(M)$ will be called *parallel* with respect to ∇ if

$$\nabla_Y X = 0$$

for all $Y \in \mathfrak{X}(M)$. We will write this condition as $\nabla X = 0$.

Fixing an affine structure on M will turn out to be equivalent to equipping M with a certain type of connection. In order to show this we need to introduce the notion of curvature and torsion of a connection.

Definition 1.19 (Torsion tensor). The *torsion tensor* of a connection ∇ on TM is defined to be

$$\tau(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$$

where $X, Y \in \mathfrak{X}(M)$.

Next, we introduce the notion of curvature which measures the failure of differentiating with respect to the connection ∇ to commute.

Definition 1.20 (Curvature Tensor). The *curvature tensor* of a connection ∇ on TM is defined as

$$R(X, Y)(Z) = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$$

where $X, Y, Z \in \mathfrak{X}(M)$.

Definition 1.21. A connection will be called *flat* if $R(X, Y)Z = 0$ and *torsion free* if $\tau(X, Y) = 0$ for all $X, Y, Z \in \mathfrak{X}(M)$.

We now illustrate the link between affine structures on M and connections on TM .

Lemma 1.22. An affine structure on a smooth manifold M is equivalent to a flat, torsion free connection on TM .

Proof. Let M be an affine manifold and $\tilde{M}$ be its universal cover. Then since dev is a local diffeomorphism, we can pull back the standard flat, torsion free connection ∇ on $\mathbb{R}^n$ to obtain a flat, torsion free connection on $\tilde{M}$ denoted by $\text{dev}^* \nabla$. We now show that this connection descends to M .

We show that ∇ is invariant under deck transformations. We have that

$$\begin{aligned}\gamma^* \text{dev}^* \nabla &= (\text{dev} \circ \gamma)^* \nabla \\ &= (\text{hol}(\gamma) \circ \text{dev})^* \nabla \\ &= \text{dev}^* \text{hol}(\gamma)^* \nabla\end{aligned}$$

where $\gamma \in \pi_1(M)$. But affine maps preserve the euclidean connection as they preserve the notion of zero acceleration and so

$$\text{dev}^* \text{hol}(\gamma)^* \nabla = \text{dev}^* \nabla$$

and hence deck transformations preserve the connection and it descends to M .

Conversely, suppose that M is equipped with a flat, torsion free connection ∇ . Let $e_1, \dots, e_n$ be a basis for $T_p M$ for some $p \in M$. Then since the connection ∇ is flat, there is an open neighborhood $U \subset M$ of p such that parallel transport by ∇ is path-independent. Hence, we parallel transport e_i using ∇ to get a frame $E_1, \dots, E_n$ on U . Now, since ∇ is torsion-free and the vector fields E_i are parallel, we have that

$$[E_i, E_j] = 0.$$

Now, Theorem 9.46 of [Lee03] implies that this frame gives rise to coordinates $x^1, \dots, x^n$ on U such that $\frac{\partial}{\partial x^i} = E_i$. Now, suppose that $y^1, \dots, y^n$ are coordinates defined in the same way on V . The transformation law for the Christoffel symbols yields

$$\tilde{\Gamma}_{ij}^k = \frac{\partial y^k}{\partial x^m} \Gamma_{ij}^m + \frac{\partial^2 y^k}{\partial x^i \partial x^j}$$

where we sum over repeated indices. But since the $\frac{\partial}{\partial x^i}$ and $\frac{\partial}{\partial y^i}$ are parallel, i.e. $\nabla \frac{\partial}{\partial x^i} = \nabla \frac{\partial}{\partial y^i} = 0$, we have that $\Gamma_{ij}^m = \tilde{\Gamma}_{ij}^k = 0$ and hence

$$\frac{\partial^2 y^k}{\partial x^i \partial x^j} = 0.$$

Integrating this gives that $y^k = A_i^k x^i + b^k$ and hence our transition maps are affine maps. $\square$

Remark 1.23. In local affine coordinates, this connection is just the usual covariant derivative in $\mathbb{R}^n$.

1.2.2 Completeness of affine structures

Associated to a connection, one has the notion of *geodesics*. Given a curve $\gamma : (0, 1) \rightarrow M$ one can associate its velocity vector field $\dot{\gamma}$ which is a vector field along γ . To avoid some technicalities, we will assume that this velocity vector field has an extension to a vector field on all of M .

Definition 1.24. Let M be a smooth manifold and let $\gamma : (0, 1) \rightarrow M$ be a curve on M . Then γ is a *geodesic* if $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$.

Definition 1.25. A connection ∇ is said to be *complete* if its geodesics are defined for all time t .

Remark 1.26. The Euclidean plane $\mathbb{R}^2$ with the usual connection is complete as its geodesics are straight lines and can be continued indefinitely. However, if we remove the origin $0 \in \mathbb{R}^2$, then $\mathbb{R}^2 \setminus 0$ is no longer complete with its usual connection as a straight line cannot be continued through the origin.

For affine manifolds, there is a relationship between this notion of completeness and a different notion of completeness coming from the theory of (G, X) -manifolds.

Definition 1.27. A (G, X) -manifold M is said to be complete if its developing map is a diffeomorphism from M to X .

Remark 1.28. In the case that M is a Riemannian manifold, this notion of completeness is equivalent to completeness of the Levi-Civita connection of M .

Proposition 1.29. Let M be a (G, X) -manifold with X simply connected. Then M is complete in the sense of Definition 1.27 if and only if $M = X/\Gamma$ with $\Gamma \subset G$.

Proof. Suppose that $M = X/\Gamma$. Then covering space theory tells us that $\pi : X \rightarrow X/\Gamma$ is its universal cover and is Γ its group of deck transformations. Letting $\text{dev} = \text{id} : X \rightarrow X$ the equivariance property is now satisfied trivially by Γ .

Conversely, suppose that $\text{dev} : \tilde{M} \rightarrow X$ is a diffeomorphism. Then, by equivariance of the holonomy representation $\text{hol} : \pi_1(M) \rightarrow G$, we have

$$\text{dev}(\gamma \cdot x) = \text{hol}(\gamma)\text{dev}(x)$$

where $\gamma \in \pi_1(M)$ acts by deck transformations. Hence, deck transformations correspond to the action by the holonomy group under this identification. Since deck transformations act freely and properly on the universal cover of M , it follows as dev is a diffeomorphism that $\text{hol}(\pi_1(M)) \subset G$ acts freely and properly on X . By covering space theory $\tilde{M}/\pi_1(M) = M$ and since by completeness and equivariance of dev we have $\tilde{M}/\pi_1(M) \cong X/\text{hol}(\pi_1(M))$ the result follows. $\square$

For affine manifolds we have the following equivalence of completeness in the sense of Definition 1.27 and geodesic completeness of the associated flat, torsion-free connection.

Theorem 1.30. Let M be an affine manifold and let ∇ be its associated flat, torsion-free connection. Then ∇ is geodesically complete if and only if $\text{dev} : \tilde{M} \rightarrow \mathbb{R}^n$ is a diffeomorphism.

Proof. If dev is a diffeomorphism and D is the standard Euclidean connection on $\mathbb{R}^n$ then dev^*D is complete since there is a one-to-one correspondence between geodesics on $\tilde{M}$ and on $\mathbb{R}^n$. Now suppose that the descension ∇ of dev^*D to M is incomplete then there would be a geodesic on M that is incomplete. But this would lift to an incomplete geodesic on $\tilde{M}$ but $(\tilde{M}, \text{dev}^*D)$ is complete.

Conversely, suppose that ∇ is complete. Since if M is complete so is its universal cover, we may assume M is simply connected. Define the exponential map by

$$\exp_p : U \subset T_p M \rightarrow M$$

$$\exp_p(tv) = \gamma(t)$$

where γ is a geodesic with initial conditions $(p, v) \in M \times T_p M$. It is well known, for instance see [Gol22], that M is complete if and only if $\exp_p$ is defined on all of $T_p M$ for all $p \in M$. The following diagram can be seen to commute;

$$\begin{array}{ccc} T_p M & \xrightarrow{D(\text{dev})_p} & T_{\text{dev}(p)} \mathbb{R}^n \\ \exp_p \downarrow & & \downarrow \exp_{\text{dev}(p)} \\ M & \xrightarrow{\text{dev}} & \mathbb{R}^n \end{array}$$

Indeed, let $\gamma : (0, 1) \rightarrow M$ be a geodesic with initial conditions $(p, v) \in M \times T_p M$. Then

$$\text{dev} \circ \exp_p(tv) = \text{dev}(\gamma(t)).$$

On the other hand,

$$\exp_{\text{dev}(p)} D(\text{dev})_p(tv) = \exp_{\text{dev}(p)} \frac{d}{dt} \Big|_{t=0} \text{dev}(\gamma(t)) \quad (1.1)$$

But $\text{dev}(\gamma(t))$ is a geodesic since $\nabla = \text{dev}^* D$ and hence

$$\exp_{\text{dev}(p)} \frac{d}{dt} \Big|_{t=0} \text{dev}(\gamma(t)) = \text{dev}(\gamma(t)).$$

Since $D(\text{dev})_p$ and $\exp$ are bijections it follows that dev is also a bijection and hence a diffeomorphism. $\square$

1.2.3 Parallel Volume

We finish the discussion of connections on affine manifolds with the notion of a *parallel volume form*. Given a connection on TM one can also obtain a connection on the k -forms on M through the following Koszul-like formula

$$(\nabla_X \omega)(X_1, \dots, X_k) = X(\omega(X_1, \dots, X_k)) - \sum_{i=1}^k \omega(X_1, \dots, \nabla_X X_i, \dots, X_k)$$

where $\omega \in \Omega^k(M)$ and $X, X_i \in \mathfrak{X}(M)$. It is conjectured in [Mar63] that if an affine manifold possesses a *parallel volume form* then it is complete. This is now referred to as the *Markus conjecture*. By parallel volume form we mean a volume form $\omega \in \Omega^n(M)$ such that $\nabla \omega \equiv 0$.

Corollary 1.68 implies that M has parallel volume if and only if every element of its linear holonomy group is volume preserving. Another characterisation which arises from this observation is the following. If M is orientable, from

$$\pi_1(M) \xrightarrow{\text{Linhol}} \text{GL}(\mathbb{R}^n) \xrightarrow{\log \circ |\det|} \mathbb{R}$$

we see that there is an additive homomorphism $\nu_M \in \text{Hom}(\pi_1(M), \mathbb{R}) \cong H^1(M, \mathbb{R})$ called the *volume obstruction* and M possesses a parallel volume form if and only if $\nu_M = 0$.

This question is still open in full generality, however it has been solved in some specific cases. One of the main tasks of this thesis is to show that the Markus conjecture is true in the case of nilpotent holonomy. This will prove a powerful tool in classifying affine structures as completeness implies that the affine manifold is a quotient of its model space.

1.2.4 Affine Structures on the Torus

The standard affine structure on the torus is obtained from viewing the torus as $\mathbb{R}^n/\mathbb{Z}^n$ where $\mathbb{Z}^n$ acts by translations. This structure is easily seen to be a complete integral affine manifold. Indeed, the action is an action by the integral affine group and is both free and proper. Covering space theory tells us that $\pi : \mathbb{R}^n \rightarrow \mathbb{R}^n/\mathbb{Z}^n$ is a covering space with the group of deck transformations via our $\mathbb{Z}^n$ -action. In [AM55] it is shown that all complete affine structures are obtained in this manner. However, even for the 2-torus, not all affine structures are complete. We now give an example of an incomplete structure on $\mathbb{T}^2$.

Example 1.31. Let $\lambda > 1$ be a real number and consider the action of $\mathbb{Z}$ generated by multiplication by λ on $\mathbb{R}^2 - \{0\}$. This action is a free and proper action by the affine group and so the quotient $M = \mathbb{R}^2 - \{0\}/\langle \lambda \rangle$ is an affine manifold.

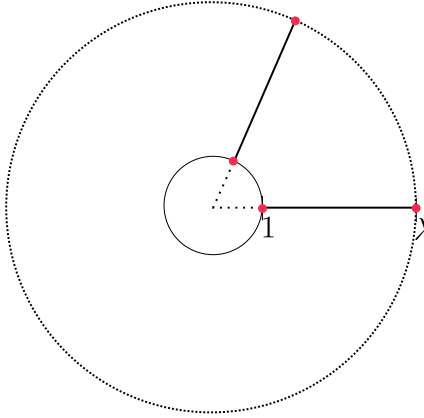


Figure 1.4: Visualisation of the Hopf manifold. The points 1 and λ are identified.

The map

$$\begin{aligned} \rho : \mathbb{R}^2 &\rightarrow \mathbb{R}^2 - \{0\} \\ (x, y) &\mapsto e^x(\cos y, \sin y) \end{aligned}$$

is a covering map. Basing our fundamental group at $[(1, 0)] \in M$, our group is generated by the loops γ_1 and γ_2 where γ_1 is the projection of the line segment joining the point $(1, 0) \mapsto (\lambda, 0)$ and γ_2 is the loop going around the unit circle once. These lift to curves $\bar{\gamma}_1$ and $\bar{\gamma}_2$ in the universal cover satisfying

$$\bar{\gamma}_1(0) = (0, \ln(1)) = (0, 0), \quad \bar{\gamma}_1(1) = (0, \ln(\lambda)),$$

$$\bar{\gamma}_2(0) = (0, 0), \quad \bar{\gamma}_2(1) = (0, 2\pi),$$

and from this we see that γ_1 and γ_2 act on (x, y) by

$$\gamma_1 \cdot (x, y) = (\ln(\gamma) + x, y)$$

$$\gamma_2(x, y) = (x, y + 2\pi).$$

Now, we define our developing pair by

$$\text{dev}(x, y) = \rho(x, y),$$

$$\text{hol}(\gamma_1) \cdot (x, y) = (\lambda x, \lambda y),$$

$$\text{hol}(\gamma_2) = \text{id}_{\mathbb{R}^2}.$$

We check that these satisfy the equivariant property;

$$\begin{aligned}\text{dev}(\gamma_1 \cdot (x, y)) &= \text{dev}((\ln(\lambda) + x, y)) \\ &= e^{\ln(\lambda)} e^x (\cos y, \sin y) \\ &= \lambda \text{dev}((x, y)) \\ &= \text{hol}(\gamma_1) \text{dev}((x, y)),\end{aligned}$$

$$\begin{aligned}\text{dev}(\gamma_2 \cdot (x, y)) &= \text{dev}((x, y + 2\pi)) \\ &= e^x (\cos(y + 2\pi), \sin(y + 2\pi)) \\ &= e^x (\cos y, \sin y) \\ &= \text{hol}(\gamma_2) \text{dev}((x, y)).\end{aligned}$$

It is now clear that this structure is incomplete as $0 \notin \text{dev}(\mathbb{R}^2)$. Indeed, it is typical that the fixed-point of a radiant structure is not in the developing image as shown in Theorem 6.5.2 [Gol22].

1.3 Group (Co)homology

A theme throughout this thesis will be that the holonomy representation of an affine manifold will encode many important properties of the affine structure. In order to illustrate this, we will need to develop some of the theory of affine representations. These results will require some machinery from the theory of group (co)homology. The aim of this section is to provide a brief overview of these theories although it will be far from a complete discussion. We will also assume a basic familiarity with modules over a ring. We refer the reader to [Wei94] and [Bro12] for a more in-depth review.

We start with the notion of the group ring.

Definition 1.32 (Group Ring). Let G be any group. The group ring $\mathbb{Z}[G]$ is the free abelian group generated by the elements of G . The multiplication is defined by

$$\left(\sum_{h \in G} m_h h \right) \left(\sum_{k \in G} n_k k \right) = \sum_{g \in G} \left(\sum_{hk=g} m_h n_k \right) g.$$

Definition 1.33. A $\mathbb{Z}[G]$ -module M is called trivial if $g \cdot m = m$ for all $m \in M$ and $g \in G$.

Remark 1.34. We can always view an abelian group A as a trivial $\mathbb{Z}[G]$ -module. We will often use this to view $\mathbb{Z}$ as a $\mathbb{Z}[G]$ -module.

Definition 1.35. A morphism of $\mathbb{Z}[G]$ -modules M and N is a homomorphism of the underlying abelian groups

$$\varphi : M \rightarrow N$$

such that $\varphi(g \cdot m) = g \cdot \varphi(m)$ for all $g \in G$ and $m \in M$.

Example 1.36 (Augmentation map). Let M be a $\mathbb{Z}[G]$ -module. Then there is a map

$$\epsilon : M \rightarrow \mathbb{Z}$$

$$\sum_{g \in G} m_g g \mapsto \sum_{g \in G} m_g$$

where $\mathbb{Z}$ is viewed as a trivial $\mathbb{Z}[G]$ -module.

Definition 1.37. The kernel of this map $\ker \epsilon := IG$ is called the *augmentation ideal*.

Definition 1.38. A *chain complex* of $\mathbb{Z}[G]$ -modules is a series of $\mathbb{Z}[G]$ -modules and $\mathbb{Z}[G]$ morphisms

$$\cdots \rightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \rightarrow \cdots$$

such that $d_n \circ d_{n+1} = 0$. This implies that $\text{Im } d_{n+1} \subseteq \ker d_n$. We will denote this complex by $C_\bullet$.

Definition 1.39. The *homology* of a chain complex $C_\bullet$ is defined as

$$H_n(C_\bullet) = \frac{\ker d_n}{\text{Im } d_{n+1}}.$$

If M is a $\mathbb{Z}[G]$ -module, then we can apply $\text{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, M)$ and since a module morphism from the trivial $\mathbb{Z}[G]$ -module $\mathbb{Z}$ to M is determined by the image of $1 \in \mathbb{Z}$, we have that

$$\text{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, M) \cong \{m \in M \mid gm = m, \forall g \in G\}.$$

The functor $\text{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, -)$ maps between the category of $\mathbb{Z}[G]$ -modules and abelian groups. This is called the *invariants* functor. This is always left exact but is not right exact in general. The idea of group cohomology is to measure the failure of this functor to be exact. The invariants functor is an example of a contravariant *additive functor* which means that it preserves direct sums. For these types of functors, assuming the category you are working in has enough injectives/projectives, you can always associate a cohomology/homology theory which measures the failure of the functor to be exact. For those already familiar with some homological algebra, the group cohomology of a $\mathbb{Z}[G]$ -module M is the right derived functor of the invariants functor $\text{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, -)$. However, we will give a more concrete definition of this in what follows.

Definition 1.40. Let M be a $\mathbb{Z}[G]$ -module. A *free resolution* of M by $\mathbb{Z}[G]$ -modules is an exact sequence

$$\cdots \rightarrow F_1 \rightarrow F_0 \rightarrow M \rightarrow 0$$

where each of the F_i is a free $\mathbb{Z}[G]$ -module.

Remark 1.41. Recall that a free R -module M is an R -module such that $M \cong \bigoplus_{i \in I} R$.

The group cohomology of a $\mathbb{Z}[G]$ -module can be calculated by taking a resolution of the trivial $\mathbb{Z}[G]$ -module $\mathbb{Z}$ by free $\mathbb{Z}[G]$ -modules, applying $\text{Hom}_{\mathbb{Z}[G]}(-, M)$ to this resolution and calculating the homology of the resulting chain complex.

Definition 1.42 (Group Cohomology). Let M be a $\mathbb{Z}[G]$ -module. Then we can define the group cohomology $H^n(G, M)$ as follows. Take a free resolution

$$\cdots \rightarrow F_2 \rightarrow F_1 \rightarrow F_0 \rightarrow \mathbb{Z} \rightarrow 0$$

of $\mathbb{Z}$ by $\mathbb{Z}[G]$ -modules. Delete the trailing $\mathbb{Z}$ from the resolution to form the *deleted resolution*

$$\cdots \rightarrow F_2 \rightarrow F_1 \rightarrow F_0 \rightarrow 0.$$

We apply $\text{Hom}_{\mathbb{Z}[G]}(-, M)$ to this resolution to obtain a new complex

$$\cdots \xleftarrow{d^{n+1}} \text{Hom}_{\mathbb{Z}[G]}(F_n, M) \xleftarrow{d^n} \cdots \xleftarrow{d^2} \text{Hom}_{\mathbb{Z}[G]}(F_1, M) \xleftarrow{d^1} \text{Hom}_{\mathbb{Z}[G]}(F_0, M) \leftarrow 0.$$

The cohomology group $H^i(G, M)$ is now the homology of this complex. Concretely,

$$H^i(G, M) = \frac{\ker d^{n+1}}{\text{Im } d^n}.$$

Definition 1.43 (Group Homology). Let M be a $\mathbb{Z}[G]$ -module. The *group homology* $H_i(G, M)$ is defined in the same way as the group cohomology except instead of applying $\text{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, -)$ to the deleted resolution, we apply $- \otimes_{\mathbb{Z}[G]} M$ and take the homology of the resulting complex.

We will make use of the following proposition which gives explicit formula for the groups $H^0(G, M)$ and $H_0(G, M)$.

Proposition 1.44. Let M be a G -module. Then

1. $H_0(G, M) = \frac{M}{(IG)M}$ where IG is the augmentation ideal. This is the largest quotient of M on which G acts trivially since $(IG)M$ is generated by elements of the form $gm - m$.
2. $H^0(G, M) := M^G = \{m \in M \mid gm = m, \forall g \in G\}$.

Proof. We first show that $H_0(G, M) = \frac{M}{(IG)M}$. We take the deleted free resolution and apply $- \otimes_{\mathbb{Z}[G]} M$ and obtain

$$\cdots \rightarrow F_2 \otimes M \rightarrow F_1 \otimes M \xrightarrow{d} F_0 \otimes M \rightarrow 0.$$

We will use that $H_0(G, M)$ is the left derived functor of $\mathbb{Z} \otimes_{\mathbb{Z}[G]} -$, see for example [Bro12], so the degree zero cohomology is simply the functor itself i.e.

$$H_0(G, M) = \mathbb{Z} \otimes_{\mathbb{Z}[G]} M.$$

Now we have the short exact sequence

$$0 \rightarrow IG \rightarrow \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$$

and so $\mathbb{Z} \cong \frac{\mathbb{Z}[G]}{IG}$. Now

$$\begin{aligned} H_0(G, M) &= \mathbb{Z} \otimes_{\mathbb{Z}[G]} M \\ &= \frac{\mathbb{Z}[G]}{IG} \otimes_{\mathbb{Z}[G]} M \\ &= \frac{M}{(IG)M}. \end{aligned}$$

To show $H^0(M, G) = M^G$ we note that $H^0(M, G) = \text{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, M)$. Now any $f \in \text{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, M)$ is determined by $f(1)$. We define

$$\begin{aligned} \varphi : \text{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, M) &\rightarrow M^G \\ f &\mapsto f(1). \end{aligned}$$

Since $\mathbb{Z}$ is viewed as a trivial $\mathbb{Z}[G]$ -module

$$\begin{aligned} f(1) &= f(g \cdot 1) \\ &= gf(1) \end{aligned}$$

and hence this does map into M^G . Injectivity and surjectivity is clear and hence this is an isomorphism. $\square$

1.4 Representation Theory and Affine Manifolds

1.4.1 Affine representations

We now turn our attention to affine manifolds before specializing to the integral affine case. An affine manifold is a (G, X) -manifold where $X = \mathbb{R}^n$ and $G = \text{Aff}(\mathbb{R}^n)$. An important tool in classifying affine structures is the representation theory of groups. Often, by studying properties of the holonomy representation of an affine manifold M , we are able to obtain information about the affine structure of M . Of fundamental importance is the case where the holonomy group is nilpotent which we shall see in Chapter 3. We will use these tools to determine conditions for the affine structure to be complete. We begin with the notion of an *affine representation*.

Definition 1.45. An affine representation of a group G is a group homomorphism

$$\alpha : G \rightarrow \text{Aff}(\mathbb{R}^n).$$

Example 1.46. Let M be a (G, X) -manifold with $G = \text{Aff}(\mathbb{R}^n)$ and $X = \mathbb{R}^n$. Then the holonomy representation is an affine representation of $\pi_1(M)$.

Definition 1.47. Let G be any group and let A be an abelian group. We will call A a G -module if there is a group homomorphism

$$\rho : G \rightarrow \text{Aut}(A).$$

Remark 1.48. A G -module A is a module in the usual sense over the group ring $\mathbb{Z}[G]$ which consists of formal integer sums of elements of G .

Remark 1.49. An affine representation α splits into a linear component and a translation component, i.e. $\alpha(g) = (\lambda(g), \mu(g))$ where $\lambda : G \rightarrow \text{GL}(\mathbb{R}^n)$ and $\mu : G \rightarrow \mathbb{R}^n$. The linear component makes $\mathbb{R}^n$ into a G -module. We will denote this G -module by E when we want to emphasize that we are viewing $\mathbb{R}^n$ as a G -module.

Analogous to irreducible representations in the usual sense, there is a notion of irreducibility for affine representations.

Definition 1.50. An affine representation $\alpha : G \rightarrow \text{Aff}(E)$ is said to be *irreducible* if α leaves a proper affine subspace $F \subset E$ invariant.

Remark 1.51. Note that unlike standard representations, for affine representations we do not allow an origin to be fixed in the notion of irreducibility as there is no concept of an origin for affine spaces.

We now wish to illustrate that completeness of the affine structure has implications on the group theoretic properties of the affine holonomy group. We will use freely some of the properties of Eilenberg Mac-Lane spaces which can be found in [Hat02] although we will recall the definition.

Definition 1.52. Let G be a group and let $n \geq 1$ be an integer. A path-connected topological space X is called an *Eilenberg–MacLane space* of type $K(G, n)$ if

$$\pi_i(X) \cong \begin{cases} G & \text{if } i = n, \\ 0 & \text{if } i \neq n. \end{cases}$$

Equivalently, X has a single nontrivial homotopy group, which occurs in degree n and is isomorphic to G .

Theorem 1.53. Let M be a compact complete affine manifold. Then the affine holonomy representation is irreducible.

Proof. Since M is complete we have that $M = \mathbb{R}^n/\Gamma$ where Γ is the affine holonomy group. Now, if Γ is not irreducible, then there is an invariant affine subspace $V \subset \mathbb{R}^n$. Since V is invariant under Γ and Γ acts freely and properly on $\mathbb{R}^n$ it follows that V/Γ is an affine manifold with universal cover V . Since both $\mathbb{R}^n$ and V are contractible, it follows that V/Γ and M are Eilenberg Mac-Lane $K(\Gamma, 1)$ spaces.

Let $i : V \rightarrow \mathbb{R}^n$ be the inclusion. The induced map

$$i_* : \pi_1(V/\Gamma) \rightarrow \pi_1(M)$$

is an isomorphism of fundamental groups and hence since both spaces are $K(\Gamma, 1)$ this is a homotopy equivalence by Whitehead's theorem. Since $\mathbb{R}^n/\Gamma$ and V/Γ are both compact, it follows that $\dim V/\Gamma = \dim \mathbb{R}^n/\Gamma$ and hence $V = \mathbb{R}^n$. $\square$

We will need some more machinery from the theory of group cohomology. This leads us to the following definitions.

Definition 1.54 (Crossed Homomorphism). Let G be a group and E a G -module. A crossed homomorphism for $\lambda : G \rightarrow \text{Aut}(E)$ is a group homomorphism

$$\mu : G \rightarrow E$$

that satisfies $\mu(gh) = \mu(g) + \lambda(g)\mu(h)$.

Proposition 1.55. Let $\alpha = (\lambda, \mu)$ be an affine representation on E of a group G . Then μ is a crossed homomorphism for λ .

Proof. Let $g, h \in G$ and $\alpha = (\mu, \lambda)$ be an affine representation. Then

$$\begin{aligned} \alpha(gh) &= (\mu(gh), \lambda(gh)) \\ &= \alpha(g) \cdot \alpha(h) \\ &= (\mu(g), \lambda(g)) \cdot (\mu(h), \lambda(h)) \\ &= (\mu(g) + \lambda(g)\mu(h), \lambda(g)\lambda(h)) \\ &\Rightarrow \mu(gh) = \mu(g) + \lambda(g)\mu(h) \end{aligned}$$

and so μ is a crossed homomorphism for λ . □

Definition 1.56 (Principal Crossed Homomorphism). A *principal crossed homomorphism* for λ is a crossed homomorphism of the form $\mu(g) = y - \lambda(g)y$ for some $y \in E$.

Remark 1.57. If μ is a principal crossed homomorphism for λ determined by some $y \in E$, we will often write $\mu = D_y$. This comes from the fact that principal crossed homomorphisms are sometimes also called *principal derivations*.

Lemma 1.58. The group cohomology $H^1(G, E)$ is isomorphic to

$$H^1(G, E) \cong \frac{\{\text{crossed homomorphisms for } \lambda\}}{\{\text{principal crossed homomorphisms for } \lambda\}}$$

Proof. This is a standard result in group cohomology. See for example [Tom23]. □

1.4.2 Radiant affine manifolds

Of importance to completeness of the affine structure is the case that our affine representation has a fixed-point. It turns out that such structures will always be incomplete.

Definition 1.59. An affine representation with a fixed-point is called *radiant*.

Proposition 1.60. Let $y \in E$. Then y is a *stationary point* of the action of α if and only if $\mu(g) = y - \lambda(g)y$ for all $g \in G$.

Proof. Suppose $y \in E$ is a stationary point of α . Then $\alpha(g)y = y$ for every $g \in G$. This gives that

$$\begin{aligned} (\lambda(g) + \mu(g))y &= \lambda(g)y + \mu(g) \\ &= y \\ \Rightarrow \mu(g) &= y - \lambda(g)y \end{aligned}$$

Conversely, suppose that $\mu(g) = y - \lambda(g)y$. Then $y = \mu(g) + \lambda(g)y = \alpha(g)y$ □

Definition 1.61 (Radiance Obstruction). Let α be an affine representation. The radiance obstruction of α is the cohomology class

$$c_\alpha = [\mu] \in H^1(G, E)$$

where μ is the translational part of α .

Lemma 1.62. The radiance obstruction vanishes, e.g. $c_\alpha = 0$ if and only if α is conjugate to its linear part by a translation, if and only if α has a stationary point.

Proof. By Lemma 1.60, $y \in E$ is stationary if and only if $\mu(g) = D_y$ is a principal crossed homomorphism for λ if and only if $c_\alpha = [\mu] = 0$.

Now suppose that $\mu = D_y$ for some $y \in E$ so that $\mu(g) = y - \lambda(g)y$. We define the translation

$$\begin{aligned} T_y : E &\rightarrow E \\ x &\mapsto x - y. \end{aligned}$$

Now, conjugating α by this translation yields

$$\begin{aligned} T_y \circ \alpha(g) \circ T_y^{-1}(x) &= T_y \circ \alpha(g)(x + y) \\ &= T_y \circ (\lambda(g)x + \lambda(g)y + \mu(g)) \\ &= T_y \circ (\lambda(g)x + \lambda(g)y + y - \lambda(g)y) \\ &= T_y \circ (\lambda(g)x + y) \\ &= \lambda(g)x + y - y \\ &= \lambda(g)x. \end{aligned}$$

Conversely, suppose that $T_y \circ \alpha(g) \circ T_y^{-1} = \lambda(g)$ for all $g \in G$ and $x \in E$. Then

$$\begin{aligned} T_y \circ \alpha(g) \circ T_y(x) &= T_y \circ \alpha(g)(x + y) \\ &= T_y \circ (\lambda(g)(x + y) + \mu(g)) \\ &= \lambda(g)(x + y) + \mu(g) - y \\ &= \lambda(g)x \end{aligned}$$

and so

$$\begin{aligned} \mu(g) &= \lambda(g)(x) - \lambda(g)(x) - \lambda(g)(y) + y \\ &= y - \lambda(g)y \end{aligned}$$

as required. □

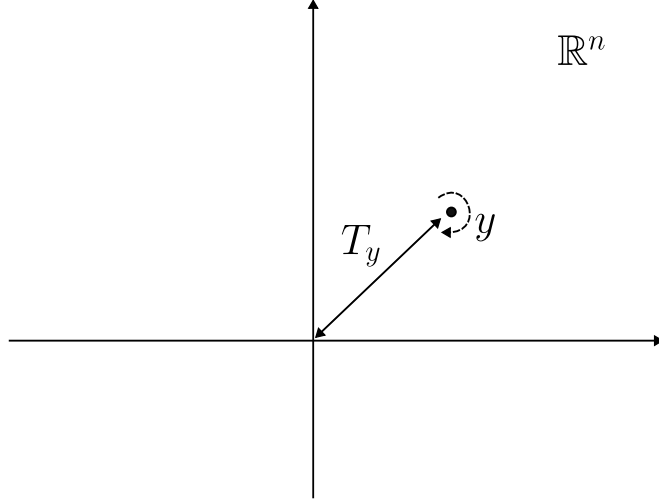


Figure 1.5: Conjugating a fixed-point y under the affine action by a translation to the origin

Remark 1.63. This lemma will allow us to alter the development map by a translation (which alters the holonomy representation by a conjugation) to assume that our fixed point is at the origin.

Definition 1.64 (Contragradient Representation). Let E be a G -module by $\lambda : G \rightarrow \text{Aut}(V)$. We define the contragradient representation on E^* by

$$\begin{aligned}\lambda^* : G &\rightarrow GL(E^*) \\ g &\mapsto \lambda^*g\end{aligned}$$

such that $\lambda^*g(f)(x) = f(\lambda(g^{-1})x)$ for all $x \in E$ and $f \in E^*$.

Remark 1.65. Once a representation $\lambda : G \rightarrow \text{Aut}(E)$ has been fixed, we will often write the action of g on an element x of E or E^* as $g \cdot x$. It is clear from if x is an element of E or E^* whether this action should be taken as the action of $\lambda(g)$ or λ^*g .

Remark 1.66. This induces a representation on $\wedge^k E^*$ by $(g \cdot \omega)(x_1, \dots, x_n) = \omega(g^{-1} \cdot x_1, \dots, g^{-1} \cdot x_n)$ for $\omega \in \wedge^k E^*$ and $x_i \in E$.

We will make use of the following lemma when looking at the relationship between the holonomy representation and parallel volume forms.

Lemma 1.67. Let $F : V \rightarrow V$ be a linear map from a vector space V of dimension n to itself and let $\omega \in \wedge^n(V^*)$. Then

$$F^*\omega = \det(A)\omega$$

where A is any matrix representing the linear transformation F .

Proof. As $\dim(\wedge^n(V^*)) = 1$ and F is a linear map, it follows that our map corresponds to multiplication by some scalar $\lambda \in \mathbb{R}$. Let $\varphi : V \rightarrow \mathbb{R}^n$ be an isomorphism identifying our vector

space V with $\mathbb{R}$. Then we have

$$\begin{aligned} F^* \varphi^* \det &= \lambda \varphi^* \det \\ \Rightarrow (\varphi^{-1})^* F^* \varphi^* \det &= \lambda \det \\ \Rightarrow (\varphi^{-1} F \varphi)^* \det &= \lambda \det. \end{aligned}$$

If we let $A = \varphi^{-1} F \varphi$ and $\{e_1, \dots, e_n\}$ the standard basis of $\mathbb{R}^n$ then

$$\begin{aligned} A^* \det(e_1, \dots, e_n) &= \det(Ae_1, \dots, Ae_n) \\ &= \lambda \det(e_1, \dots, e_n) \\ &= \lambda \end{aligned}$$

which gives us that $\det(A) = \lambda$ as required. $\square$

This leads us to the following elementary but important property of the action of the holonomy representation on volume forms.

Corollary 1.68. Let M be an affine manifold and $\lambda : G \rightarrow GL(\mathbb{R}^n)$ be the linear part of the holonomy representation. Then M has a parallel volume form with respect to the affine connection if and only if $\det(\lambda(g)) = 1$ for all $g \in G$.

Proof. Suppose ω is a parallel volume form on M . Then the parallel transport of ω around a loop in M leaves ω unchanged, e.g. the linear holonomy action preserves ω . This gives us that

$$\begin{aligned} g \cdot \omega &= \omega \\ \Rightarrow \omega(g^{-1} \cdot x_1, \dots, g^{-1} \cdot x_n) &= \omega(x_1, \dots, x_n) \end{aligned}$$

for all $x_i \in M$. But by Lemma 1.67

$$\omega(g^{-1} \cdot x_1, \dots, g^{-1} \cdot x_n) = \det(\lambda(g^{-1})) \cdot \omega(x_1, \dots, x_n)$$

and so the volume is preserved if $\det(\lambda(g)) = 1$

Conversely, suppose that the determinant of the linear holonomy is 1 for all $g \in G$. If we pick a point $m_0 \in M$ then locally there exists a volume form ω_0 on a chart U of M with respect to the affine connection, corresponding to a constant, top degree form. We now claim that we can uniquely extend such a form. Indeed, as parallel transport provides a linear isomorphism between tangent spaces we can attempt to define a volume form on M by parallel transporting ω_0 .

Pick another point $m \in M$. We claim that the parallel transport of ω_0 along a curve γ from m_0 to m does not depend on the choice of curve. If γ' is another curve then the parallel transport $(P_{m_0 \rightarrow m}^{\gamma'})^{-1} \circ P_{m_0 \rightarrow m}^{\gamma} = g$ for some g in the linear holonomy group. It follows that

$$\begin{aligned} g^* \circ \omega_0 &= ((P_{m_0 \rightarrow m}^{\gamma'})^{-1} \circ P_{m_0 \rightarrow m}^{\gamma})^* \circ \omega_0 \\ &= \det(\lambda(g)) \cdot \omega_0 \\ &= \omega_0 \\ \Rightarrow (P_{m_0 \rightarrow m}^{\gamma})^* \circ \omega_0 &= (P_{m_0 \rightarrow m}^{\gamma'})^* \circ \omega_0 \end{aligned}$$

and so the parallel transport from m to m_0 does not depend on the choice of path. We note that this defines a continuous maps as in each affine chart the parallel transport identifies the tangent spaces via the identity map and on overlaps they differ by an element of $GL(\mathbb{R}^n)$ as the derivative of an affine map extracts the linear part. $\square$

Definition 1.69 (Radiant Manifold). An affine manifold is said to be radiant if its affine holonomy representation has a fixed-point.

Remark 1.70. By Lemma 1.62 we may conjugate our developing map by a translation to assume that the fixed-point is at the origin.

When M is radiant the above remark implies that the radiant vector field $R = \sum_i x_i \frac{\partial}{\partial x_i}$ is fixed by the affine action. There is then a unique vector field $\tilde{X}$ on the universal cover $\tilde{M}$ that is related to R by the developing map. Since this is preserved by deck transformations (by equivariance of developing map and since the representation is radiant) it descends to a vector field X on M .

Definition 1.71. The vector field X on M is called the radiant vector field of M .

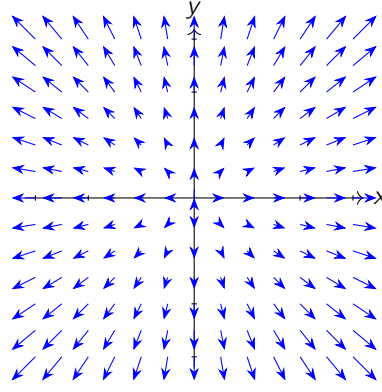


Figure 1.6: Radiant Vector Field on $\mathbb{R}^2$

1.5 Integral Affine Manifolds

One of the main objects of interest in this thesis is integral affine manifolds. These are (G, X) -structures whereby $G = \text{Aff}_{\mathbb{Z}}(\mathbb{R}^n)$ and $X = \mathbb{R}^n$. They show up in symplectic geometry as the base space of a regular Lagrangian torus fibration inherits such a structure as we explain in Chapter 2. They are also important objects in the study of the SYZ conjecture in physics which relates a six-dimensional Calabi-Yau manifold X to its so-called mirror $\tilde{X}$ by a Lagrangian fibration over a common base space B [Gro08]. In this chapter we give some basic properties of these manifolds as well as some examples.

Proposition 1.72. Let M be a smooth manifold. An integral affine structure on M is equivalent to a full-rank lattice $\Lambda \subset T^*M$ that is locally spanned by closed one-forms.

Proof. Given an integral affine structure on M , we can define a lattice in T^*M as follows. Let $(U, x_1, \dots, x_n)$ be a chart on M . We define $\Lambda|_U = \text{span}_{\mathbb{Z}}(dx_1, \dots, dx_n)$. Carrying out this construction in each affine chart defines a full-rank lattice in T^*M . On overlaps, these lattices will differ by the differential of an integral affine map $x \mapsto Ax + b$ with $A \in \text{GL}(\mathbb{Z}^n)$ and $b \in \mathbb{R}^n$. But the differential of such a map is given by the linear part A and hence is an automorphism of the lattice.

Conversely, given a full-rank lattice $\Lambda \subset T^*M$ that is locally spanned by closed one-forms, by shrinking charts if necessary we can assume that our lattice is spanned by exact one-forms

$(dx_1, \dots, dx_n)$ in an open neighborhood U . Since the $dx_i|_p$ span a full-rank lattice in T_p^*M , they are linearly independent and hence if we define coordinates by $(U, x_1, \dots, x_n)$ by the inverse function theorem, the map $\bar{x} = (x_1, \dots, x_n)$ is a local diffeomorphism. On the overlap of two charts $U_i \cap U_j$, we know that the pullback of the lattice in U_i to U_j must be an automorphism of the lattice. But these maps are given exactly by elements of the integral affine group $\text{Aff}_{\mathbb{Z}}(\mathbb{R}^n)$. $\square$

We now give some examples of integral affine manifolds.

Example 1.73. Let $\mathbb{Z}^n$ act on $\mathbb{R}^n$ by translations. Then the quotient $(\mathbb{R}/\mathbb{Z})^n$ inherits an integral affine structure and is the standard integral affine torus. We note that this integral affine structure cannot be equivalent to any of the structures obtained from Example 1.31 as these are incomplete. In fact, it is also true that a closed integral affine manifold cannot be radiant [FGH81].

Example 1.74. Let B be an integral affine manifold and let $f : M \rightarrow B$ be a local diffeomorphism where M is any smooth manifold. Then there is a unique integral affine structure on M such that f is an integral affine map as defined in Definition 1.1. The developing map is an example of such a map and in this sense the integral affine structure on any integral affine manifold can be realised as the pullback of the standard integral affine structure on $\mathbb{R}^n$.

Example 1.75. Let M be an integral affine manifold and let Γ be a discrete group acting freely and properly on M by integral affine diffeomorphisms. Then the quotient M/Γ inherits an integral affine structure and the projection map $\pi : M \rightarrow M/\Gamma$ is an integral affine map.

We can of course define an integral affine structure by specifying a full-rank lattice in T^*M that is locally spanned by closed one-forms. Importantly, the cotangent bundle of any manifold M inherits a symplectic structure which gives rise to a volume form ω on T^*M . If $\Lambda, \Lambda' \subset T^*M$ are two lattices in T^*M such that they give M an integral affine structure, then the volume form descends to the quotient and the induced volume of T^*M/Λ and T^*M/Λ' must be the same if Λ and Λ' determine equivalent integral affine structures. This will be discussed more in Chapter 2.

Example 1.76 (Radiant integral affine manifold). If we drop the compactness assumption, it is possible to equip a manifold M with an integral affine structure that is radiant. Let $M = \mathbb{R}^2 - \{0\}$ and

$$i : M \hookrightarrow \mathbb{R}^2$$

be the inclusion. Then this is a local diffeomorphism and hence, equipping $\mathbb{R}^2$ with the standard integral affine structure, this induces an integral affine structure on M . There is a covering map

$$\begin{aligned} \pi : (0, \infty) \times \mathbb{R} &= \tilde{M} \rightarrow M, \\ (r, \theta) &\mapsto (r \cos \theta, r \sin \theta) \end{aligned}$$

and the composition $\widetilde{\text{dev}} = i \circ \pi : \tilde{M} \rightarrow \mathbb{R}^2$ is the developing map for the integral affine structure induced on M . This developing map fails to be injective and surjective, hence this affine structure is not complete and is therefore not affinely diffeomorphic to the standard integral affine structure on $\mathbb{R}^2$. By considering $\theta \mapsto \theta + 2\pi$ one notes that $\widetilde{\text{dev}}(r, \theta + 2\pi) = \text{id}_{\mathbb{R}^2} \widetilde{\text{dev}}(r, \theta)$ and so the holonomy representation is the trivial one in this case.

We can modify the affine structure so that it becomes radiant as follows. For any $A \in \text{GL}(\mathbb{Z}^2)$, there is a free and proper action of $\mathbb{Z}$ on $\tilde{M}$ by integral affine diffeomorphisms which acts by covering the multiplication by A^n on M and translating a sheet as shown in [BF07]. It follows that the quotient of $\tilde{M}$ by this action is an integral affine manifold. Its holonomy representation satisfies

$$\text{hol}(\gamma) = (A, 0)$$

where γ is the generator of $\pi_1(M)$. It follows that this integral affine structure is radiant.

Example 1.77 (Mapping tori). Let M be an integral affine manifold and let $f : M \rightarrow M$ be an integral affine diffeomorphism. Then, equipping $\mathbb{R}$ with its standard integral affine structure, we can form the quotient manifold

$$M_f = \frac{M \times \mathbb{R}}{\sim}$$

where $(m, t) \sim (m', t')$ if $(m, t) = (f^{-n}(m'), t' + n)$ for some $n \in \mathbb{Z}$ where f^n denotes composition of f or f^{-1} depending on if n is positive. We note that even if f is not free, the $\mathbb{Z}$ -action defining the quotient is free as it is a translation in the second component. The resulting manifold is called the *mapping torus* of f . We note that this is a fiber bundle over S^1 through projection onto the second component.

Remark 1.78. If Σ is compact and admits a $\mathbb{R}P^n$ -structure and f is a projective automorphism, then the mapping torus of f admits a radiant affine structure [Gol22].

Lagrangian Fibrations and Integral Affine Manifolds

2.1 Symplectic Geometry

In this section we will recall some of the basics of symplectic geometry and Lagrangian submanifolds.

Symplectic geometry has its roots in classical mechanics as the phase space of a dynamical system naturally has the structure of a symplectic manifold. In symplectic geometry it is often global properties that are of interest. In fact, it turns out that all symplectic manifolds are locally equivalent in contrast to Riemannian geometry where curvature plays the role of a local invariant. In this chapter we will give an extremely brief introduction to symplectic geometry. We will then describe the relationship between integral affine structures and symplectic geometry. Indeed, in Chapter 1 we have seen that an integral affine structure is equivalent to a lattice in the cotangent bundle. It will turn out that the cotangent bundle of any manifold inherits the structure of a symplectic manifold and this will serve as the starting point for our discussion.

We start with some of the basic definitions and properties of symplectic manifolds.

Definition 2.1 (Symplectic Manifold). A symplectic manifold (M, ω) consists of a smooth manifold M and a closed 2-form ω such that ω is non-degenerate meaning the bi-linear form ω_p is non-degenerate on $T_p M$ for all $p \in M$.

Non-degeneracy of the symplectic form gives a canonical way to identify the tangent and cotangent bundle of a symplectic manifold. This is done via the map

$$\tilde{\omega}|_p : T_p M \rightarrow T_p^* M$$

$$v \mapsto \omega_p(v, \cdot)$$

for $v \in T_p M$. This is an isomorphism by non-degeneracy of the symplectic form. This identification leads to the notion of a Hamiltonian vector field.

Definition 2.2 (Hamiltonian Vector Field). To any $H \in C^\infty(M)$ we can associate a unique Hamiltonian vector field X_H by

$$\iota_{X_H} \omega := \omega(X_H, \cdot) = dH$$

For mathematicians there is generally no distinguished choice of Hamiltonian function H . However, physicists often pick a Hamiltonian function based on physical criteria and are generally interested in studying the flow of this Hamiltonian vector field which, for instance, recovers Hamilton's equations.

2.1.1 The Cotangent Bundle

One of the most important examples of symplectic manifolds is the cotangent bundle, which naturally inherits the structure of a symplectic manifold. In fact, Darboux's theorem states that every symplectic manifold looks locally like a cotangent bundle. This class of symplectic manifolds will play a crucial role throughout this thesis and we will see now exactly how the cotangent bundle obtains a symplectic structure.

Example 2.3 (Cotangent Bundle). Let Q be any smooth manifold and $M = T^*Q$ its cotangent bundle. We will now illustrate that there is a canonical one-form $\theta \in \Omega^1(T^*Q)$ and show that the exterior derivative $-d\theta$ defines a symplectic structure on T^*Q .

Let $\pi : T^*Q \rightarrow Q$ be the canonical projection map. Then for $X_p \in T_p(T^*Q)$ we define $\langle \theta_p, X_p \rangle := \langle p, d_p\pi(X_p) \rangle$ where $\langle \cdot, \cdot \rangle$ denotes contraction of a vector and covector. This canonical one-form satisfies the following important property.

Lemma 2.4. The canonical one-form θ is the unique one-form such that for any other one-form $\alpha \in \Omega^1(Q)$

$$\alpha = \alpha^*\theta$$

where we view α as a section $\alpha : Q \rightarrow T^*Q$.

Proof. Let $w \in T_qQ$. Then we have that

$$\begin{aligned} \langle (\alpha^*\theta)_q, w \rangle &= \langle \theta_{\alpha_q}, d_q\alpha(w) \rangle \\ &= \langle \alpha_q, w \rangle \end{aligned}$$

Uniqueness is obtained by noting that every tangent vector $v \in T_p(T^*Q)$ can be written in the form $d_q\alpha(w)$ except for those in the kernel of $d_p\pi$. But the vectors not in this kernel form an open, dense set and so by continuity we get uniqueness. $\square$

We can also find an expression for this canonical one-form in coordinates.

Let $(U, q_1, \dots, q_n)$ be a coordinate chart on Q . We denote the vector fields corresponding to these coordinates by

$$\frac{\partial}{\partial q_1}, \dots, \frac{\partial}{\partial q_n} \in \mathfrak{X}(U).$$

These vector fields define functions on each fibre of $T^*Q|_U$ by

$$\begin{aligned} p_i &: T_x^*Q|_U \rightarrow \mathbb{R} \\ \eta &\mapsto \langle \eta, \frac{\partial}{\partial q_i} |_x \rangle \end{aligned}$$

where $\langle \cdot, \cdot \rangle$ is the canonical pairing between vectors and covectors. The set of functions

$$\pi^*q_1, \dots, \pi^*q_n, p_1, \dots, p_n \in C^\infty(T^*Q|_U)$$

is a set of coordinates on $T^*Q|_U$. We will often abuse notation and write the π^*q_i as simply q_i .

Lemma 2.5. In local cotangent coordinates $q_1, \dots, q_n, p_1, \dots, p_n$ on T^*Q , the canonical one-form is given by

$$\theta|_{T^*U} = \sum_j p_j dq_j.$$

Proof. We let $\alpha = \sum_j \alpha_j dq_j$ be a 1-form on U . We have that $\alpha^* p_j = \alpha_j$ and $\alpha^* q_j = 0$ which gives that

$$\begin{aligned} \alpha^* \theta|_{T^*U} &= \alpha^* \sum_j p_j dq_j \\ &= \sum_j \alpha_j dq_j \\ &= \alpha. \end{aligned}$$

□

Putting this together we obtain a symplectic form on T^*Q .

Theorem 2.6. The 2-form $-d\theta$ defines a symplectic form on T^*Q .

Proof. In local coordinates we have that $\omega = \sum_j dq_j \wedge dp_j$ which is clearly closed and non-degenerate. □

We will also wish to speak about maps between the symplectic manifolds. The correct notion of "isomorphism" in this category is the following.

Definition 2.7. Let (M, ω) and (N, ω') be two symplectic manifolds. Then a symplectomorphism between M and N is a diffeomorphism

$$\varphi : M \rightarrow N$$

such that $\omega = \varphi^* \omega'$.

2.1.2 Lagrangian Submanifolds

The goal of this thesis is to classify Lagrangian fibrations over compact surfaces. In this section we will introduce the concept of Lagrangian submanifolds and prove some basic results regarding these.

Definition 2.8 (Symplectic Orthogonal). Let (M, ω) be a symplectic manifold and let $N \subset M$ be a submanifold. The symplectic orthogonal at a point $p \in M$ is

$$(T_p N)^\omega := \{v \in T_p M \mid \omega(v, w) = 0, \forall w \in T_p N\}$$

- If $(T_p N)^\omega \subset T_p N$ then $T_p N$ is said to be *coisotropic*.
- If $T_p N \subset (T_p N)^\omega$ then $T_p N$ is said to be *isotropic*.
- If $(T_p N)^\omega = T_p N$ then $T_p N$ is said to be *Lagrangian*.
- If $(T_p N)^\omega \cap T_p N = \{0\}$ then $T_p N$ is said to be *symplectic*.

A submanifold N of a symplectic manifold M is said to be coisotropic (isotropic, Lagrangian, symplectic) if its tangent space is a coisotropic (isotropic, Lagrangian, symplectic) subspace of $T_p M$ at every point $p \in M$.

Remark 2.9. We note that a submanifold $\iota : N \rightarrow M$ is isotropic if $\iota^* \omega = 0$ and furthermore, it is Lagrangian if we also have $\dim N = \frac{1}{2} \dim M$.

We will make use of the following lemma.

Lemma 2.10. The graph $\Gamma_\alpha \subset T^*Q$ of a one-form α is Lagrangian if and only if α is closed.

Proof. It is clear that $\dim \Gamma_\alpha = \frac{1}{2} \dim T^*Q$. We compute

$$\begin{aligned} \alpha^* \omega &= -\alpha^* d\theta \\ &= -d\alpha^* \theta \\ &= -d\alpha \end{aligned}$$

and so $\alpha^* \omega = 0$ if and only if α is closed completing the proof by the previous remark. $\square$

The cotangent bundle T^*Q also gives rise to another Lagrangian submanifold.

Example 2.11. Let $\pi : T^*Q \rightarrow Q$ and fix $x \in Q$. Then $\pi^{-1}(x) = T_x^*Q$ is a Lagrangian submanifold inside (T^*Q, ω) . It is easy to see that $\dim T_x^*Q = \frac{1}{2} \dim T^*Q$. Letting $i : T_x^*Q \hookrightarrow T^*Q$ be the inclusion and considering local coordinates $(q_1, \dots, q_n, p_1, \dots, p_n)$ around x and writing $\omega = \sum_j dq_j \wedge dp_j$ it follows that $\omega|_{T_x^*Q} = 0$ as $i^*(\sum_j dq_j \wedge dp_j) = \sum_j d(i^*q_j) \wedge d(i^*p_j)$ and $d(i^*q_j) = 0$ as i^*q_j is constant.

2.2 Lagrangian Fibrations

One of the goals of this thesis is to classify the Lagrangian fibrations over closed surfaces. In this section we will give an overview of these objects and give some examples and basic results surrounding them. We will then describe how the base space of a Lagrangian fibration inherits an integral affine structure.

2.2.1 Definition of Lagrangian Fibration

Definition 2.12. A Lagrangian fibration $\pi : (M, \omega) \rightarrow B$ is a surjective map whose regular fibres are Lagrangian submanifolds of (M, ω) .

Definition 2.13. A Lagrangian fibration is called *regular* if every fibre is regular. This means that for every point $q \in \pi^{-1}(p)$, the map $d\pi_q$ is surjective where $p \in B$.

Remark 2.14. For most of this thesis we will assume that π is a surjective submersion and a proper map so that by a theorem of Ehresmann in [Ehr51], the fibration $\pi : (M, \omega) \rightarrow B$ is a fiber bundle and the fibres are compact. We will sometimes refer to these as *Lagrangian torus fibrations* as in this case the fibres turn out to be tori as we shall soon see.

Example 2.15. We saw in Lemma 2.10 that the fibres of $\pi : (T^*Q, \omega_{can}) \rightarrow Q$ are Lagrangian submanifolds and hence this is a Lagrangian fibration.

Example 2.16. Let $Q = (\mathbb{R}/\mathbb{Z})^n = \mathbb{T}^n$ be the n -torus. Then we have that $T^*(\mathbb{T}^n) = \mathbb{T}^n \times \mathbb{R}^n$ and the map

$$\pi : T^*(\mathbb{T}^n) \rightarrow \mathbb{R}^n$$

is a Lagrangian fibration with fibre $\mathbb{T}^n$.

Definition 2.17 (Shift by a one-form.). Let $\pi : (T^*Q, \omega_{can}) \rightarrow Q$ and fix $\alpha \in \Omega^1(Q)$. We define the *shift by α* to be the map

$$s_\alpha : T^*Q \rightarrow T^*Q$$

$$\beta_q \mapsto \beta_q + \alpha_q.$$

Proposition 2.18. This defines a symplectomorphism of T^*Q if and only if $d\alpha = 0$.

Proof. This map is a fibrewise translation and hence is obviously a diffeomorphism. We now check when it preserves the symplectic form. We calculate

$$\begin{aligned} s_\alpha^* \omega_{can} &= -s_\alpha^* d\theta \\ &= -ds_\alpha^* \theta. \end{aligned}$$

Thus it remains to see how s_α acts on the canonical one-form. We compute

$$\begin{aligned} s_\alpha^* \theta(v)|_{\beta_q} &= \theta_{\beta_q + \alpha_q}(ds_\alpha(v)) \\ &= (\beta_q + \alpha_q)(d(\pi \circ s_\alpha)(v)) \\ &= (\beta_q + \alpha_q)(v) \\ &= \theta_{\beta_q}(v) + \alpha_q(v) \end{aligned}$$

and so $s_\alpha^* \theta = \theta + \alpha$. Now

$$\begin{aligned} -ds_\alpha^* \theta &= -d\theta - d\alpha \\ &= \omega_{can} - d\alpha \end{aligned}$$

and hence this defines a symplectomorphism if and only if $d\alpha = 0$. $\square$

Remark 2.19. We now note that given any Lagrangian section of T^*Q , i.e. a section determined by a closed one-form, we can shift this section to the zero section since shifts by closed one-forms are symplectomorphisms. This implies that if we consider T^*Q as an abstract Lagrangian bundle it has no marked zero section and so the fibres do not have a canonical vector space structure, only the structure of an affine space.

We now give the notion of equivalence of Lagrangian bundles.

Definition 2.20. Let $M \rightarrow B$ and $N \rightarrow B$ be two Lagrangian fibrations over B . Then a *Lagrangian equivalence* between $M \rightarrow B$ and $N \rightarrow B$ is given by

$$\begin{array}{ccc} M & \xrightarrow{\Phi} & N \\ & \searrow & \downarrow \\ & & B \end{array}$$

where Φ is a fibrewise symplectomorphism i.e. a symplectomorphism that is also a bundle map.

Remark 2.21. There is no loss in generality in assuming that Φ covers the identity on B . Indeed if $f : B \rightarrow B$ is a diffeomorphism then there exists a fibrewise symplectomorphism M and N covering f if and only if there exists a fibrewise symplectomorphism between M and the pullback bundle f^*N covering the identity on B .

2.3 Integral Affine Structure on the Base

We now discuss how the base space of a Lagrangian fibration inherits an integral affine structure. This structure will be an invariant of the fibration.

2.3.1 Torus Bundles

Throughout this section let Q be a manifold and let G be a Lie group.

Definition 2.22. Suppose that $\dim Q = \dim G$. We say that Q is a *principal homogeneous G -space* if there is a free, transitive action of G on Q .

Remark 2.23. For any $q \in Q$ we have that $Q = G \cdot q \cong G$ as the action is transitive. For any $q' \in Q$ there exists a $g \in G$ such that $g \cdot q = q'$. Therefore, any two such identifications differ by an element of G .

Definition 2.24. If $G \cong (\mathbb{R}/\mathbb{Z})^n$ and Q is a principal homogeneous G -space, we call Q an *affine torus*. If G is a vector space we call Q an *affine vector space*.

Now we consider the case that $G := V$ is a vector space. Suppose now that the action is transitive but *not necessarily free*. We define

$$V_q := \{v \in V \mid v \cdot q = q\}$$

.

Proposition 2.25. V_q is a discrete subgroup of V independent of $q \in Q$.

Proof. As a topological space, $Q = V/V_q$. Consider the map

$$\begin{aligned} \phi_q : V &\rightarrow Q \\ v &\mapsto v \cdot q \end{aligned}$$

Taking the differential $(d\phi_q)_0 : T_0V \rightarrow T_0Q$ and applying the rank-nullity theorem we find that $\dim V_q = 0$ and hence it is a discrete subgroup. We now show it is independent of q .

2.3. INTEGRAL AFFINE STRUCTURE ON THE BASE

For any $q' \in Q$, by transitivity of the action, there exists $w \in V$ such that $w \cdot q = q'$ and so for $v \in V_q$, $w \in V_{q'}$

$$\begin{aligned} (w^{-1}vw) \cdot q &= q \\ \Rightarrow (w^{-1}v) \cdot q' &= q \\ \Rightarrow v \cdot q' &= w \cdot q \\ \Rightarrow v \cdot q' &= q \end{aligned}$$

and hence $V_q = V_{q'}$ □

Remark 2.26. This implies that V_q is a lattice, i.e. $V_q \cong \mathbb{Z}^n$. We therefore have that $V/V_q \cong \mathbb{R}^n/\mathbb{Z}^k \cong \mathbb{R}^{n-k} \times \mathbb{T}^k$.

Remark 2.27. It follows from our previous definitions that Q is the product of an affine torus and an affine vector space.

Definition 2.28 (Group Bundle). A group bundle $\pi : \mathcal{G} \rightarrow B$ is a fibre bundle such that each fibre is equipped with a group structure and there are charts $\pi^{-1}(U) \cong U \times G$ that are fibrewise isomorphisms with a fixed group G .

We can define the notion of an action of a group bundle on any fibre bundle over the same base.

Definition 2.29. Let $\mathcal{G} \rightarrow B$ be a group bundle and $M \rightarrow B$ be a fibre bundle. An action of $\mathcal{G} \rightarrow B$ on $M \rightarrow B$ is a smooth map

$$\varphi : \mathcal{G} \times_B M \rightarrow M$$

that is a fibrewise group action.

Example 2.30. Let $\mathcal{T} \rightarrow B$ be a torus bundle. Then an affine torus bundle $M \rightarrow B$ is a fiber bundle with a free, transitive action of $\mathcal{T} \rightarrow B$. We therefore have by Remark 2.26 that if $M \rightarrow B$ is a fiber bundle with compact fibres and $E \rightarrow B$ is a vector bundle with a transitive action on $M \rightarrow B$ such that $\dim M = \dim E$, then M inherits the structure of an affine torus bundle.

Suppose that we have an action as in the above example. Then if we let

$$\Lambda = \bigsqcup_p \Lambda_p$$

be the bundle of stabilisers of the action of $E \rightarrow B$ on $M \rightarrow B$, i.e. $\Lambda_p \subset E_p$ is the isotropy group at $p \in B$, then we can form a torus bundle

$$\mathcal{T} := E/\Lambda \rightarrow B.$$

In this way $M \rightarrow B$ becomes an affine torus bundle since $\mathcal{T}$ now acts freely on M as we have quotiented out the fixed points.

Suppose now that we have a global section $\sigma : B \rightarrow M$. Then $M \rightarrow B$ becomes a torus bundle as we can use the global section to pick out an identity - see Lemma 2.38.

Theorem 2.31. Let (M, ω) be a symplectic manifold and let $\pi : M \rightarrow B$ be a Lagrangian fibration with compact connected fibres. Then, there is a canonical, fibrewise transitive action of the vector bundle $T^*B \rightarrow B$ on $M \rightarrow B$. Therefore, $M \rightarrow B$ can be canonically equipped with the structure of an affine torus bundle.

Proof. Let $VM \subset TM$ be the vertical bundle i.e. $V_p M = \ker d_p \pi$. Since $\pi : M \rightarrow B$ is a Lagrangian fibration it follows that $VM \subset TM$ is a Lagrangian subbundle as it is tangent to the fibres of the fibration. Let $\alpha \in \Omega^1(B)$ be any 1-form and let $X_\alpha \in \mathfrak{X}$ be the vector field defined by

$$\iota_{X_\alpha} \omega = -\pi^* \alpha.$$

Let $Y \in \mathfrak{X}(M)$ be a vertical vector field i.e. $d\pi(Y(p)) = 0$ for all $p \in M$. Then we have

$$\begin{aligned} \omega(X_\alpha, Y) &= \iota_Y \iota_{X_\alpha} \omega \\ &= -\iota_Y \pi^* \alpha \\ &= 0 \end{aligned}$$

as Y is vertical. Now as $VM^\omega = VM$, this implies that X_α is vertical. We note that the value of X_α at $m \in M$ depends only on the value of $\alpha_{\pi(m)}$ as it is defined through pulling back α . We have thus assigned to each $\alpha_{\pi(m)}$ with $m \in M$, a vector in the vertical bundle VM by $X_\alpha(m)$ depending only on the value of $\alpha_{\pi(m)}$. Moreover, this assignment is linear since

$$\begin{aligned} \iota_{X_{a\alpha+b\beta}} \omega &= -\pi^*(a\alpha + b\beta) \\ &= -a\pi^*(\alpha) - b\pi^*(\beta) \end{aligned}$$

for $a, b \in \mathbb{R}$ and $\alpha, \beta \in \Omega^1(B)$. We thus have a isomorphism

$$V_m M \cong T_{\pi(m)}^* B$$

and putting together all these isomorphisms yields a bundle isomorphism

$$VM \cong \pi^* T^* B$$

covering the identity. Now, we let F_α^t denote the flow of X_α and we will denote by $F_\alpha := F_\alpha^1 : M \rightarrow M$ the flow at time 1. This gives a fibrewise action on $M \rightarrow B$ as the X_α are vertical. We get a bundle map

$$\begin{aligned} T^*B \times_B M &\rightarrow M, \\ (\alpha_b, m) &\mapsto F_\alpha(m) \end{aligned}$$

and we claim that this is a vector bundle action. Transitivity is clear as F_α is a fibrewise diffeomorphism. It remains to show that the action is commutative i.e. that the flows of the X_α commute. We let $\alpha, \beta \in \Omega^1(B)$ be two one-forms with corresponding vector fields X_α, X_β . Then, using the formula from Cartan calculus,

$$\begin{aligned} \iota_{[X_\alpha, X_\beta]} \omega &= (L_{X_\alpha} \iota_{X_\beta} - \iota_{X_\beta} L_{X_\alpha}) \omega \\ &= -L_{X_\alpha} \pi^* \beta - \iota_{X_\beta} d\iota_{X_\alpha} \omega \\ &= -L_{X_\alpha} \pi^* \beta + \iota_{X_\beta} \pi^* d\alpha \\ &= -(d\iota_{X_\alpha} \pi^* \beta + \iota_{X_\alpha} \pi^* d\beta) + \iota_{X_\beta} \pi^* d\alpha \\ &= 0 \end{aligned}$$

since $d\pi(X_\alpha) = d\pi(X_\beta) = 0$. Now, by non-degeneracy of ω , it follows that $[X_\alpha, X_\beta] = 0$ for all $\alpha, \beta \in \Omega^1(B)$ and hence the flows commute. We therefore have a vector bundle action of $T^*B \rightarrow B$ on $M \rightarrow B$ and hence $M \rightarrow B$ has the structure of an affine torus bundle. $\square$

2.3. INTEGRAL AFFINE STRUCTURE ON THE BASE

It is through this action that the base space of a Lagrangian fibration inherits an integral affine structure. Indeed the bundle of stabilisers for this action $\Lambda \subset T^*B$ is a discrete sub-bundle of the cotangent space and it will turn out that this will locally be spanned by closed one forms. Since by Proposition 1.72 this is equivalent to the base space being an integral affine manifold, this will be exactly how B inherits an integral affine structure.

We in fact wish to show that Λ is a Lagrangian submanifold of T^*B . We will first need the following lemma.

Lemma 2.32. Let $\alpha \in \Omega^1(B)$ be a one-form and let X_α be its corresponding vector field on M and denote by F_α^t its flow. Then

$$(F_\alpha^t)^*\omega = \omega - t\pi^*d\alpha$$

and so F_α^1 is a symplectomorphism if and only if $d\alpha = 0$.

Proof. We start by integrating

$$\int_0^t \frac{\partial}{\partial t} (F_\alpha^t)^*\omega = (F_\alpha^t)^*\omega - \omega.$$

On the otherhand, calculating this derivative yields

$$\begin{aligned} \frac{\partial}{\partial t} (F_\alpha^t)^*\omega &= (F_\alpha^t)^*L_{X_\alpha}\omega \\ &= (F_\alpha^t)^*d\iota_{X_\alpha}\omega \\ &= -(F_\alpha^t)^*\pi^*d\alpha \\ &= -(\pi \circ F_\alpha^t)^*d\alpha \\ &= -\pi^*d\alpha. \end{aligned}$$

Now $\int_0^t -\pi^*d\alpha = -t\pi^*d\alpha$ and so

$$(F_\alpha^t)^*\omega - \omega = -t\pi^*d\alpha$$

as required. $\square$

We can now show the following proposition regarding the bundle of stabilisers Λ .

Proposition 2.33. The lattice $\Lambda \subset T^*B$ is a Lagrangian submanifold.

Proof. Since by assumption our fibres are compact, by Remark 2.26 we must have that $\Lambda_p \cong \mathbb{Z}^n$. Now let $U \subseteq B$ be an open set on the base such that $\Lambda|_U \cong U \times \mathbb{Z}^n$. Now, the image of any section of $\Lambda|_U$ is given by the graph of a one-form $\alpha : U \rightarrow T^*U$ such that its corresponding time one flow F_α^1 is the identity on $\pi^{-1}(U)$. But by Lemma 2.32 this implies that α is closed and by Lemma 2.10 the image of each section is a Lagrangian submanifold of T^*B . $\square$

We are now able to see how the base space of a Lagrangian fibration inherits an integral affine structure.

Corollary 2.34. The base space of any Lagrangian fibration $M \rightarrow B$ inherits an integral affine structure.

Proof. By Proposition 2.33, $\Lambda \subset T^*B$ is a discrete lattice in T^*B locally spanned by closed one-forms. By Proposition 1.72 this is equivalent to an integral affine structure on B $\square$

The lattice Λ will be an important object of study throughout the remainder of this thesis. We introduce some terminology for it now.

Definition 2.35 (Period Lattice). Let B be an integral affine manifold and let $\Lambda \subset T^*B$ be the corresponding lattice in the cotangent bundle. Then we will call Λ the period lattice associated to the integral affine structure on B .

Theorem 2.36. The symplectic form on T^*B descends to T^*B/Λ and the projection map $\tau : T^*B/\Lambda \rightarrow B$ is a Lagrangian fibration.

Proof. The local sections of Λ are by definition, given by closed one-forms. Locally, a section of Λ looks like

$$\alpha = \sum_i m_i dx_i$$

where $m_i \in \mathbb{Z}$. Two points $x, y \in T_p^*B$ are identified if $x = y + \alpha(p)$ for some local section α .

But this is just a shift by a closed one form on B . By Proposition 2.18, the shift by a closed one-form is a symplectomorphism and hence preserves the symplectic form. It follows that the symplectic form descends to the quotient. This means that there is a unique symplectic form $\tilde{\omega}$ on T^*B/Λ such that

$$\pi^*\tilde{\omega} = \omega$$

where $\pi : T^*B \rightarrow T^*B/\Lambda$ is the projection map. Since the fibres of $T^*B \rightarrow B$ are Lagrangian it follows that the fibres of $T^*B/\Lambda \rightarrow B$ are too. $\square$

Remark 2.37. We have thus constructed a torus bundle $\tau : T^*B/\Lambda \rightarrow B$ which acts fibrewise on our original Lagrangian fibration $M \rightarrow B$. Indeed, this torus bundle is itself a Lagrangian fibration and inherits a symplectic form from T^*B . The torus bundle $\tau : T^*B/\Lambda \rightarrow B$ always has a global section, namely the projection of the zero section. However, our original Lagrangian fibration $M \rightarrow B$ has only the structure of an *affine torus bundle*. It becomes a torus bundle itself only if it admits a global section $\sigma : B \rightarrow M$ in which case it identifies with the torus bundle $T^*B/\Lambda \rightarrow B$. There are in general, obstructions to the existence of such a global section.

If we assume that the obstructions to the existence of a global section vanish, then $M \rightarrow B$ itself becomes a torus bundle and moreover, we can identify it with $T^*B/\Lambda \rightarrow B$ in a way that respects the group structure.

Lemma 2.38. Let $\sigma : B \rightarrow M$ be a global section of $M \rightarrow B$ and $\alpha \in \Omega^1(B)$ be a one-form. Then this determines a unique, fibre bundle isomorphism $A : T^*B/\Lambda \rightarrow M$ that is T^*B/Λ -equivariant such that the following diagram commutes.

$$\begin{array}{ccc} T^*B/\Lambda & \xrightarrow{A} & M \\ \hat{F}_\alpha \downarrow & & \downarrow F_\alpha \\ T^*B/\Lambda & \xrightarrow{A} & M \end{array}$$

Here $\hat{F}_\alpha$ is determined by the action of T^*B/Λ on itself (it is namely a Lagrangian fibration over B).

Proof. We define $A : T^*B/\Lambda \rightarrow M$ by

$$\begin{aligned} A_b : T_b^*B/\Lambda_b &\rightarrow M_b \\ [\alpha_b] &\mapsto F_\alpha(\sigma(b)). \end{aligned}$$

This map is injective as we have quotiented out the fixed point set of this action and hence a bijection by dimension arguments. We saw that this action is a fibrewise diffeomorphism in the proof of Theorem 2.31 and so this defines a bundle isomorphism. We now show equivariance. This follows from the following calculation.

$$\begin{aligned} A \circ \hat{F}_\alpha([\beta]) &= A([\alpha] + [\beta]) \\ &= F_{\alpha+\beta}(\sigma(b)) \\ &= F_\alpha \circ F_\beta(\sigma(b)) \\ &= F_\alpha \circ A([\beta]) \end{aligned}$$

for $b \in B$ and $\alpha, \beta \in \Omega^1(B)$. □

We note that this gives only a fibrewise, equivariant bundle isomorphism between $\tau : T^*B/\Lambda$ and $M \rightarrow B$. However, we have not yet discussed when the symplectic structure is preserved. We discuss this now.

Proposition 2.39. The cohomology class $[\sigma^*\omega] \in H^2(B)$ does not depend on the choice of σ .

Proof. Since the action of T^*B on M is transitive, any other section $\hat{\sigma}$ differs by an action of T^*B i.e. $\hat{\sigma} = F_\alpha \circ \sigma$. It follows that

$$\begin{aligned} \hat{\sigma}^*\omega &= (F_\alpha \circ \sigma)^*\omega \\ &= \sigma^*(F_\alpha^*\omega) \\ &= \sigma^*(\omega - \pi^*d\alpha) \\ &= \sigma^*\omega - d\alpha. \end{aligned}$$
□

Lemma 2.40. Suppose that $\sigma : B \rightarrow M$ is a section with $[\sigma^*\omega] = 0 \in H^2(B)$. Then $A : T^*B/\Lambda \rightarrow M$ is a symplectomorphism.

Proof. We can assume without loss of generality that $\sigma^*\omega = 0$ as otherwise we can shift it to the zero section by a closed form which is a symplectomorphism. It is enough to show that the map $\hat{A} : T^*B \rightarrow M$ covering A is a local symplectomorphism since the symplectic form descends to the quotient. The map $\hat{A}$ is determined by the condition

$$\hat{A} \circ \alpha = F_\alpha \circ \sigma$$

for every $\alpha \in \Omega^1(B)$. We show that the symplectic form is preserved. We compute the pullback

$$\begin{aligned} \alpha^* \hat{A}^* \omega &= (F_\alpha \circ \sigma)^* \omega \\ &= \sigma^* F_\alpha^* \omega \\ &= \sigma^* (\omega - d\pi^* \alpha) \\ &= -d\sigma^* \pi^* \alpha \\ &= -d\alpha \\ &= \alpha^* (-d\theta) \\ &= \alpha^* \omega. \end{aligned}$$

Since this holds for any $\alpha \in \Omega^1(B)$ it follows that $\hat{A}^* \omega = \omega$. $\square$

Remark 2.41. We note that the compactness assumption is used so that the flows F_α are defined for all time. If we drop the assumption that our fibres are compact and instead assume only that the vector fields X_α defined in Theorem 2.31 are complete, then most of the above discussion holds with some slight modifications. This allows us to obtain the following classification result for regular Lagrangian fibrations with fibre $\mathbb{R}^n$.

Proposition 2.42. Let $\rho : (E, \omega) \rightarrow B$ be a regular Lagrangian fibration with fibre $\mathbb{R}^n$ and suppose that the vector fields X_α are complete for all $\alpha \in \Omega^1(B)$. Then $\rho : E \rightarrow B$ is Lagrangian equivalent to $\pi : (T^*B, \omega_{can} + \rho^* \beta) \rightarrow B$ where β is a closed two-form on B . The cohomology class $[\beta] \in H^2(B, \mathbb{R})$ determines up to Lagrangian equivalence the fibration.

Proof. Since the fibres are contractible, obstruction theory tells us that there is a section $s : B \rightarrow E$, for example see [DK01]. Now, let ω be the symplectic form on E and let $\beta = s^* \omega$. Proposition 2.18 tells us that $(E, \omega - \rho^* \beta)$ is a Lagrangian bundle. But now

$$\begin{aligned} s^*(\omega - \rho^* \beta) &= s^* \omega - s^* \rho^* s^* \omega \\ &= s^* \omega - (s \circ \rho \circ s)^* \omega \\ &= s^* \omega - s^* \omega \\ &= 0 \end{aligned}$$

and hence s is a Lagrangian section of $(E, \omega - \rho^* \beta)$. Therefore this section gives a symplectomorphism $(T^*B, \omega_{can}) \cong (E, \omega - \rho^* \beta)$ by the same arguments as Lemma 2.38. Adding the closed two-form β to both bundles gives the required result. Since (T^*B, ω_{can}) is symplectomorphic to $(T^*B, \omega_{can} + \pi^* \beta)$ if and only if β is exact it follows that $[\beta] \in H^2(B, \mathbb{R})$ determines the Lagrangian fibration. $\square$

2.3.2 (Global) Action-Angle Coordinates

In [Dui80], Duistermaat studied when a Lagrangian torus fibration can globally admit a type of coordinate system called *action-angle coordinates*. He was motivated by a class of dynamical

2.3. INTEGRAL AFFINE STRUCTURE ON THE BASE

systems called *completely integrable Hamiltonian systems*. Roughly speaking, these are dynamical systems in which there is sufficient symmetry such that the equations of motion can be exactly solved. The techniques developed in this paper have been widely used and expanded upon in the theory of integrable systems. We will now give an outline of some of the key details.

In this section, we will assume that there exists a global Lagrangian section $\sigma : B \rightarrow M$ and that the cohomology class $[\sigma^*\omega] = 0 \in H^2(B)$ so that $M \rightarrow B$ and $T^*B/\Lambda \rightarrow B$ are fibrewise symplectomorphic unless stated otherwise. If we assume also that the base space B is simply connected then we have that the linear holonomy representation of B is trivial and so the group bundle $\Lambda \rightarrow B$ is trivial. Fixing a basepoint $b \in B$ we pick some isomorphism $\Lambda_b \cong \mathbb{Z}^n$ and then for any other $b' \in B$, we identify $\Lambda_b \cong \Lambda_{b'}$ by picking a path between them. Since by Corollary 2.34, B is itself an integral affine manifold, it hence possesses a flat, torsion free connection and so we use this connection to parallel transport a basis of Λ_b to a basis of $\Lambda_{b'}$.

Since B is assumed to be simply connected, this gives an isomorphism

$$\Lambda \cong B \times \mathbb{Z}^n$$

that depends on our choice of base point. Any two such isomorphisms will differ by an element of $GL(n, \mathbb{Z})$. Let $\beta_1, \dots, \beta_n$ be a global frame for $B \times \mathbb{Z}^n$. We note that the β_i will be closed since sections of $\Lambda \subset T^*B$ are given by closed one-forms. Since Λ is a full rank lattice, this also defines a global frame for T^*B and hence we have trivializations

$$T^*B \cong B \times \mathbb{R}^n$$

and

$$M \cong T^*B/\Lambda \cong B \times (\mathbb{R}/\mathbb{Z})^n.$$

Definition 2.43. We define *angle coordinates* on $M \cong B \times (\mathbb{R}/\mathbb{Z})^n$ by

$$s : M \cong B \times (\mathbb{R}/\mathbb{Z})^n \rightarrow (\mathbb{R}/\mathbb{Z})^n$$

where s is the projection map.

Recall that we can associate vector fields on M to the $\beta_i \in \Omega^1(B)$ by $\iota_{X_{\beta_i}}\omega = -\pi^*\beta_i$ and a quick calculation using Cartan's formula shows that these vector fields are symplectic i.e. $L_{X_{\beta_i}}\omega = 0$ since both β_i and ω are closed. Denoting their corresponding flow by F_i^t , in terms of the angle coordinates we have that

$$s_j(F_i^t(x)) = \begin{cases} s_j(x), & i \neq j, \\ s_j(x) + t, & i = j \end{cases}$$

for $x \in M$. Indeed, a point $x \in M$ identifies with

$$x = (p, a_1\beta_1(p), \dots, a_n\beta_n(p))$$

under $M \cong B \times \mathbb{R}^n$ where $p \in B$ and $a_i \in \mathbb{R}$. The flows of X_{β_i} preserve the fibres and are linear maps. Hence, the flow of β_i acts on x by

$$F_i^t(x) = (p, a_1\beta_1(p), \dots, a_i\beta_i(p) + t\beta_i(p), \dots, a_n\beta_n(p))$$

and since s_j is projection onto the j^{th} component, the result follows.

Moreover, since B is simply connected the β_i are in fact exact i.e. $\beta_i = dI_i$. Since the β_i form a global frame, $dI_p = (dI_1, \dots, dI_n)_p$ is full rank at each $p \in M$ and hence we can use them to define coordinates on the base.

Definition 2.44. We define *action-angle coordinates* on M by $(\pi^*I_1, \dots, \pi^*I_n, s_1, \dots, s_n)$.

Remark 2.45. It is clear from the above discussion that the action coordinates I_i define an integral affine structure in B . Indeed, the differentials of these span a full-rank lattice in T^*B and hence by Proposition 1.72 they equip B with an integral affine structure.

Remark 2.46. In these action-angle coordinates we can write the symplectic form as $\omega = \sum_i dI_i \wedge ds_i$.

Remark 2.47. For a regular Lagrangian torus fibration $M \rightarrow B$, even if there does not exist global action-angle coordinates, we can cover B in charts such that in each chart the fibration does admit global action-angle coordinates.

2.3.3 Completely integrable Hamiltonian systems

We will give a very brief introduction to completely integrable Hamiltonian systems as historically, they were important in the development of much of this theory and they also provide us with many examples of Lagrangian torus fibrations.

Definition 2.48. Let (M^{2n}, ω) be a symplectic manifold. A *completely integrable Hamiltonian system* (M, ω, F) , is a map

$$F = (f_1, \dots, f_n) : M \rightarrow \mathbb{R}^n$$

such that the components satisfy

1. $\{f_i, f_j\} = 0$ for all i, j .
2. $df_1 \wedge \dots \wedge df_n \neq 0$ on an open, dense set of M .

Remark 2.49. The bracket $\{\cdot, \cdot\}$ is the *Poisson* bracket on M . It is defined as

$$\{f, g\} = \omega(X_f, X_g)$$

with $f, g \in C^\infty(M)$.

Remark 2.50. The condition $\{f_i, f_j\} = 0$ ensures that the fibres of the map F are Lagrangian submanifolds of M . The second condition is a regularity condition on the fibres of F .

The following theorem shows the relationship between completely integrable Hamiltonian systems and Lagrangian fibrations.

Theorem 2.51 (Liouville-Arnold). Let (M, ω, F) be a completely integrable Hamiltonian system. Suppose that $x \in \mathbb{R}^n$ is a regular value of F and that $F^{-1}(x)$ is compact and connected. Then there exists an open neighborhood $U \subset \mathbb{R}^n$ of x such that

$$F|_{F^{-1}(U)} : M \rightarrow U$$

is a regular Lagrangian torus fibration.

Remark 2.52. This allows us to associate a Lagrangian fibration to an open neighborhood of a

2.3. INTEGRAL AFFINE STRUCTURE ON THE BASE

regular fibre of the completely integrable Hamiltonian system. Conversely, given a Lagrangian fibration admitting global action-angle coordinates, the action coordinates determine a completely integrable Hamiltonian system.

Nilpotent Holonomy and Completeness

3.1 Nilpotent Holonomy

3.1.1 Nilpotent Groups

In this section, we will see that the condition that the holonomy group of an affine manifold M be nilpotent will be useful in determining whether the affine structure is complete. The main goal of this chapter is to prove the Markus conjecture in this case. The conjecture is posed in [Mar63] and states that for a compact affine manifold, completeness of the affine structure is equivalent to parallel volume. This conjecture is yet to be solved in full generality. This chapter is follows closely the paper [FGH81] and many of the results we state were first proved there.

Definition 3.1 (Central Series). Let G be a group. Then a central series for G is a finite collection of normal subgroups such that

$$1 = G_0 \leq G_1 \leq \cdots \leq G_s = G$$

and

$$\frac{G_{i+1}}{G_i} \leq Z\left(\frac{G}{G_i}\right)$$

where $Z\left(\frac{G}{G_i}\right)$ denotes the center.

Definition 3.2 (Nilpotent Group). A group G is said to be nilpotent if it admits a central series.

Remark 3.3. The *nilpotent class* of G is the length of the shortest central series.

Example 3.4. An important example of nilpotent groups are the upper triangular matrices. For example, the group

$$H(R) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \right\}$$

is called the Heisenberg group and is nilpotent of class 2. This group notably shows up in quantum mechanics.

For nilpotent groups, we have the following results regarding their group cohomology.

Lemma 3.5. Let G be a nilpotent group and E a G -module. If $H^0(G, E) = 0$ then also $H^i(G, E) = 0$ for all $i \geq 0$.

Proof. The proof of this can be found in [Hir78]. □

A consequence of this is the following lemma.

Lemma 3.6. If G is a nilpotent group and E is a G -module, then the following statements are equivalent.

1. $H^0(G, E) = 0$
2. $H^0(G, E^*) = 0$
3. $H_0(G, E) = 0$
4. $H_0(G, E^*) = 0$

Proof. See [FGH81]. □

3.1.2 Unipotence and the Fitting submodule

In the case that the affine holonomy group is nilpotent, we will be able to show that completeness is equivalent to a property called unipotence of the linear holonomy. We will now outline some consequences of unipotence.

Definition 3.7 (Unipotent). A G -module E is called unipotent if $(g - I)^n = 0$ for every $g \in G$ where $n = \dim(E)$.

Definition 3.8. An affine representation is unipotent if its linear part defines a unipotent G -module.

Although E may not be unipotent itself, we can consider the smallest submodule (possibly the zero module) of E which is unipotent. We shall denote this submodule by E_U .

Remark 3.9. The submodule E_U is called the *Fitting submodule* of E .

Often, it will be useful to quotient out by the unipotent part of a G -module. The following lemma gives some insight on why this is a useful idea.

Lemma 3.10. Let E_U be the Fitting submodule of E . Then

$$H^0(G, E/E_U) = 0$$

Proof. Take E_U to be the Fitting submodule and assume $E_U \neq E$ otherwise the result is trivial. Suppose that $\dim E_U = r$ so that $(g - 1)^r \cdot y = 0$ for all $y \in E_U$.

As $H^0(G, E/E_U)$ is the set of stationary points of the G -module E/E_U we can take $x + E_U \in H^0(G, E/E_U)$ not the identity so that

$$\begin{aligned} g \cdot (x + E_U) &= x + E_U \\ \Rightarrow (g - 1) \cdot x &\in E_U \end{aligned}$$

for all $g \in G$.

Now we claim that the G -submodule spanned by x and E_U is unipotent of dimension $r + 1$. As $x \notin E_U$ this submodule is strictly bigger than E_U . Now, letting $y \in E_U$ we have that

$$\begin{aligned} (g - 1)^{r+1} \cdot (x + y) &= (g - 1)^r \cdot ((g - 1)x + (g - 1)y) \\ &= 0 \end{aligned}$$

since $(g - 1)x + (g - 1)y \in E_U$ and $\dim E_U = r$. Hence this is a unipotent submodule strictly bigger than E_U which contradicts maximality. $\square$

The following splitting lemma will prove important in our characterization of completeness.

Lemma 3.11. If G is a nilpotent group and E a G -module with Fitting submodule E_U , then there exists a unique submodule F such that $E = E_U \oplus F$.

Proof. Let β be the induced representation $\beta : G \rightarrow GL(E/E_U)$ on $GL(E/E_U)$. By Lemma 3.10, $H^0(G, E/E_U) = 0$.

We will make use of the following claim which we resolve in Proposition 3.12.

Claim. A submodule $F \subset E$ is complementary to E_U e.g. $E = E_U \oplus F$ if and only if $F = T(E/E_U)$ where $T : E/E_U \rightarrow E$ is an equivariant linear map with $P \circ T = 1_{E/E_U}$ with P the canonical projection map.

Now let $S : E/E_U \rightarrow E$ be a linear map satisfying $P \circ S = 1_{E/E_U}$. Then any such T as above can be uniquely written as $T = R + S$ with $R : E/E_U \rightarrow E_U$. It is T -equivariant if and only if

$$\begin{aligned} R + S &= g \circ (R + S) \circ \beta(g)^{-1} \\ \Rightarrow R &= g \circ R \circ \beta(g)^{-1} + g \circ S \circ \beta(g)^{-1} - S. \end{aligned} \tag{3.1}$$

If we can show that there is a unique such T , then we have shown that E splits uniquely into $E = E_U \oplus F$. This will follow if we can show there is a unique R satisfying the above.

To do this we define a representation on $\text{Hom}(E/E_U, E)$ by

$$\begin{aligned} \gamma : G &\rightarrow \text{Hom}(E/E_U, E) \\ \gamma(g)(R) &= g \circ R \circ \beta(g)^{-1} \end{aligned}$$

and we define a crossed homomorphism for γ by $\mu(g) = g \circ S \circ \beta(g)^{-1} - S$. This defines a crossed homomorphism as

$$\begin{aligned} \mu(gh) &= g \circ h \circ S \circ \beta(h)^{-1} \circ \beta(g)^{-1} - S \\ &= \gamma(g)(h \circ S \circ \beta(h)^{-1}) - S \\ &= \gamma(g)(\mu(h) + S) - S \\ &= \gamma(g)(\mu(h)) + g \circ S \circ \beta(g)^{-1} - S \\ &= \gamma(g)(\mu(h)) + \mu(g). \end{aligned}$$

As P is equivariant and $P \circ S = 1_{E/E_U}$ we get

$$\begin{aligned} P \circ \mu(g) &= P \circ g \circ S \circ \beta(g)^{-1} - P \circ S \\ &= \beta(g) \circ P \circ S \circ \beta(g)^{-1} - 1 \\ &= \beta(g) \circ \beta(g)^{-1} - 1 \\ &= 0. \end{aligned}$$

Now we can write (3.1) as

$$R = \gamma(g) \circ R + \mu(g)$$

which by Proposition 1.60 is saying that R is a fixed point of the affine representation induced by γ and μ . We wish now to show that this affine representation has a unique fixed point. This is equivalent by Lemma 3.5 and Lemma 3.6 to $H^0(G, \text{Hom}(E/E_U)) = 0$.

To this end we suppose that $R : E/E_U \rightarrow E$ is fixed under all $\gamma(g)$ for $g \in G$. Then, letting $d = \dim(E_U)$ for any d elements $g_1, \dots, g_d \in G$ we have

$$(1 - g_1) \circ \dots \circ (1 - g_d)|_{E_U} = 0$$

as they act unipotently on E_U . It now follows by equivariance that

$$\begin{aligned} (1 - g_1) \circ \dots \circ (1 - g_d) \circ R &= (1 - g_1) \circ \dots \circ (R - g_d \circ R) \\ &= (1 - g_1) \circ \dots \circ (R - R \circ \beta(g_d)) \\ &= (1 - g_1) \circ \dots \circ R \circ (1 - \beta(g_d)) \\ &\Rightarrow R \circ (1 - \beta(g_1)) \circ \dots \circ (1 - \beta(g_d)) = 0 \end{aligned}$$

so that R vanishes on vectors of the form

$$(1 - \beta(g_1)) \circ \dots \circ (1 - \beta(g_d))x$$

for $x \in E/E_U$. Thus if all vectors are in this span, then the only such R fixed under the action of γ is the zero map. As $1 - \beta(g)$ is invertible on E/E_U (it acts without unipotent part), this is equivalent to showing that E/E_U is spanned by vectors of the form $(1 - \beta(g))x$ with $x \in E/E_U$. By Proposition 1.44, this is equivalent to $H_0(G, E/E_U) = 0$ which is true by Lemma 3.10 completing the proof. $\square$

This proves our result once we have cleared up that the stated claim is indeed true which we show in the following proposition.

Proposition 3.12. A submodule $F \subset E$ is complementary to E_U e.g. $E = E_U \oplus F$ if and only if $F = T(E/E_U)$ where $T : E/E_U \rightarrow E$ is an equivariant linear map with $P \circ T = 1_{E/E_U}$ with P the canonical projection map.

Proof. Suppose $F = T(E/E_U)$, with T being G -equivariant so that F is a G -module. If $T(x + E_U) = e \in E$, then $P(e) = e + E_U = x + E_U \Rightarrow x \in E_U$ and so by linearity $e = 0$ and so $F \cap E_U = \{0\}$. Clearly, $E_U \oplus F \subseteq E$. For any $x \in E$, $x - T(x + E_U) \in E_U$. We can thus write $x = (x - T(x + E_U)) + T(x + E_U) \in E_U + F$.

Conversely, suppose that F is a G -submodule and $F \cap E_U = \{0\}$ and $F \oplus E_U = E$. Then it's easy to see that $P|_F : F \rightarrow E/E_U$ is bijective. It is G -equivariant as P is and F is a G -submodule. We can thus define T as $T = P^{-1}|_F$. $\square$

An important consequence of this theorem is that for *indecomposable affine representations* of a nilpotent group the affine representation is always unipotent. We recall the definition of an indecomposable affine representation now and then prove this result.

Definition 3.13 (Decomposable affine representation). Let E be a G -module. An affine representation $\alpha : G \rightarrow \text{Aff}(E)$ is said to be *decomposable* if there exists a decomposition $E = E_1 \oplus F$ with $E_1 \neq E$ such that the decomposition is invariant under the linear part of α and some coset $x + E_1$ is invariant under the entire affine holonomy α where $x \in E$.

Definition 3.14 (Indecomposable affine representation). An affine representation $\alpha : G \rightarrow \text{Aff}(E)$ is called *indecomposable* if it is not decomposable.

Lemma 3.15. Let $E_U \subset E$ be the Fitting submodule of an affine representation $\alpha : G \rightarrow \text{Aff}(E)$. If G is nilpotent then there is a coset of E_U that is invariant under α .

Proof. Let $\mu : G \rightarrow E$ be the translational component of the affine representation $\alpha = (\lambda, \mu)$. We check that

$$\mu' : G \xrightarrow{\mu} E \xrightarrow{\pi} E/E_U$$

is a crossed homomorphism for $\lambda' = \pi \circ \lambda$. Letting $g, h \in G$ we compute

$$\begin{aligned} \mu'(gh) &= \pi \circ \mu(gh) \\ &= \pi \circ (\mu(g) + \lambda(g)\mu(h)) \\ &= \pi \circ \mu(g) + \pi \circ \lambda(g)\mu(h) \\ &= \mu'(g) + \lambda'(g)\mu'(h). \end{aligned}$$

Now, it follows that $\alpha' = (\lambda', \mu')$ is an affine representation of G on E/E_U . Furthermore, the following diagram commutes

$$\begin{array}{ccc} E & \xrightarrow{\pi} & E/E_U \\ \alpha(g) \downarrow & & \downarrow \alpha'(g) \\ E & \xrightarrow{\pi} & E/E_U \end{array}$$

for all $g \in G$. Hence π is a G -equivariant map with respect to the representations α and α' . Now Lemma 3.10 says that $H^0(G, E/E_U) = 0$ and hence by Lemma 3.5 we have that $H^1(G, E/E_U) = 0$ since G is nilpotent. Namely, the radiance obstruction $[c_{\alpha'}] \in H^1(G, E/E_U)$ vanishes and hence the affine representation α' fixes some $x + E_U$ where $x \in E$ and so α fixes that coset. $\square$

We can now prove the following theorem relating indecomposable affine representations to unipotency.

Lemma 3.16. Let $\alpha : G \rightarrow \text{Aff}(E)$ be an indecomposable affine representation of a nilpotent group G . Then α is unipotent.

Proof. This follows as a corollary of our previous two lemmas. Lemma 3.11 tells us that there is a unique $F \subset E$ such that $E = E_U \oplus F$ where F is a G -submodule of E and E_U is the Fitting submodule. Now, Lemma 3.15 tells us that there is some coset of E_U invariant under the linear part of α . But this by definition means that α is decomposable unless $E_U = E$ which completes the proof as E_U is the unipotent part of E . $\square$

3.2 Unipotent Representations and Completeness

3.2.1 Expansions in the linear holonomy

The aim of this section is to show that for a compact affine manifold, the holonomy group being unipotent is equivalent to completeness of the affine structure. Of geometrical importance to this will be the concept of an *expansion* in the linear holonomy.

Definition 3.17 (Expansion). Let $T : V \rightarrow V$ be a linear map. Then T is said to be an expansion of V if for every eigenvalue λ of T we have $|\lambda| > 1$.

We now wish to prove Lemma 3.30 involving expansions in the linear holonomy group. However, in order to do so we will need to recall some of the theory of algebraic groups and the representation theory of nilpotent groups.

3.2.2 Algebraic groups and representation theory of Lie algebras

We will need some tools from the theory of algebraic groups and the representation theory of Lie algebras. This discussion will be extremely brief and we refer the reader to [Hum75] for more background on algebraic groups and [Hal03] for the representation theory of Lie algebras.

We start with the definition of an algebraic group.

Definition 3.18 (Algebraic Group). An *algebraic group* is a matrix group that can be defined using polynomials.

If G is an algebraic group, then there is a unique irreducible component passing through the identity denoted by G° . The topology on G is taken to be the Zariski topology.

Remark 3.19. This unique irreducible component is called the *identity component* of G .

Proposition 3.20. Let G be an algebraic group. Then G° is a normal subgroup of finite index in G whose cosets are connected and irreducible components of G .

Proof. See section 7.3 of [Hum75]. $\square$

Proposition 3.21. Let G be an algebraic group and H a normal subgroup. Then

$$\overline{[G, H]} = [\overline{G}, \overline{H}]$$

where $\overline{H}$ denotes the Zariski closure.

Proof. This follows from Proposition 2.4 of [Bor91]. $\square$

This proposition leads to the following corollary.

Corollary 3.22. Let G be an algebraic group and N a normal subgroup that is nilpotent. Then $\overline{N}$ is also nilpotent.

Proof. By Proposition 3.21, a central series for N will pass to a central series of $\overline{N}$. □

The next result we will state is an important result from the representation theory of Lie algebras. It will guarantee us a basis for our representation such that the matrices of the representation are upper triangular.

Definition 3.23 (Weight Space). Let $\mathfrak{g}$ be a Lie algebra and $\pi : \mathfrak{g} \rightarrow \mathfrak{gl}_V$ be a representation on a vector space V . For $\lambda \in \mathfrak{g}^*$ we define the weight space of $\mathfrak{g}$ with respect to λ to be

$$V_\lambda^\mathfrak{g} = \{v \in V \mid \pi(g)v = \lambda(g)v, \quad \forall g \in \mathfrak{g}\}.$$

Remark 3.24. If $V_\lambda^\mathfrak{g} \neq 0$ we will say that λ is a weight for π .

Lemma 3.25 (Lie's theorem). Let $\mathfrak{g}$ be a solvable Lie algebra and let π be a representation of $\mathfrak{g}$ on a finite dimensional, non-zero vector space V . Then there exists a weight $\lambda \in \mathfrak{g}^*$ for π .

Proof. This result can be found for instance in [Kac10]. □

Corollary 3.26. Let V be a vector space. If $\pi : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is a finite dimensional representation of a solvable Lie algebra, then there is a basis of V such that all linear transformations in $\pi(\mathfrak{g})$ are represented by upper triangular matrices.

We will need a slightly stronger theorem than Lie's theorem which gives a decomposition of our vector space into generalized weight spaces.

Definition 3.27 (Generalized Weight Space). Let π be a representation of a Lie algebra $\mathfrak{g}$ over a finite dimensional vector space V . Then for $\lambda \in \mathfrak{g}^*$, we can associate to λ

$$V_\lambda = \{v \in V \mid (\pi(g) - \lambda(g))^n v = 0 \text{ for all } g \in \mathfrak{g}\}$$

where n depends on $g \in \mathfrak{g}$ and $v \in V$. If $V_\lambda \neq 0$ then we call it the *generalized weight space* associated to λ .

Remark 3.28. This tells us that $\pi(g) - \lambda(g)$ is a nilpotent operator so that $\pi(g) = \lambda(g) + N(g)$ with $N(g)$ nilpotent.

Lemma 3.29. Let $\mathfrak{g}$ be a nilpotent Lie algebra over $\mathbb{C}$. Let π be a representation of $\mathfrak{g}$ on a finite dimensional complex vector space V . Then there are finitely many generalized weight spaces V_{λ_k} , which are invariant under the action of the representation and $V = \bigoplus_k V_{\lambda_k}$.

Proof. See Proposition 3.2 in [War13]. □

We now have the required machinery to start our proof.

Lemma 3.30. Let G be a nilpotent group and $E = \mathbb{R}^n$ a G -module. Suppose that the linear holonomy group does not contain an expansion of E . Then for each $n \in \mathbb{N}_{>0}$, there is a C^r map $\varphi : E \rightarrow \mathbb{R}$ for any $r \in \mathbb{Z}_{>0}$ which satisfies the following.

1. $\varphi > 0$ almost everywhere.
2. φ is G -invariant.
3. There exists some $a > 0$ such that

$$\varphi(e^t x) = e^{ta} \varphi(x)$$

for all $t \in \mathbb{R}$ and $x \in E$.

Proof. If we can prove this for a normal subgroup G_0 of finite index, then we will be able to construct our map φ as follows. Let $g_1 G_0, \dots, g_k G_0$ be the left cosets of G_0 and φ_0 a map satisfying properties 1 and 3 defined on G_0 . Now define

$$\varphi(x) = \sum_{i=1}^k \varphi_0(g_i x).$$

This will be G -invariant as G_0 is a normal subgroup of G and multiplication by an element of G will simply permute the cosets.

As we want to eventually apply some results from complex Lie Theory, we complexify the dual space E^* as $F = \mathbb{C} \otimes E^*$. The contragradient representation of G on E^* extends to a representation on F in the obvious way. We will denote this as $\rho : G \rightarrow GL(F)$. Now, $\rho(G)$ is a nilpotent subgroup of $GL(F)$ that passes through the identity.

Let H be the identity component of the algebraic closure of $\rho(G)$. As $\rho(G)$ is nilpotent, by Corollary 3.22 so is its Zariski closure and hence its identity component as subgroups of nilpotent groups are nilpotent. The subgroup H will be a normal subgroup of finite index by Proposition 3.20 and so $G_0 = \rho^{-1}(H)$ will also be a normal subgroup of finite index.

We can now obtain a Lie algebra representation of the Lie algebra of H on F by

$$di : \mathfrak{h} \rightarrow \mathfrak{gl}(F)$$

where i is the inclusion map. Importantly, by construction on the underlying vector spaces $\rho(G_0) \subseteq di(\mathfrak{h})$. As H is a nilpotent Lie group, its Lie algebra is nilpotent hence solvable so we can apply Lemma 3.29 to obtain a decomposition $F = \bigoplus_{k=1}^m F_k$ which is $di(\mathfrak{h})$ -invariant and hence ρ -invariant by construction when restricted to G_0 .

Again, Lemma 3.29 yields that there is a basis B_k of each F_k that represents the operators $di(h)|_{F_k}$ as upper triangular complex matrices

$$di_k(h) = \lambda_k(h)I + N_k(h)$$

where di_k is the restriction to F_k and $N_k(h)$ is strictly upper triangular nilpotent matrix and $\lambda_k(h)$ are the eigenvalues of $\rho(h)$. The first basis vector in each B_k is killed by the strictly upper triangular matrix so that if $f_k \in F_k$ is the first basis vector in B_k

3.2. UNIPOTENT REPRESENTATIONS AND COMPLETENESS

$$di_k(h)f_k = \lambda_k(h)f_k$$

for all $h \in H$.

Now we define the following group morphisms

$$\begin{aligned}\varphi_k &: G_0 \rightarrow \mathbb{R} \\ \varphi_k(g) &= \log |\lambda_k(g)|\end{aligned}$$

for $k = 1, \dots, m$.

This defines a map into $\mathbb{R}^m$ by

$$\varphi = (\varphi_1, \dots, \varphi_m) : G_0 \rightarrow \mathbb{R}^m.$$

If G_0 does not contain any expansions of E then the image $\varphi(G_0)$ is disjoint from the set

$$P = \{y \in \mathbb{R}^m : y_k > 0, k = 1, \dots, m\}.$$

As it is a subgroup, it follows that its linear span $\text{span}\{\varphi(G_0)\}$ is also disjoint from P as if a point in the image was in the strictly negative orthant, its inverse would lie in P . Therefore there exists a vector in the orthogonal complement of $\varphi(G_0)$ that lies in the closure of P . We denote this by $v = (v_1, \dots, v_m)$ and it satisfies

$$v_k \geq 0 \text{ for all } k,$$

$$\sum_{k=1}^m v_k = a \geq 0,$$

$$\sum_{k=1}^m v_k \varphi_k(g) = 0 \text{ for all } g \in G_0.$$

If we scale our v by a positive scalar $c > 0$ it will satisfy the same properties so that we can arrange that

$$v_k = 0 \text{ or } v_k > 2r \text{ for each } k \text{ with } r \text{ coming from the lemma.}$$

Now for each k we have a vector $f_k \in \text{Hom}(\mathbb{C} \otimes E, \mathbb{C})$ satisfy $di_k(h)f_k = \lambda_k(h)f_k$. We now include E into $\mathbb{C} \otimes E$ and construct the map

$$\begin{aligned}\psi &: E \rightarrow \mathbb{R} \\ \psi(x) &= \prod_{k=1}^m |f_k(x)|^{v_k}.\end{aligned}$$

Now ψ is C^r differentiable as $|f_k(x)|^{v_k} = (u_k(x)^2 + w_k(x)^2)^m$ with $m > r$. Properties 1) and 3) are satisfied by our construction of v_k . We now show that it is G_0 -invariant.

$$\psi(g^{-1}x) = \prod_{k=1}^m |f_k(g^{-1}x)|^{v_k}$$

noting this is the contragradient representation and also that f_k is an eigenvector

$$\begin{aligned}\psi(g^{-1}x) &= \prod_{k=1}^m |g \circ f_k(x)|^{v_k} \\ &= \prod_{k=1}^m |\lambda_k(g) f_k(x)|^{v_k} \\ &= \prod_{k=1}^m |\lambda_k(g)|^{v_k} \prod_{k=1}^m |f_k(x)|^{v_k}\end{aligned}$$

Now, recalling that $\varphi_k(g) = \log |\lambda_k(g)| \Rightarrow e^{\varphi_k(g)} = |\lambda_k(g)|$ so we obtained

$$\begin{aligned}\prod_{k=1}^m |\lambda_k(g)|^{v_k} \prod_{k=1}^m |f_k(x)|^{v_k} &= e^{\sum_{k=1}^m v_k \varphi_k(g)} \psi(x) \\ &= \psi(x)\end{aligned}$$

completing the proof. □

3.2.3 Consequences of expansions in linear holonomy

The previous theorem leads to several important results regarding completeness of affine manifolds.

Theorem 3.31. Let M be a compact affine manifold. Let $E_0 \subset E$ be a proper linear subspace invariant under the affine holonomy and denote by Λ the linear holonomy. If the image of the linear holonomy group $\Lambda_1 \subseteq GL(E/E_0)$ is nilpotent, then some element of Λ_1 expands E/E_0 .

Proof. Let $q : E \rightarrow E/E_0$ be the canonical projection. Denote by R the radiant vector field on E/E_0 . If $\{y_1, \dots, y_m\}$ are coordinates on E/E_0 , then this appears as $R = \sum_i y_i \frac{\partial}{\partial y_i}$. Then in each affine chart there is a vector field X_i that is π -related to R . As M is compact, we can cover it in finitely many charts and using a partition of unity subordinate to that covering form the vector field

$$X = \sum_i \mu_i X_i$$

on M .

Now suppose that Λ_1 does not contain any expansions of E/E_0 . Then by Lemma 3.30, there is a Λ_1 -invariant C^1 map $\Psi : E/E_0 \rightarrow \mathbb{R}$. As the flow along the radiant vector field at a point z is given by $\phi_t(z) = e^t z$, our map satisfies

$$\begin{aligned}d\Psi_z R(Z) &= \frac{d}{dt} \psi(e^t z)|_{t=0} \\ &= \frac{d}{dt} e^{ta} \psi(z)|_{t=0} \\ &= a \Psi(z)\end{aligned}$$

for some $a > 0$ and all $z \in E/E_0$.

Now the composition

$$\tilde{f} : \tilde{M} \rightarrow E \rightarrow E/E_0 \rightarrow \mathbb{R}$$

3.2. UNIPOTENT REPRESENTATIONS AND COMPLETENESS

is invariant under deck transformations. Indeed, let $\gamma \in \pi_1(M)$ then for $x \in \tilde{M}$

$$\Psi \circ \pi \circ \text{dev}(\gamma \star x) = \Psi \circ \pi \circ \text{hol}(\gamma) \circ \text{dev}(x)$$

and this is invariant as E_0 is invariant under the holonomy group.

This implies that $\tilde{f}$ covers a map $f : M \rightarrow \mathbb{R}$. As by Proposition 1.11, we can give charts in terms of the development pair, in affine coordinates we have that $f = \Psi \circ \pi$. Now

$$\begin{aligned} df_y X(y) &= d(\Psi \circ \pi)_y X(y) \\ &= \frac{d}{dt} \Psi \circ \pi(\gamma(t))|_{t=0} \\ &= \frac{d}{dt} \Psi(e^t \tilde{y})|_{t=0} \\ &= a\Psi(\tilde{y}) \end{aligned}$$

where $\tilde{y} = \pi(y)$ and $\gamma(t)$ is an integral curve of X starting at y . This gives that

$$df_y X(y) = af(y).$$

It follows that if $\alpha : I \rightarrow M$ is an integral curve of X starting at $p \in M$ then

$$\begin{aligned} f(\alpha(t)) &= \Psi \circ \pi(\alpha(t)) \\ &= \Psi(\overline{\alpha(t)}) \\ &= \Psi(e^t \tilde{p}) \\ &= e^{ta} \Psi(\tilde{p}) \\ &= e^{ta} f(\alpha(c)). \end{aligned}$$

Now as $\Psi > 0$ almost everywhere, there exists $x_0 \in M$ with $f(x_0) > 0$ and the integral curve α_0 through x_0 is defined for all t as M is compact. Then $\lim_{t \rightarrow \infty} f(\alpha_0(t)) = \lim_{t \rightarrow \infty} e^{ta} f(\alpha_0(c)) \rightarrow \infty$. But as f is a continuous function and M is compact it must be bounded and so we reach a contradiction. $\square$

This proof leads to several important results. The first result follows almost immediately.

Corollary 3.32. Let M be a compact radiant manifold with nilpotent linear holonomy Λ . Then Λ contains an expansion of E .

Proof. We can take $E_0 = 0$ in Theorem 3.31 and the result follows immediately. $\square$

The next corollary follows from considering the Fitting submodule of E .

Corollary 3.33. Let M be a compact affine manifold with nilpotent affine holonomy group. If $F \neq 0$ then some element of the linear holonomy group expands F .

Proof. Recall from Lemma 3.11, that nilpotence of the affine holonomy gives a unique decomposition $E = E_U \oplus F$ which is invariant under the action of the holonomy group. If we take $E_0 = E_U$ in Theorem 3.31, then $F \cong E/E_U$ as a group module with respect to the linear holonomy group. $\square$

3.3 Parallel volume

In [Mar63] it is conjectured that parallel volume is equivalent to completeness for affine manifolds. Using the previous results, we can now give a partial resolution of this conjecture for the case that our manifold has nilpotent holonomy. Orientable integral affine manifolds always possess a parallel volume form since their linear holonomy is in $SL(\mathbb{Z}^2)$ and so the following results will prove important in our classification of such structures.

Theorem 3.34. ([FGH81]) Let M be a compact affine manifold with nilpotent affine holonomy group. Suppose that there is a parallel volume form on M . Then the linear holonomy is unipotent.

Proof. Again we will exploit that there is a unique decomposition $E = E_U \oplus F$ by Lemma 3.11. Now let g be any element of the linear holonomy group. As there is a parallel volume form by Lemma 1.67, $\det(g) = 1$. But by the unique decomposition

$$1 = \det(g) = \det(g|_{E_U}) \det(g|_F).$$

But $g|_{E_U}$ is a unipotent operator and so has determinant 1. This forces $\det(g|_F) = 1$ so that F cannot contain an expansion. But now Corollary 3.33 forces $F = 0$ and so $E = E_U$. $\square$

Proposition 3.35. Let M be an affine manifold and suppose that M is compact and has unipotent affine holonomy representation. Then there exists a collection of n regular, involutive distributions $\mathfrak{F}_i$ on M , parallel one-forms ω_i on the leaves of $\mathfrak{F}_i$ vanishing on $\mathfrak{F}_{i-1}$ and vector fields $X_i \in \mathfrak{X}(M)$ such that

1. The X_i are tangent to the distributions $\mathfrak{F}_i$.
2. The one-forms $\omega_i(X_i) = 1$ on each leaf of $\mathfrak{F}_i$.

Proof. Let $\text{hol}(\pi_1(M)) = \Gamma$ be the affine holonomy group and $\text{Lin}(\Gamma)$ the linear holonomy group.

Now, suppose that the linear affine holonomy representation is unipotent. Then $\text{Lin}(\gamma)$ is upper-triangular with respect to some suitable basis and hence there exists a linear flag

$$0 = F_0 \subset F_1 \subset \cdots \subset F_n = \mathbb{R}^n$$

preserved by the linear affine holonomy group. Furthermore, the induced action on the quotients F_k/F_{k-1} is trivial and so the restriction to F_k preserves a linear functional $l_k : F_k \rightarrow \mathbb{R}$ with kernel F_{k-1} as shown in Section 17.5 of [Hum12].

This determines a family of involutive distributions $\mathfrak{F}_k \subset TM$ as follows. The differential of the developing map $d\text{dev}_p$ is an isomorphism for each $p \in \tilde{M}$. We define $\tilde{\mathfrak{F}}_i$ by

$$\tilde{\mathfrak{F}}_i = d\text{dev}^{-1}(F_i).$$

Now, by equivariance of the developing map we have

$$\begin{aligned} d\text{dev} \circ d\gamma(\tilde{\mathfrak{F}}_i) &= d\text{hol}(\gamma) \circ d\text{dev}(\tilde{\mathfrak{F}}_i) \\ &= \text{Lin}(\text{hol}(\gamma)) \circ d\text{dev}(\tilde{\mathfrak{F}}_i) \\ &= d\text{dev}(\tilde{\mathfrak{F}}_i) \end{aligned}$$

as F_i is invariant under the linear holonomy. Hence the $\tilde{\mathfrak{F}}_i$ descend to distributions $\mathfrak{F}_i$ on M as they are invariant under deck transformations. Since dev is a local diffeomorphism there exists a cover by affine charts $\{U_i\}_{i \in I}$ such that the $\mathfrak{F}_i$ are constant distributions and hence they are integrable.

Furthermore, each linear map l_i determines a parallel one-form ω_i defined on $\mathfrak{F}_i$ and vanishing on $\mathfrak{F}_{i-1}$. Indeed, we define these on $\tilde{M}$ by

$$\tilde{\omega}_i|_p(v) = l_i \circ d\text{dev}_p(v)$$

for $v \in \tilde{\mathfrak{F}}_i|_p$. Since the l_i are invariant under the linear holonomy, the $\tilde{\omega}_i$ descend to one-forms ω_i on M which are parallel as they are constant tensors in affine charts where dev is a diffeomorphism.

We can construct vector fields X_i on M , tangent to $\mathfrak{F}_i$ such that $\omega_i(X_i) = 1$ using partitions of unity. Locally there exist affine charts $\{U_k\}_{k \in I}$ such that the distributions $\mathfrak{F}_i$ are constant as they are the pull back of a constant distribution on $\mathbb{R}^n$ by an isomorphism. In each of these charts U_k , we can find a vector $v \in \mathfrak{F}_i|_p$ such that $\omega_i|_p(v) = 1$ where $p \in U_k$. We can extend this to a vector field X_k in each chart such that $\omega_i(X_k(p)) = 1$ on all of U_k .

Now, we pick a partition of unity $\{\rho_k\}_{k \in I}$ subordinate to the cover $\{U_k\}_{k \in I}$ and construct a vector field

$$X_i = \sum_k \rho_k X_k.$$

This vector field is tangent to the $\mathfrak{F}_i$ and satisfies $\omega_i(X_i) = 1$ on the leaves of $\mathfrak{F}_i$. These lift to vector fields $\tilde{X}_i$ on the universal cover which are complete as M is compact. $\square$

This will be used in conjunction with the previous theorem in order to show that for a compact affine manifold, unipotency of the affine holonomy representation implies completeness of the affine structure.

Theorem 3.36. Let M be an affine manifold. If M is compact and has unipotent linear holonomy representation then M is complete.

Proof. We pick a basepoint x_0 in M . Let $p : \tilde{M} \rightarrow M$ be the universal cover of M with $p(\tilde{x}_0) = x_0$. We now choose a developing map $\text{dev} : \tilde{M} \rightarrow \mathbb{R}^n$ such that $\text{dev}(\tilde{x}_0) = 0$. Let

$$0 = F_0 \subset F_1 \subset \cdots \subset F_n = \mathbb{R}^n$$

be the flag preserved by the linear holonomy and $\mathfrak{F}_i$ the corresponding involutive distribution. Let ω_i and X_i be the corresponding one-forms and vector fields on M satisfying the properties from Proposition 3.35. The X_i lift to vector fields $\tilde{X}_i$ on $\tilde{M}$ which are complete as M is compact and the ω_i lift to one forms $\tilde{\omega}_i$ on $\tilde{M}$ with $\tilde{\omega}_i(\tilde{X}_i) = 1$ on the leaves of $\tilde{\mathfrak{F}}_i$. We note that this is clear from the construction of these objects in Proposition 3.35.

We first show surjectivity of the developing map. Let $\gamma_i(t)$ be an integral curve of $\tilde{X}_i$ starting at the point $\tilde{x}_i$ and let $\text{dev}(\tilde{x}_i) = v_i \in \mathbb{R}^n$. From integrating

$$\frac{d}{dt} \text{dev} \circ \gamma_i(t) = d\text{dev}_{\gamma_i(t)} \circ \tilde{X}(\gamma_i(t))$$

from 0 to t we obtain

$$\text{dev} \circ \gamma_i(t) = \text{dev} \circ \gamma_i(0) + \int_0^t d\text{dev} \circ \tilde{X}_i(\gamma_i(s)) ds.$$

Applying our linear functional l_i to this we obtain

$$\begin{aligned} l_i(\text{dev} \circ \gamma_i(t)) &= l_i(\text{dev} \circ \gamma_i(0)) + \int_0^t l_i \circ d\text{dev} \circ \tilde{X}_i(\gamma_i(s)) ds \\ &= l_i(\text{dev} \circ \gamma_i(0)) + \int_0^t \tilde{\omega}_i(\tilde{X}_i)_{\gamma_i(s)} ds \\ &= l_i(\text{dev} \circ \gamma_i(0)) + \int_0^t ds \\ &= l_i(\text{dev} \circ \gamma_i(0)) + t. \end{aligned} \tag{*}$$

Now we can start our inductive step. We start with some $v \in \mathbb{R}^n$ and we wish to show that there is a point $x \in \tilde{M}$ with $\text{dev}(x) = v$. We will do so by flowing along the vector fields $\tilde{X}_i$ in a suitable manner.

Taking $i = n$ we start at $\tilde{x}_0 \in \tilde{M}$ such that $\text{dev}(\tilde{x}_0) = 0$. We flow along $\tilde{X}_n$ for time $t = l_n(v)$ until the point $\tilde{x}_1 \in \tilde{M}$ with $\text{dev}(\tilde{x}_1) = v_1$. From (*) we obtain

$$l_n(v_1) = l_n(v) + 0$$

and so $v - v_1 \in \ker l_n = F_{n-1}$.

We can repeat the same process taking $i = n - 1$ i.e. we flow along $\tilde{X}_{n-1}$ for $t = l_{n-1}(v - v_1)$ ending at $\tilde{x}_2 \in \tilde{M}$ with $\text{dev}(\tilde{x}_2) = v_2$. Then

$$\begin{aligned} l_{n-1}(v_2) &= l_{n-1}(v - v_1) + l_{n-1}(v_1) \\ &= l_{n-1}(v) \end{aligned}$$

and hence we have that $v - v_2 \in \ker l_{n-1} = F_{n-2}$.

We continue this process inductively until we reach $F_0 = \{0\}$ at which point we have $v - v_n = 0$ where $v_n = \text{dev}(\tilde{x}_n)$. Hence we have showed surjectivity of the developing map.

To show injectivity we consider a path

$$\gamma_0 : [a, b] \rightarrow \tilde{M}$$

which develops to a closed loop $\delta : [a, b] \rightarrow \mathbb{R}^n$. We can deform γ_0 to a new path γ_1 such that it develops entirely inside F_{n-1} . We define a path $\gamma_t : [a, b] \rightarrow \tilde{M}$ where $\gamma_t(s)$ is the image of $\gamma_0(s)$ at time $-t l_n(\delta(s))$ under the flow of $\tilde{X}_n$. Fix $s \in [a, b]$ and let $\alpha : [0, 1] \rightarrow \mathbb{R}^n$ be an integral curve of $\tilde{X}_n$ with $\alpha(0) = \gamma_0(s)$. Then,

$$\alpha'(t) = \tilde{X}_n(\alpha(t))$$

and applying the differential of the developing map to this yields

$$d\text{dev} \circ \alpha'(t) = d\text{dev}(\tilde{X}_n(\alpha(t))).$$

We can apply our linear functional l_n to this and obtain

$$\begin{aligned} l_n(d\text{dev} \circ \alpha'(t)) &= l_n(d\text{dev}(\tilde{X}_n(\alpha(t)))) \\ &= \tilde{\omega}_i(\tilde{X}_n(\alpha(t))) \\ &= 1. \end{aligned}$$

Now, integrating this equation between $t = 0$ and $t = -tl_n(\delta(s))$ yields

$$\begin{aligned} l_n(\text{dev} \circ \gamma_t(s)) &= l_n(\text{dev} \circ \gamma_0(s)) - tl_n(\delta(s)) \\ &= l_n(\delta(s)) - tl_n(\delta(s)) \end{aligned}$$

since $\text{dev} \circ \gamma_0(s) = \delta(s)$ by definition of $\delta(s)$.

At $t = 1$ this yields that $\text{dev} \circ \gamma_t(s) \in \ker l_n = F_{n-1}$ and this holds for all $s \in [a, b]$. Repeating this process inductively, we get a curve that develops entirely in $F_0 = \{0\}$. Since by construction, our original curve is path homotopic to this deformed curve that develops onto 0, it follows as dev depends only on the homotopy class of the curve, that dev is injective. $\square$

We summarize the previous results into the following theorem.

Theorem 3.37. Let M be a compact affine manifold with nilpotent affine holonomy representation. Then the following are equivalent

1. M is a complete affine manifold.
2. The linear holonomy is unipotent.
3. M has a parallel volume form.
4. The affine holonomy is irreducible.
5. The affine holonomy is indecomposable.

Proof. We have that $2) \Rightarrow 1)$ by Theorem 3.36. We have that $3) \Leftrightarrow 2)$ by Theorem 3.34. Now, $1) \Rightarrow 4)$ by Theorem 1.53 and by definition $4) \Rightarrow 5)$. Now $5) \Rightarrow 2)$ by Lemma 3.16 and we are done. $\square$

Remark 3.38. We note that if the fundamental group of M is nilpotent, then its holonomy representation will also be nilpotent as this is preserved by group morphisms.

CHAPTER FOUR

Classification of Integral Affine Closed Surfaces

As we have seen in the previous chapter, the integral affine structure induced by a regular Lagrangian torus fibration is an invariant of the base space of the fibration. Moreover, given an integral affine structure on a manifold B , we can construct a Lagrangian fibration by taking the quotient of T^*B by the lattice inducing the integral affine structure. We have seen in Chapter 2 that after having fixed an integral affine structure on B , all Lagrangian fibrations over B locally look like $T^*B/\Lambda \rightarrow B$ where $\Lambda \subset T^*B$ is the period lattice inducing the integral affine structure.

In order to classify the regular Lagrangian torus fibrations over a manifold B , the first step is to classify the integral affine structures on B . The goal of this chapter is to carry this out for compact surfaces. We will follow closely the ideas used in [Sep10] and [Mis96].

4.1 Classification of integral affine circles

It is easy to classify the integral affine structures on the circle S^1 and so for completeness we will also give the classification of these.

The fundamental group of S^1 is the integers $\mathbb{Z}$ and hence by Theorem 3.37 any affine structure will be complete if it possesses a parallel volume form. The holonomy representation

$$\text{hol} : \pi_1(S^1) \rightarrow \text{Aff}_{\mathbb{Z}}(\mathbb{R})$$

is determined by the image of $1 \in \mathbb{Z}$ and so the possible affine holonomy groups are

$$A = \langle (1, a) \rangle \quad \text{or} \quad B = \langle (-1, b) \rangle$$

with $a, b \in \mathbb{R}$. We note that $B \cong \mathbb{Z}/2\mathbb{Z}$ and in fact, this affine structure would be radiant as $\frac{b}{2}$ is a fixed point. However, Corollary 3.32 says that the linear holonomy would contain an expansion but all the eigenvalues of the elements of B have modulus one. Hence B cannot occur as the affine holonomy group of an integral affine structure over S^1 .

Now, since the affine holonomy group A preserves a parallel volume form, by Theorem 3.37, this affine structure is complete. The integral affine structures on S^1 are therefore of the form $S^1 = \mathbb{R}/A$. By Lemma 1.12, complete integral affine structures are equivalent when their affine holonomy groups are conjugate. The following lemma classifies the integral affine structures on S^1 .

Lemma 4.1. The integral affine structures on S^1 are determined by a positive real number $a > 0$ and are of the form $\mathbb{R}/A$ where

$$A = \langle (1, a) \rangle.$$

Proof. Our previous discussion showed that all integral affine structures on S^1 are of the form $\mathbb{R}/A$ where the translational component is any real number. It remains to check when two structures are equivalent. We note that the only possible linear affine holonomy is $\{\pm 1\}$ and that if it is 1, conjugating A leaves it unchanged. We therefore compute the conjugation by an element with negative linear holonomy. Letting $b \in \mathbb{R}$ we compute

$$\begin{aligned} (-1, b)(1, a)(-1, b)^{-1} &= (-1, b)(1, a)(-1, b) \\ &= (1, -a) \end{aligned}$$

but this generates the same affine holonomy group as $(1, a)$ completing the proof since if $a \neq 0$ the action is clearly free. $\square$

4.2 Some basic results

We now move on to classifying the integral affine structures over compact surfaces which requires a bit more work. We first discuss our approach to the classification problem and state some important results in the theory of affine manifolds. The first task is to determine the possible compact surfaces that can admit an affine structure. The following theorem classifies the compact, connected 2-manifolds.

Theorem 4.2. Every compact, connected 2-manifold is homeomorphic to the sphere, the connected sum of tori or the connected sum of $\mathbb{R}P^2$.

Proof. This can be found in [Hat02]. $\square$

We make use of the following theorem from Milnor relating to the existence of a flat connection on closed orientable surfaces.

Theorem 4.3 (Milnor [Mil58]). Let M^2 be a closed orientable surface of genus g . Then if $g \geq 2$ it does not possess a connection with zero curvature.

Corollary 4.4. The possible orientable closed surfaces admitting an integral affine structure are the sphere S^2 or the torus $\mathbb{T}^2$.

Proof. This follows immediately from the classification of closed 2-surfaces and the previous theorem. $\square$

However, the following simple argument shows that the sphere S^2 does not admit such a structure.

Proposition 4.5. The sphere S^2 does not admit an affine structure.

Proof. Suppose there was such an affine structure. Then

$$\text{dev} : S^2 \rightarrow \mathbb{R}^2$$

is a local diffeomorphism and hence an open map. However dev is also continuous, and so $\text{dev}(S^2)$ is a compact subset of $\mathbb{R}^2$ and hence closed. Now, as $\text{dev}(S^2) \neq \emptyset$, by connectivity $\text{dev}(S^2) = \mathbb{R}^2$ but $\mathbb{R}^2$ is not compact. $\square$

We have the following immediate corollary.

Corollary 4.6. The only closed, orientable surface admitting an (integral) affine structure is $\mathbb{T}^2$.

We will deal with the non-orientable case by passing to the orientable double cover.

Lemma 4.7. Every closed, non-orientable surface has a double cover that is orientable. Furthermore, the genus of the double cover is given by

$$g = \bar{g} - 1$$

where $\bar{g}$ is the number of connected sums of $\mathbb{R}P^2$.

Proof. This is proved in [Hat02]. $\square$

This lemma leads to the following corollary.

Corollary 4.8. The only closed surfaces admitting an (integral) affine structure are the torus $\mathbb{T}^2$ and the Klein bottle $\mathbb{K}^2$.

Proof. As any covering space of an integral affine structure inherits an integral affine structure by Lemma 1.7, it follows that the orientable double cover of a non-orientable integral affine surface must be $\mathbb{T}^2$. Furthermore, it must have non-orientable genus $\bar{g} = 1$ or 2 . If $\bar{g} = 1$ our surface is homeomorphic to $\mathbb{R}P^2$ which is covered by S^2 and hence admits no affine structure. If $\bar{g} = 2$ we get the Klein bottle which is covered by $\mathbb{T}^2$ and hence it can admit such a structure. $\square$

4.3 Classification of Integral Affine structures on the torus

4.3.1 Completeness and freeness of the action

We now look at how we can classify the integral affine structures on the torus. This was first obtained in [Sep10]. The following observation will greatly simplify the classification process.

Lemma 4.9. Any integral affine structure on the torus is complete.

CHAPTER 4. CLASSIFICATION OF INTEGRAL AFFINE CLOSED SURFACES

Proof. We have that the fundamental group $\pi_1(\mathbb{T}^2) = \mathbb{Z} \times \mathbb{Z}$ is abelian and hence nilpotent. As the torus is orientable, our linear holonomy group $\Gamma \subseteq \mathrm{SL}(2, \mathbb{Z})$ by Corollary 1.68 and now by Theorem 3.37 this is complete. $\square$

From the above discussion, classifying complete integral affine structures on a compact surface M amounts to classifying injective homomorphisms from the fundamental group into the group $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ which induce a free and proper action on $\mathbb{R}^2$.

Indeed, Proposition 1.29 tells us that every complete integral affine manifold is the quotient of a subgroup $\Gamma \subset \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^n)$ acting freely and properly on $\mathbb{R}^n$. If dev is a diffeomorphism, then if $\gamma \in \pi_1(M)$ and $\mathrm{hol}(\gamma)$ acts trivially on $\mathbb{R}^n$, equivariance implies

$$\begin{aligned} \mathrm{dev}(\gamma \cdot x) &= \mathrm{hol}(\gamma)\mathrm{dev}(x) \\ &= \mathrm{dev}(x) \end{aligned}$$

and hence injectivity of dev implies γ acts trivially.

We now wish to do this for the torus. We will establish some lemmas to help us determine when an integral affine transformation acts freely on $\mathbb{R}^2$.

Lemma 4.10. If $g = (A, b) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ acts freely on $\mathbb{R}^2$, then

$$\text{if } \det(A) = 1 \text{ then } \mathrm{Tr}(A) = 2$$

and,

$$\text{if } \det(A) = -1 \text{ then } \mathrm{Tr}(A) = -1.$$

Proof. Since the action by g is free, this implies

$$Ax + b = x$$

has no solution. But now,

$$(A - I)x = -b$$

has no solutions for all $x \in \mathbb{R}^2$ and so $\det(A - I) = 0$. Calculating this determinant yields

$$1 - \mathrm{Tr}(A) + \det(A) = 0$$

and substituting in the appropriate values of $\det(A)$ gives the required results. $\square$

We note also that 1 is an eigenvalue of such an A as by the above proof $\det(A - 1 \cdot I) = 0$. Using this result we can find a nice basis to express the linear part of our element $g \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$.

Lemma 4.11. Let $g = (A, b) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ act freely. Then there is an integral basis of $\mathbb{R}^2$ such that

$$A = \begin{pmatrix} 1 & n \\ 0 & \pm 1 \end{pmatrix}$$

4.3. CLASSIFICATION OF INTEGRAL AFFINE STRUCTURES ON THE TORUS

Proof. By our previous lemma, A must have 1 as an eigenvalue. We can take an eigenvector e_1 such that e_1 is integral, i.e. its components are integers. Now as $\text{Tr}(A) = 2$ or 0, it follows that we can pick our second basis vector such that

$$A = \begin{pmatrix} 1 & n \\ 0 & \pm 1 \end{pmatrix}$$

since we need $A \in \text{GL}(2, \mathbb{Z})$. Furthermore, by changing orientation if necessary, we may assume $n \geq 0$. $\square$

Lemma 4.12. Let $H \subset \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)^+$ be a subset of the elements of determinant one and suppose H acts freely on $\mathbb{R}^2$. Then there is an integral basis such that for all $(A, b) \in H$ we can write

$$A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

for $n \in \mathbb{Z}$.

Proof. Let $(A, a), (B, b) \in H$. Now by Lemma 4.11, we can write

$$A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

with $n \neq 0$. Now by Lemma 4.10, we have that

$$\begin{aligned} \text{Tr}(A^k B) &= a + nck + d \\ &= \text{Tr}(B) + nck \\ &= 2. \end{aligned}$$

As $n \neq 0$ and this holds for all $k \in \mathbb{Z}$, it follows that $c = 0$.

But now, $\det B = ad = 1$ and $\text{Tr}(B) = a + d = 2$ and so $a = d = 1$ giving the required result as $b \in \mathbb{Z}$. $\square$

We have the following corollary of this result.

Corollary 4.13. For any two elements $(A, a), (B, b) \in H$, there exists relatively prime $(k, l) \neq (0, 0)$ such that $A^k B^l = 1$.

Proof. We simply take $k = \frac{m}{(m, n)}$ and $l = \frac{-n}{(m, n)}$ and then

$$\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}^k \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix}^l = 1$$

$\square$

4.3.2 Finding the possible generators

We have now built up the required machinery to begin the classification of the integral affine structures on the torus $\mathbb{T}^2$.

Theorem 4.14. Any integral affine structure on $\mathbb{T}^2$ has its affine holonomy group generated by one of the following subgroups of $\text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$

1. $\mathcal{A}_{a,b,c,d}$ - These integral affine structures are the quotient of $\mathbb{R}^2$ by a lattice of pure translations. The generators in $\text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ are given by

$$\left(1, \begin{pmatrix} a \\ c \end{pmatrix}\right) \text{ and } \left(1, \begin{pmatrix} b \\ d \end{pmatrix}\right)$$

with $a, b, c, d \in \mathbb{R}$ and $ad - bc \neq 0$.

2. $\mathcal{B}_{n,a,b}$ - The generators in $\text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ are given by

$$\left(1, \begin{pmatrix} a \\ 0 \end{pmatrix}\right) \text{ and } \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ b \end{pmatrix}\right)$$

with $a, b > 0$ and $n \in \mathbb{N}$.

Proof. Let $\pi_1(\mathbb{T}^2) = \langle \tilde{C}, \tilde{D} \rangle$ and denote by C, D the images of these generators under the affine holonomy representation. We wish to show that these generators reduce to the form of either $\mathcal{A}_{a,b,c,d}$ or $\mathcal{B}_{n,a,b}$. We can do this by

1. Changing the generators $\tilde{C}, \tilde{D}$ of the $\pi_1(\mathbb{T}^2)$.
2. Conjugating the images of these generators by some $g \in \text{Aff}_{\mathbb{Z}}^+(\mathbb{R})$ - this corresponds to pre-composing the developing map by g which preserves the affine structure.

If both $C = \left(1, \begin{pmatrix} a \\ c \end{pmatrix}\right)$ and $D = \left(1, \begin{pmatrix} b \\ d \end{pmatrix}\right)$ are pure translations, i.e. the linear component is the identity, injectivity forces that our translational vectors are linearly independent, otherwise $\text{hol}(k\tilde{C} + m\tilde{D}) = 0$ for some $k, m \neq 0 \in \mathbb{R}$. In this case our lattice in $\mathbb{R}^2$ would be one-dimensional. This gives the condition that $ad - bc \neq 0$ and completes the first class of lattices.

Now suppose that C, D are not pure translations. Then by Corollary 4.13, there exists relatively prime $k, l \in \mathbb{Z}$ such that $C^k D^l$ is a translation. We can therefore change generators in $\mathbb{Z}^2$ to $\tilde{A} = \tilde{C}^k \tilde{D}^l$ and $\tilde{B} = \begin{pmatrix} a \\ b \end{pmatrix}$. In the basis $\tilde{C}, \tilde{D}$ we have that the matrix

$$M = \begin{pmatrix} k & a \\ l & b \end{pmatrix}$$

has $\det M = kb - la$ and since k, l are relatively prime by Bezout's theorem we can always pick $a, b \in \mathbb{Z}$ such that this determinant is 1 and hence an automorphism of our lattice.

We denote the images of our new generators under the affine holonomy representation by

$$A' = \left(l, a = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}\right) \text{ and } B' = \left(B, b = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}\right).$$

Now by Lemma 4.12, we can find a basis of $\mathbb{R}^2$ such that $B = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ with $n > 0$. Now $\text{hol}(\pi_1(\mathbb{T}^2))$ is abelian so we require that $A'B' = B'A'$ which gives the condition

$$Ba = a$$

this implies that

$$\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} a_1 + na_2 \\ a_2 \end{pmatrix} \\ = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

which gives $a_2 = 0$ as $n > 0$ and so $a = \begin{pmatrix} a_1 \\ 0 \end{pmatrix}$. We note that $a_1 \neq 0$ as otherwise the action is not free. For freeness, we also require that $A^l B^k$ acts freely for all $k, l \in \mathbb{Z}$. We check

$$\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}^k \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} kb_1 + la_1 \\ kb_2 \end{pmatrix} = \begin{pmatrix} x + kny + kb_1 + la_1 \\ y + kb_2 \end{pmatrix} \\ = \begin{pmatrix} x \\ y \end{pmatrix}$$

If $b_2 = 0$ then there is a fixed point for $y = \frac{-kb_1 - la_1}{kn} \in \mathbb{R}$ so we must have $b_2 \neq 0$. Conversely, if $b_2 \neq 0$ and the action is not free, this requires $k = 0$ and so $x + la_1 = a_1$ for $l \neq 0$ but $a_1 \neq 0$ and hence there are no solutions. Now, to show that this takes the form of $\mathcal{B}_{n,a,b}$, we conjugate by

$$T = \left(I, \begin{pmatrix} 0 \\ -\frac{b_1}{n} \end{pmatrix} \right)$$

this fixes A' and so we compute

$$T^{-1}B'T = \left(B, \begin{pmatrix} 0 \\ b_2 \end{pmatrix} \right)$$

and so we may assume $b_1 = 0$. We can also make the change $A' \mapsto A'^{-1}$ and $B' \mapsto B'^{-1}$ and assume $a_1, b_2 > 0$ which completes the proof. $\square$

4.3.3 When are two structures equivalent?

This gives the possible integral affine structures on the 2-torus. However we still have to tackle the question of when two lattices give the same integral affine structure. This occurs when the subgroups of the integral affine group determined by the holonomy representation are conjugate to each other by Lemma 1.12.

We have the following classification result.

Lemma 4.15. Lattices from different series do not induce equivalent integral affine structures.

Proof. Lattices in series $\mathcal{A}_{a,b,c,d}$ are generated by pure translations and hence cannot be conjugated into lattices of the form $\mathcal{B}_{n,a,b}$. $\square$

Theorem 4.16. Lattices $\mathcal{A}_{a,b,c,d}$ and $\mathcal{A}_{a',b',c',d'}$ induce equivalent integral affine structures if and only if

$$X = GX'H$$

where $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $X' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ with $G, H \in GL(2, \mathbb{Z})$.

Proof. Performing lattice automorphisms in $\pi_1(\mathbb{T}^2)$ and conjugating in the image of the generators in $\text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ preserves the integral affine structure.

Let $\pi_1(\mathbb{T}^2) = \langle v_1, v_2 \rangle$ with $\text{hol}(v_1) = \left(I, \begin{pmatrix} a \\ c \end{pmatrix} \right)$ and $\text{hol}(v_2) = \left(I, \begin{pmatrix} b \\ d \end{pmatrix} \right)$ and set $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

Let $H \in GL(2, \mathbb{Z})$ and $g = (G, w) \in \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. We note that conjugating a pure translation (I, v) by g gives the element (I, Gv) . We compute

$$\text{hol}(Hv_1) = \left(I, XH \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right)$$

and

$$\text{hol}(Hv_2) = \left(I, XH \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right).$$

This corresponds to an automorphism of our lattice in $\pi_1(\mathbb{T}^2)$. We can also conjugate generators by $g = (G, w) \in \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ which gives

$$g \circ \text{hol}(Hv_1) \circ g^{-1} = \left(I, GXH \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right)$$

and

$$g \circ \text{hol}(Hv_2) \circ g^{-1} = \left(I, GXH \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

and so it follows that the integral affine structures are equivalent if and only if $X' = GXH$. $\square$

Theorem 4.17. Lattices in $\mathcal{B}_{n,a,b}$ never induce the same integral affine structure.

Proof. Suppose that $\mathcal{B}_{n,a,b} = \langle (1, \bar{a}), (B, \bar{b}) \rangle$ and $\mathcal{B}_{n',a',b'} = \langle (1, \bar{a}'), (B', \bar{b}') \rangle$ induce equivalent integral affine structures. We must have that

$$g\tilde{A}g^{-1} = \tilde{A}'^k \tilde{B}'^r \tag{4.1}$$

$$g\tilde{B}g^{-1} = \tilde{A}'^l \tilde{B}'^s \tag{4.2}$$

where $\tilde{A}, \tilde{B}$ and $\tilde{A}', \tilde{B}'$ are the images of the generators of $\pi_1(\mathbb{T}^2)$ of the structures $\mathcal{B}_{n,a,b}$ and $\mathcal{B}_{n',a',b'}$ and $g = (G = [g_{ij}], \bar{h}) \in \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. The matrix

$$M = \begin{pmatrix} k & l \\ r & s \end{pmatrix} \in GL(2, \mathbb{Z})$$

corresponds to an automorphism of our lattice. This implies that

$$GBG^{-1} = \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$$

4.3. CLASSIFICATION OF INTEGRAL AFFINE STRUCTURES ON THE TORUS

where $B = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$. Computing this condition yields

$$\begin{pmatrix} * & * \\ g_{21}g_{22} - ng_{21}^2 - g_{21}g_{22} & * \end{pmatrix} = \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$$

and so $g_{21}g_{22} - ng_{21}^2 - g_{21}g_{22} = 0$ which gives that $g_{21} = 0$.

Now, inspecting the linear part of $g\tilde{A}g^{-1}$ as the linear part of $\tilde{A}$ is the identity this must also be a translation i.e. $\tilde{A}'^k \tilde{B}'^r$ is a translation. Again, inspecting the linear components we must have that $r = 0$. Our matrix M corresponding to the automorphism of our lattice now looks like

$$M = \begin{pmatrix} k & l \\ 0 & s \end{pmatrix}$$

and so $k, s = \pm 1$ as this matrix needs to have determinant ± 1 . Now from Equation 4.1 we obtain

$$\begin{aligned} (G, \bar{h}) \left(I, \begin{pmatrix} a \\ 0 \end{pmatrix} \right) (G^{-1}, -G^{-1}\bar{h}) &= \left(I, G \begin{pmatrix} a \\ 0 \end{pmatrix} \right) \\ &= \left(I, \begin{pmatrix} g_{11}a \\ 0 \end{pmatrix} \right) \\ &= \left(I, \begin{pmatrix} ka' \\ 0 \end{pmatrix} \right) \end{aligned}$$

where the last equality comes from the RHS of Equation 4.1. This now gives that

$$g_{11}a = ka' \Rightarrow g_{11}a = \pm a'$$

as $k = \pm 1$. But $a, a' > 0$ by the classification and so $g_{11} = k$ and $a = a'$.

Next we inspect Equation 4.2. The matrix part gives that

$$GBG^{-1} = B'^s.$$

Computing this gives

$$\begin{aligned} \begin{pmatrix} g_{11} & g_{12} \\ 0 & g_{22} \end{pmatrix} \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} g_{11} & -g_{12}g_{11}g_{22} \\ 0 & g_{22} \end{pmatrix} &= \begin{pmatrix} 1 & g_{11}g_{22}n \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & n's \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

Where the last equality comes again from the RHS of Equation 4.2. Comparing matrix entries yields $g_{11}g_{22}n = n's$. Using that $g_{11} = \pm 1$, $g_{22} = \pm 1$ and $s = \pm 1$ we have that $n = \pm n'$. But both $n, n' \in \mathbb{N}$ by the classification and so we must have $n = n'$. This now gives that $s = g_{11}g_{22}$. This was the linear part of Equation 4.2 so we now analyze the translational component.

Comparing the translational parts yields

$$-GBG^{-1}\bar{h} + G\bar{b} + \bar{h} = s\bar{b}' + l\bar{a}'. \quad (4.3)$$

We can factorise the LHS as

$$-GBG^{-1}h + G\bar{b} + \bar{h} = G(I - B)G^{-1}\bar{h} + G\bar{b}.$$

CHAPTER 4. CLASSIFICATION OF INTEGRAL AFFINE CLOSED SURFACES

Now we simplify

$$G(I - B)G^{-1}\bar{h} = \begin{pmatrix} x \\ 0 \end{pmatrix}$$

where $x \in \mathbb{R}$ is arbitrary depending on $\bar{h}$. Now Equation 4.3 becomes

$$\begin{pmatrix} x \\ 0 \end{pmatrix} + G\bar{b} = s\bar{b}' + I\bar{a}'.$$

Expanding this we are left with

$$\begin{pmatrix} x + g_{12}b \\ g_{22}b \end{pmatrix} = \begin{pmatrix} Ia' \\ sb' \end{pmatrix}.$$

This gives $g_{22}b = sb'$ and again as $b, b' > 0$ and $g_{22}, s = \pm 1$ we have $b = b'$ and now we are finished as we showed $a = a'$, $b = b'$ and $n = n'$.

□

This completes the classification of the integral affine structures on the torus $\mathbb{T}^2$. We note that lattices in the series $\mathcal{A}_{a,b,c,d}$ induce a Euclidean structure on the torus - i.e. their holonomy representation is a subgroup of $O(n) \rtimes \mathbb{R}^n$. This implies that there is a Riemannian metric preserved by this structure. However, the integral affine structures of $\mathcal{B}_{n,a,b}$ never induce a Euclidean structure on the torus.

Next, we will look at the classification of integral affine structures over the Klein bottle, hence completing the classification of all integral affine structures over closed surfaces.

4.4 Classification of Integral Affine structures on the Klein Bottle

4.4.1 Completeness of integral affine structures on K^2

We now wish to classify the integral affine structures on the Klein bottle. We would hope to proceed as before by calculating all of the injective homomorphisms from $\Gamma = \pi_1(K^2) \rightarrow \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ that induce a free and proper action on $\mathbb{R}^2$. However, in order to do this we would need to know that every integral affine structure on K^2 is complete. The following lemma takes care of this fact.

Lemma 4.18. Let N be an integral affine manifold and suppose that there is a regular covering $M \rightarrow N$ by a compact orientable integral affine manifold whose fundamental group is nilpotent. Then any integral affine structure on N is complete.

Proof. We can pull back the integral affine structure on N to M . A regular covering induces an injection of fundamental groups $\iota : \pi_1(M) \rightarrow \pi_1(N)$. We can induce the holonomy representation on M by $\text{hol} \circ \iota$ where hol is the affine holonomy representation induced by the integral affine structure on N . As M and N share a common universal cover X , the integral affine structure on N induces a developing map $\text{dev} : X \rightarrow \mathbb{R}^n$ and this induced holonomy representation will remain equivariant with respect to this developing map as the action of $\pi_1(\mathbb{T}^2)$ on the universal cover is a restriction of the action of $\pi_1(K^2)$.

From Proposition 1.11 this defines an integral affine structure on M which has the same developing map as that of N . Therefore, if the integral affine structure on N is incomplete then the integral affine structure on M will also be. However, M is a compact orientable integral affine manifold with nilpotent fundamental group and hence by Theorem 3.37 is complete. It follows that any integral affine structure on N must also be complete. $\square$

Throughout this section we let $\pi_1(K^2) = \Gamma$ have presentation

$$\Gamma = \langle a, b \mid aba = b \rangle$$

and we note that $\langle a, b^2 \rangle \subset \Gamma$ is abelian. Indeed, the oriented double cover of K^2 is the torus $\mathbb{T}^2$ and so there is an injection of fundamental groups $\pi_1(\mathbb{T}^2) \hookrightarrow \pi_1(K^2)$. This abelian subgroup can be realised as the aforementioned injection. As in the previous proposition, the induced integral affine structure on $\mathbb{T}^2$ via pullback will have holonomy representation $\text{hol} \circ \iota$ where ι is the injection of fundamental groups and hol is the holonomy representation of the affine structure on $\mathbb{T}^2$ - indeed this will be equivariant with respect to the same developing map on their shared universal cover. This shows the following proposition.

Proposition 4.19. Let K^2 be equipped with an integral affine structure with holonomy representation

$$\text{hol} : \Gamma \rightarrow \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2).$$

Then if $\text{hol}(a) = g \in \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ and $\text{hol}(b) = h \in \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ then the induced holonomy representation on the torus $\mathbb{T}^2$ will be given by

$$\tilde{\text{hol}} : \pi_1(\mathbb{T})^2 = \langle x, y \rangle \rightarrow \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$$

$$x \mapsto g, \quad y \mapsto h^2$$

Remark 4.20. This implies that once we classify the injective homomorphisms from $\Gamma \rightarrow \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$, we can easily check which integral affine structure they induce on the torus.

4.4.2 Determining the linear holonomy

We now start the classification of the injective group morphisms from $\Gamma \rightarrow \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. We will denote also by $a = (A, \bar{x})$ and $b = (B, \bar{y})$ the images of our generators of Γ under the holonomy representation where $A, B \in GL(2, \mathbb{Z})$ and $\bar{x}, \bar{y} \in \mathbb{R}^2$.

We will make use of the following in our classification

1. We must have that $\det A = 1$ and $\det B^2 = 1$ as these are generators from $\pi_1(\mathbb{T}^2)$.
2. Since b reverses orientation we in fact have $\det B = -1$
3. The action given by the composition $\pi_1(\mathbb{T}^2) \hookrightarrow \Gamma \rightarrow \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ must also act freely and properly as this defines the induced integral affine structure on $\mathbb{T}^2$ by Proposition 4.19.

We will now make use of Lemma 4.10 to get restrictions on the matrix B . Namely, this lemma implies that $\text{Tr } B = 0$. By Lemma 1 of [FS83], our matrix B must be conjugate to one of

$$B_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{or} \quad B_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Now we must find the matrices $A \in GL(2, \mathbb{Z})$ such that $\det A = 1$, $\text{Tr } A = 2$ which also satisfy the group relation $ABA = B$ where $B = B_1, B_2$. We perform this calculation now.

Proposition 4.21. Let $B = B_1$. Then we have the following choices for our matrix A .

1. $A = I$.
2. $A_1 = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$.
3. $A_2 = \begin{pmatrix} 1 & 0 \\ p & 1 \end{pmatrix}$.

Proof. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2, \mathbb{Z})$. The trace condition implies that $d = 2 - a$. We note

4.4. CLASSIFICATION OF INTEGRAL AFFINE STRUCTURES ON THE KLEIN BOTTLE

that $B^{-1} = B$. Calculating the group condition $ABA = B$ we obtain

$$\begin{aligned} ABA &= \begin{pmatrix} a^2 - bc & ab - bd \\ ca - cd & cb - d^2 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{aligned}$$

Comparing matrix entries we find

$$a = d$$

since $d = 2 - a$ this implies $a = d = 1$. Our matrix A now has the form

$$A = \begin{pmatrix} 1 & b \\ c & 1 \end{pmatrix}.$$

From $\det A = 1$ we find that either $b = 0$ or $c = 0$ which completes the proof. $\square$

Now we perform the same calculation for $B = B_2$.

Proposition 4.22. Let $B = B_2$. Then we have the following choices for our matrix A .

1. $A = I$.
2. $A_3 = \begin{pmatrix} 1+n & n \\ -n & 1-n \end{pmatrix}$.
3. $A_4 = \begin{pmatrix} 1-n & n \\ -n & 1+n \end{pmatrix}$.

Proof. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2, \mathbb{Z})$. Again we must have that $d = 2 - a$ by the trace condition. Calculating the group law $ABA = B$ yields

$$\begin{aligned} ABA &= \begin{pmatrix} ab + ac & b^2 + ad \\ ad + c^2 & bd + cd \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \end{aligned}$$

Comparing entries yields that $a(b + c) = 0$ and $d(b + c) = 0$. However, the trace condition implies that a and d can't both be zero and so $b = -c$. Our matrix A now has the form

$$A = \begin{pmatrix} a & b \\ -b & d \end{pmatrix}.$$

Calculating the determinant condition and using $d = 2 - a$ yields

$$\begin{aligned} b^2 &= 1 - ad \\ &= (a - 1)^2 \end{aligned}$$

and so

$$b = a - 1 \quad \text{or} \quad b = 1 - a.$$

The first case yields

$$A = \begin{pmatrix} n+1 & n \\ -n & 1-n \end{pmatrix}$$

where $n = a - 1$ and the second case yields

$$A = \begin{pmatrix} 1-n & n \\ -n & 1+n \end{pmatrix}$$

where $n = 1 - a$. The case where $n = 0$ corresponds to the identity and this completes our proof. $\square$

This gives the possible choices of A and B that satisfy the group law and Lemma 4.10. We must now deduce the translational component in each case and check that the induced action is free and proper.

4.4.3 Translational component and freeness of the action

To find the translational components we impose the group law $aba = b$. When checking freeness, we can often use Proposition 4.19 to take shortcuts as we know that a, b^2 must generate one of the integral affine structures on the torus and hence act freely.

We will now compute the translational components and check whether the action is free.

Lemma 4.23. Let $B = B_1$. Then the possible choices of A that induce an integral affine structure on K^2 are $A = I, A_1$.

Proof.

Case 1 ($A = I, B = B_1$). Let $a = (I, \bar{x})$ and $b = (B_1, \bar{b})$ with $\bar{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $\bar{b} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ be our generators in $\text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. They must satisfy the relation $aba = b$. Calculating this we find

$$(B, B\bar{x} + \bar{x} + \bar{b}) = (B, \bar{b}).$$

The translational component gives $B\bar{x} = -\bar{x}$ and so $\bar{x}$ lives in the -1 -eigenspace. This implies that $x_1 = 0$. We are free to conjugate both our generators as this preserves the integral affine structure and so we will do so by $g = \left(I, \begin{pmatrix} 0 \\ \frac{y_2}{2} \end{pmatrix}\right)$. Conjugating a leaves it unchanged so we calculate

$$g^{-1} \circ b \circ g = \left(B, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$$

and so we may assume that $y_2 = 0$ by performing this conjugation. Our generators now look like

$$a = \left(I, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(B, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right).$$

Since B alone clearly does not act freely we can't have that $y_1 = 0$ and if $x_2 = 0$ then our lattice is one-dimensional and hence the holonomy representation is not injective. Therefore, $x_2, y_1 \neq 0$. Furthermore, by changing generators $a \mapsto a^{-1}$ and $b \mapsto b^{-1}$ we can assume $x_2, y_1 > 0$. It remains to check that this action is free. We must check that $a^q b^n$ acts freely for all $q, n \in \mathbb{Z}$ not both zero. However, we now note that as $B^{2k} = I$ for $k \in \mathbb{Z}$, by the torus classification of Theorem 4.16, we have already shown that $a^q b^{2k}$ acts freely. We therefore must check the case of $a^q b^{2k+1}$. We compute that

$$a^q = \left(I, \begin{pmatrix} 0 \\ qx_2 \end{pmatrix}\right), \quad b^{2k+1} = \left(B, \begin{pmatrix} (2k+1)y_1 \\ 0 \end{pmatrix}\right).$$

4.4. CLASSIFICATION OF INTEGRAL AFFINE STRUCTURES ON THE KLEIN BOTTLE

For the action to be free we need $a^q b^{2k+1} \circ z = z$ to have no solutions for all $z \in \mathbb{R}^2$. We compute this action

$$\begin{aligned} a^q b^{2k+1} \circ z &= B \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} + \begin{pmatrix} (2k+1)y_1 \\ x_2 \end{pmatrix} \\ &= \begin{pmatrix} (2k+1)y_1 + z_1 \\ x_2 - z_2 \end{pmatrix} \end{aligned}$$

and so, for there to be a fixed point we need

$$(2k+1)y_1 + z_1 = z_1$$

which gives $(2k+1)y_1 = 0$. But now $y_1 > 0$ and $2k+1 = 0 \Rightarrow k = -\frac{1}{2} \notin \mathbb{Z}$ and so this action can have no fixed points.

The integral affine structure on K^2 is thus given by

$$a = \left(I, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(B, \begin{pmatrix} y_1 \\ 0 \end{pmatrix} \right), \quad x_2, y_1 > 0$$

and the induced integral affine structure on the torus is given by

$$a = \left(1, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b^2 = \left(1, \begin{pmatrix} 2y_1 \\ 0 \end{pmatrix} \right), \quad x_2, y_1 > 0.$$

Case 2 ($A = A_1, B = B_1$). Let $a = \left(A_1, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right)$ and $b = \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right)$. As $A_1^2 \neq I$ and $B^2 = I$, by the torus classification such a structure on K^2 would need to induce the second family from the torus classification in Theorem 4.14. By this consideration, we obtain that $x_1 = 0$. We calculate the group law $aba = b$. This gives us

$$\begin{aligned} aba &= \left(B, \begin{pmatrix} y_1 + n(y_2 - x_2) \\ y_2 \end{pmatrix} \right) \\ &= \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right) \end{aligned}$$

which imposes the restriction $n(y_2 - x_2) = 0$ and since $n \neq 0$ this implies $y_2 = x_2$. We can also assume that $x_2 > 0$ by conjugating the generators by $(-I, \vec{0})$ and that $y_1 > 0$ by changing generators $b \mapsto b^{-1}$ if necessary. Now we check freeness of the action.

We calculate that

$$a^q = \left(\begin{pmatrix} 1 & qn \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} \frac{nx_2 q(q-1)}{2} \\ qx_2 \end{pmatrix} \right)$$

and

$$b^k = \begin{cases} \left(B, \begin{pmatrix} ky_1 \\ x_2 \end{pmatrix} \right), & k \text{ odd} \\ \left(I, \begin{pmatrix} ky_1 \\ 0 \end{pmatrix} \right), & k \text{ even.} \end{cases}$$

Since the case of k being even is covered by the torus classification we must only check freeness for k being odd.

For k odd we compute

$$a^q b^k = \left(\begin{pmatrix} 1 & -qn \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} \frac{nx_2 q(q-1)}{2} + qnx_2 + ky_1 \\ (q+1)x_2 \end{pmatrix} \right)$$

and calculating this action on $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$ yields

$$(a^q b^k) \circ z = \begin{pmatrix} \frac{nx_2 q(q-1)}{2} + qnx_2 + ky_1 + z_1 - qnz_2 \\ (q+1)x_2 - z_2 \end{pmatrix}.$$

Solving the fixed point equation, the bottom entry of the vector yields

$$z_2 = \frac{q+1}{2}x_2. \quad (4.4)$$

Solving the fixed point equation in the top component yields

$$z_2 = \frac{x_2(q-1)}{2}x_2 + \frac{ky_1}{qn}.$$

Substituting Equation 4.4 for z_2 yields $ky_1 = 0$ which implies $k = 0$ as $y_1 > 0$. This now implies that a^q would need to have a fixed point but one checks

$$a^q \circ z = \begin{pmatrix} * \\ z_2 + qx_2 \end{pmatrix}$$

and so we require that $qx_2 = 0$ but $x_2 > 0$. Therefore, the only element with a fixed point occurs when $k = q = 0$ which is the identity and hence the action is free.

The integral affine structure on K^2 is

$$a = \left(A_1, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(B_1, \begin{pmatrix} y_1 \\ x_2 \end{pmatrix} \right), \quad x_2, y_1 > 0$$

and the induced integral affine structure on $\mathbb{T}^2$ is given by

$$a = \left(A_1, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b^2 = \left(I, \begin{pmatrix} 2y_1 \\ 0 \end{pmatrix} \right)$$

with $x_2, y_1 > 0$

Case 3 ($A = A_2, B = B_1$). Let $a = \left(A_1, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right)$ and $b = \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right)$. In this case the action will fail to be free. Again, by considering the induced structure on the torus it follows that $x_1 = 0$. We calculate aba again which yields

$$\begin{aligned} aba &= \left(B, \begin{pmatrix} y_1 \\ py_1 + y_2 \end{pmatrix} \right) \\ &= \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right) \end{aligned}$$

and so we must have $py_1 = 0$ and since $p > 0$ we have $y_1 = 0$. But now we calculate b^2 as

$$b^2 = \left(I, \begin{pmatrix} 2y_1 \\ 0 \end{pmatrix} \right)$$

but since $y_1 = 0$, this cannot induce an integral affine structure on the torus hence the action is not free.

□

This completes the case where $B = B_1$. Next, we tackle the case of $B = B_2$.

4.4. CLASSIFICATION OF INTEGRAL AFFINE STRUCTURES ON THE KLEIN BOTTLE

Lemma 4.24. Let $B = B_2$. Then the possible choices of A that induce an integral affine structure on K^2 are $A = I, A_4$.

Proof.

Case 1 ($A = I, B = B_2$). We calculate the group law $aba = b$ which yields

$$\begin{aligned} aba &= \left(B, \begin{pmatrix} x_1 + x_2 + y_1 \\ x_2 + y_2 + x_1 \end{pmatrix} \right) \\ &= \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right). \end{aligned}$$

This implies that $x_1 = -x_2$. We conjugate a, b by

$$c = \left(-I, \begin{pmatrix} -\frac{y_2}{2} \\ -\frac{y_2}{2} \end{pmatrix} \right)$$

which gives use the new generators

$$a = \left(I, \begin{pmatrix} -x_1 \\ x_1 \end{pmatrix} \right), \quad b = \left(B, \begin{pmatrix} -y_2 - y_1 \\ 0 \end{pmatrix} \right)$$

and by switching generators $a \mapsto a^{-1}$ if necessary, we may assume that $x_1 < 0$ so that we will switch the signs on the translational component of a . We will do the same for the translational component of b and assume $y_2 = 0$ and $y_1 > 0$ after relabeling $-y_2 - y_1 \rightarrow y_1$. This finishes our generating set which we write as

$$a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix} \right), \quad b = \left(B, \begin{pmatrix} y_1 \\ 0 \end{pmatrix} \right)$$

with $x_1, y_1 > 0$. We now check freeness of the action. Again since $B^{2k} = I$, the even case is covered by the torus classification and so we will check freeness of words $a^q b^k$ with k being odd. We calculate that

$$\begin{aligned} a^q &= \left(I, \begin{pmatrix} qx_1 \\ -qx_1 \end{pmatrix} \right) \\ b^k &= \left(B, \begin{pmatrix} \frac{k+1}{2}y_1 \\ \frac{k-1}{2}y_1 \end{pmatrix} \right), \text{ for } k \text{ odd} \\ a^q b^k &= \left(B, \begin{pmatrix} \frac{k+1}{2}y_1 + qx_1 \\ \frac{k-1}{2}y_1 - qx_1 \end{pmatrix} \right), \text{ for } k \text{ odd.} \end{aligned}$$

Now let $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \in \mathbb{R}^2$ and consider the fixed point equation $(a^q b^k) \circ z = z$. This gives

$$\begin{pmatrix} \frac{k+1}{2}y_1 + qx_1 + z_2 \\ \frac{k-1}{2}y_1 - qx_1 + z_1 \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}.$$

Solving this gives $\frac{k+1}{2}y_1 + \frac{k-1}{2}y_1 = 0 \Rightarrow ky_1 = 0$. But since $y_1 > 0$ we must have $k = 0$. So if the action is not free we must have that a^q acts non-freely for some $q \neq 0$ but this action is by translations and hence is free and so $A = I, B = B_2$ does induce an integral affine structure.

The integral affine structure on K^2 is given by

$$a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix} \right), \quad b = \left(B_2, \begin{pmatrix} y_1 \\ 0 \end{pmatrix} \right), \quad x_1, y_1 > 0$$

and the induced structure on the torus is given by

$$a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix} \right), \quad b^2 = \left(I, \begin{pmatrix} y_1 \\ y_1 \end{pmatrix} \right)$$

$x_1, y_1 > 0$.

Case 2 ($A = A_3, B = B_2$). We show that this cannot induce an integral affine structure on the torus. We conjugate A and B by

$$G = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

which puts them in the form

$$A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}$$

and so we have $B^2 = I$. Hence, the integral affine structure on $\mathbb{T}^2$ should belong to the second family. However calculating

$$b^2 = \left(I, \begin{pmatrix} y_2 \\ 2y_2 \end{pmatrix} \right)$$

we require that $2y_2 = 0 \Rightarrow y_2 = 0$ but then b^2 does not act freely and hence does not induce an integral affine structure on the torus. Therefore, this case cannot induce an integral affine structure on K^2 .

Case 3 ($A = A_4, B = B_2$). Again, we conjugate A, B by the matrix

$$G = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

and assume that

$$a = \left(A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right), \quad b = \left(B = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right).$$

Using that $B^2 = I$ and a, b^2 must induce an integral affine structure on the torus it follows that $x_1 = 0$ since the induced structure belongs to the second family. Now, calculating the group law $aba = b$ yields

$$\left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 + ny_2 + x_2(1-n) \\ y_2 \end{pmatrix} \right) = \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right)$$

and so $ny_2 + (1-n)x_2 = 0 \Rightarrow y_2 = \frac{n-1}{n}x_2$.

Our generators for the integral affine structure on K^2 are therefore

$$a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ \frac{n-1}{n}x_2 \end{pmatrix} \right)$$

and the induced structure on the torus is

$$a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b^2 = \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2y_1 + \frac{n-1}{n}x_2 \\ 0 \end{pmatrix} \right)$$

with $y_1 > 0$ and so a necessary condition is $2y_1 \neq -\frac{n-1}{n}x_2$.

In fact, we shall now show that this is sufficient. We calculate that

$$a^k = \left(\begin{pmatrix} 1 & kn \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} \frac{nk(k-1)}{2}x_2 \\ kx_2 \end{pmatrix} \right), \quad b^q = \left(B, \begin{pmatrix} qy_1 + \frac{q-1}{2}y_2 \\ y_2 \end{pmatrix} \right), \text{ for } q \text{ odd}$$

4.4. CLASSIFICATION OF INTEGRAL AFFINE STRUCTURES ON THE KLEIN BOTTLE

and

$$a^k b^q = \left(\begin{pmatrix} 1 & 1 - kn \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} qy_1 + \frac{q-1}{2}y_2 + kny_2 + \frac{nk(k-1)}{2}x_2 \\ y_2 + kx_2 \end{pmatrix} \right).$$

Letting $v_1 := qy_1 + \frac{q-1}{2}y_2 + kny_2 + \frac{nk(k-1)}{2}x_2$ and $v_2 := y_2 + kx_2$ and $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \in \mathbb{R}^2$, the fixed point equation yields

$$\begin{pmatrix} (1 - kn)z_2 \\ -2z_2 \end{pmatrix} = - \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

This yields the equation $(1 - kn)v_2 + 2v_1 = 0$. Solving this equation and using that $y_2 = \frac{n-1}{n}x_2$ we obtain

$$q(2y_1 + y_2) = 0.$$

It is quick to check that a^k acts freely so if the action is not free then $q \neq 0$, and this gives the requirement

$$2y_1 \neq -\frac{n-1}{n}x_2$$

and so this condition is sufficient and necessary.

□

This completes the classification of the integral affine structures on the Klein bottle. We summarize the results of the previous two lemmas into the following theorem.

Theorem 4.25. The possible generators for the holonomy representation of the integral affine structures on the Klein bottle are

1. $a = \left(I, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix} \right),$ with $x_2, y_1 > 0$.
2. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ x_2 \end{pmatrix} \right),$ with $x_2, y_1 > 0$.
3. $a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix} \right),$ with $x_1, y_1 > 0$.
4. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ \frac{n-1}{n}x_2 \end{pmatrix} \right),$ with $y_1 > 0$ and $2y_1 \neq -\frac{n-1}{n}x_2$.

We note that families 1) and 3) induce a Euclidean structure on the Klein bottle, i.e. they preserve some Riemannian metric as we have that the group they generate is a subset of $O(n) \times \mathbb{R}^2$.

We now wish to classify the corresponding Lagrangian fibrations for each of the integral affine structures on the two torus and the Klein bottle. Below are some pictures of the different families of integral affine structures on both the torus and the Klein bottle.

4.4.4 Fundamental domains for the action of the holonomy group on $\mathbb{R}^2$.

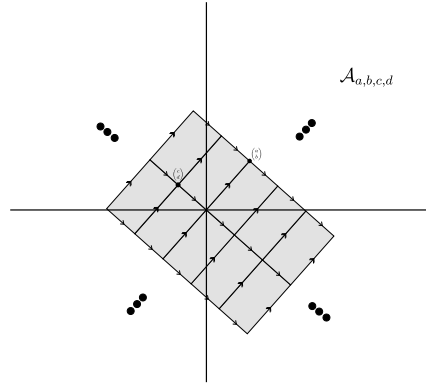


Figure 4.1: Fundamental domain for the torus lattice $\mathcal{A}_{a,b,c,d}$.

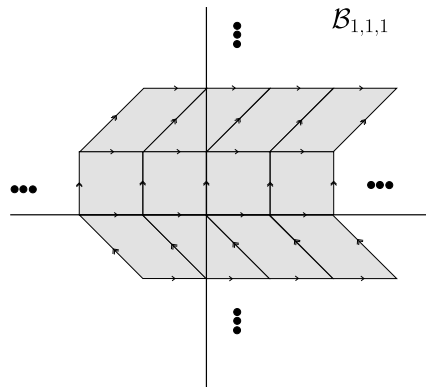


Figure 4.2: Fundamental domain for torus lattice $\mathcal{B}_{1,1,1}$.

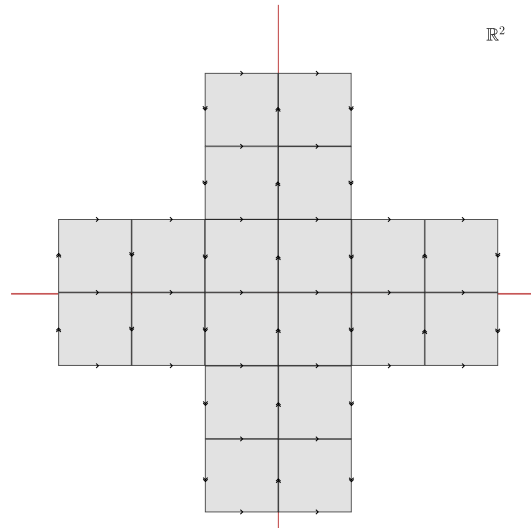


Figure 4.3: Fundamental domain for a Klein bottle lattice from family 1) with $x_2 = y_1 = 1$.

Classification of Lagrangian Fibrations over Closed Surfaces

5.1 The Classification Problem

Now that we have classified the integral affine structures on closed surfaces, we can look to classify the Lagrangian fibrations $E \rightarrow B$ which induce a given integral affine structure. The integral affine structure on the base B is one invariant of a Lagrangian fibration however it is not the only one. We have seen that there is a correspondence between integral affine structures on B and full rank lattices in T^*B locally spanned by closed one-forms. This lattice $\Lambda \subset T^*B$ gives rise to a Lagrangian fibration $\pi : T^*B/\Lambda \rightarrow B$ which is a canonical Lagrangian fibration we can associate to B once we have fixed the integral affine structure. There are in general, other Lagrangian fibrations over B which are in some sense *twistings* of this canonical fibration.

We wish to classify these fibrations up to fibrewise symplectomorphism, covering the identity. We shall see that there are obstructions both to a Lagrangian bundle over B being topologically equivalent to this canonical Lagrangian fibration $\pi : T^*B/\Lambda \rightarrow B$ as well as obstructions to a Lagrangian bundle over B being symplectically equivalent to the canonical Lagrangian bundle. Since the problem of classifying these bundles will be an issue of stitching together locally defined Lagrangian fibrations into a global one, it will be helpful to do this in the language of sheaves and their cohomology.

The idea behind sheaf cohomology is similar to that of group cohomology from Chapter 1. Sheaf cohomology will measure the failure of the global sections functor to be exact. We will not develop this theory from scratch in this thesis, but we refer the unfamiliar reader to [Bre97].

5.2 Latticed one-forms

We shall make use of the following sheaves on B .

Definition 5.1. Let B be a closed integral affine surface with period lattice $\Lambda \subset T^*B$.

1. Λ is the sheaf of sections of the period lattice $\Lambda \subset T^*B$.
2. ΛT^*B is the sheaf of Lagrangian sections of T^*B and T^*B is the sheaf of all sections of T^*B .
3. $\mathcal{Z}(T^*B/\Lambda)$ is the sheaf of Lagrangian sections of T^*B/Λ and T^*B/Λ is the sheaf of all sections of T^*B/Λ .
4. $\Lambda^2 B$ is the sheaf of closed 2-forms on B .

Remark 5.2. We note that we somewhat abuse notation and write the sheaf of sections of T^*B , T^*B/Λ and Λ by the same symbols. It will be obvious from the context whether we are considering these objects as bundles or as the sheaf of their corresponding sections.

Remark 5.3. Let $\Omega^1(B)$ denote the sheaf of differential forms on B . Then the differential

$$d : \Omega^1(B) \rightarrow \Omega^2(B)$$

descends to a well-defined differential

$$d_\Lambda : T^*B/\Lambda \rightarrow \Omega^2(B)$$

on the sections of T^*B/Λ as sections of Λ are given by closed one forms and hence vanish under this differential. This leads to the following notion of latticed forms.

Definition 5.4 (Latticed one-form). Fix a period lattice $\Lambda \subset T^*B$. A *latticed one-form* is a section of T^*B/Λ . A lattice one-form will be called closed if it determines a Lagrangian submanifold of the symplectic manifold $(T^*B/\Lambda, \tilde{\omega}_{can})$.

A normal one-form on B determines a latticed one-form by the bundle projection $\pi : T^*B \rightarrow T^*B/\Lambda$. However, the converse is not true. In general, there are obstructions to lifting a latticed one-form to a one-form on B . There is a short exact sequence of sheaves

$$0 \rightarrow \Lambda \rightarrow T^*B \rightarrow T^*B/\Lambda \rightarrow 0$$

and taking the long exact sequence in sheaf cohomology we have

$$H^0(T^*B) \xrightarrow{d_1} H^0(T^*B/\Lambda) \xrightarrow{d_2} H^1(\Lambda) \rightarrow \dots$$

Since sheaf cohomology is a derived functor of the global sections functor, $H^0(\cdot)$ consists of the global sections over B .

A section of T^*B/Λ , i.e. a latticed one-form, can be lifted to a section of T^*B if and only if it is in $\text{Im } d_1$. By exactness, this is equivalent to the latticed one-form being in $\ker d_2$. It follows, that if $H^1(\Lambda) = 0$ then every latticed one-form lifts to a normal one-form on B . This is true over contractible open sets as the sheaf cohomology of the sheaf of sections of Λ is a homotopy invariant - see Theorem 5.38.

Since the differential descends to T^*B/Λ , it makes sense to consider elements in $H^2(B, \mathbb{R})$ which arise from this differential.

Definition 5.5. Let s be a latticed one-form on B . Then the cohomology class $[ds] \in H^2(B, \mathbb{R})$ will be called a *cohomology invariant of the shift s*

We now work towards showing when adding closed two-forms to $(T^*B/\Lambda, \tilde{\omega}_{can})$ produces equivalent Lagrangian bundles. This will turn out to depend on the cohomological invariants of shifts.

Proposition 5.6. A fibrewise affine shift of T^*B/Λ is determined by a latticed one-form and any latticed one-form determines such a shift. Moreover, this shift is a Lagrangian equivalence if the latticed one-form is closed.

Proof. Let $F : T^*B/\Lambda \rightarrow T^*B/\Lambda$ be a fibrewise affine . On each fibre, this takes the form $F_b(x) = x + c$ for some $c \in T_b^*B/\Lambda_b$ and hence is determined by a smooth section of T^*B/Λ .

Conversely, given a section $\alpha : B \rightarrow T^*B/\Lambda$ we define an affine by

$$F_\alpha : T^*B/\Lambda \rightarrow T^*B/\Lambda$$

$$x_b \mapsto x_b + \alpha_b.$$

Proposition 2.18 yields that this is a Lagrangian equivalence if and only if α is closed. $\square$

Remark 5.7. The contents of Theorem 2.31 show that if $E \rightarrow B$ is a Lagrangian fibration with period lattice Λ , then a latticed one-form also determines a fibrewise of E . Indeed, since over contractible open sets, $E \rightarrow B$ identifies with $T^*B/\Lambda \rightarrow B$, the converse can also be shown to be true.

Proposition 5.8. Let $\rho : (E, \omega) \rightarrow B$ be a Lagrangian fibration with period lattice Λ . For $\alpha, \beta \in \Omega^2(B)$ the bundles $(E, \omega + \rho^*\alpha)$ and $(E, \omega + \rho^*\beta)$ are Lagrangian equivalent if and only if

$$[\alpha] - [\beta] = [d\eta] \in H^2(B)$$

where $[d\eta]$ is a cohomological invariant of the shift η .

Proof. Locally, our bundle $\rho : E \rightarrow B$ looks like the bundle $\pi : T^*B/\Lambda \rightarrow B$. On overlaps $U \cap V$ where our bundle is of this form, the transition maps are given by Lagrangian equivalences of the bundle over $U \cap V$ and it is therefore enough to consider what happens on $T^*B/\Lambda \rightarrow B$.

Proposition 2.18 implies that the bundles $(T^*B, \omega_{can} + \pi^*\alpha)$ and $(T^*B, \omega_{can} + \pi^*\beta)$ are Lagrangian equivalent if and only if $\alpha - \beta$ is an exact two-form. A latticed one-form is locally given by a usual one form, however a global lift may not exist.

However, over contractible open sets, latticed one-forms are determined by usual one-forms. Hence if the shift by a latticed one-form η is a Lagrangian equivalence of the bundles $(T^*B/\Lambda, \omega + \pi^*\alpha)$ and $(T^*B/\Lambda, \omega + \pi^*\beta)$ then over contractible open sets we require that $\alpha - \beta = d\tilde{\eta}$ where $\tilde{\eta}$ is the lift of the latticed one-form to a normal one-form on B .

Locally if $\alpha - \beta = d\tilde{\eta}$, then we can add any latticed one-form that lifts to a global one-form on B to η so that $\alpha - \beta = d(\tilde{\eta} + \varphi)$ which is still exact. Since these local lifts of η glue together globally on T^*B/Λ , and we can modify η by a global one-form φ , it follows that $[\alpha] - [\beta] = [d\eta] \in H^2(B, \mathbb{R})$. $\square$

5.2.1 Calculating Cohomological Invariant of Shifts

If we fix a period lattice Λ and consider the bundle T^*B/Λ then adding the pullback of a closed latticed one-form is a Lagrangian equivalence. Proposition 5.8 tells us that the possible

CHAPTER 5. CLASSIFICATION OF LAGRANGIAN FIBRATIONS OVER CLOSED SURFACES

Lagrangian equivalences of these bundles are determined by cohomological invariants of a shift. We now wish to calculate the cohomological invariants of shifts for the integral affine structures of Theorem 4.14. Since a class $[d\eta] \in H^2(\mathbb{T}^2, \mathbb{R})$ is determined by the value $\int_{\mathbb{T}^2} d\eta$ we are left to calculate the value of this integral. We note that this cohomology group is spanned by the class of $[dx]$ and $[dy]$.

Theorem 5.9. We have that

1. For the lattice $\mathcal{A}_{a,b,c,d}$, the cohomological invariant of shifts are determined by integer covectors $\tilde{m} = kdx + rdy$ and $\tilde{n} = ldx + sdy$ subject to no restrictions. Two invariants of shift are equivalent if the quantity $ka + lb + rc + sd$ is the same.
2. For the lattice $\mathcal{B}_{n,a,b}$, the cohomological invariant of shifts are determined by integer covectors $\tilde{m} = kdx + rdy$ and $\tilde{n} = sdy$. Two invariants of shifts are equivalent if the quantity $ka + sb$ is the same.

Proof.

Case 1 ($\mathcal{A}_{a,b,c,d}$). Recall that in this case the fundamental domain for the torus is generated by

$$g = \left(1, \begin{pmatrix} a \\ c \end{pmatrix}\right) \text{ and } h = \left(1, \begin{pmatrix} b \\ d \end{pmatrix}\right)$$

where the translational components are linearly independent. To construct a latticed one-form, we need the value of the one-form to agree with the identifications induced by the translational actions up to taking the quotient by the lattice $\Lambda \subset T^*B$. Namely, given the value of a latticed one-form $\alpha \in \Gamma(T^*B/\Lambda)$, if we fix a chart $U \subset \mathbb{T}^2$ around p and let $\alpha_p = \alpha_1 dx + \alpha_2 dy \in T_p^*\mathbb{T}^2$, pulling back α by one of our generators g or h we should get back $\alpha_p + kdx + rdy$ for some integers $k, r \in \mathbb{Z}$ as in affine coordinates (x, y) , the lattice Λ is spanned by dx and dy over $\mathbb{Z}$.

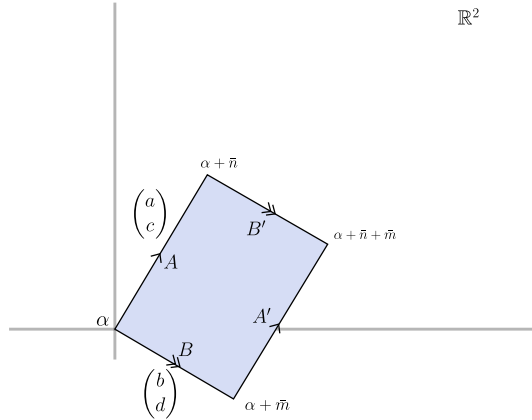


Figure 5.1: Fundamental domain for the first lattice.

The invariant we wish to calculate is $\int_{\mathbb{T}^2} d\alpha$ where α is a latticed one-form. By Stokes theorem, letting U denote a fundamental domain for $\mathbb{T}^2$,

$$\begin{aligned} \int_U d\alpha &= \int_{\partial U} \alpha \\ &= \int_A \alpha - \int_{A'} \alpha + \int_B \alpha - \int_{B'} \alpha \end{aligned}$$

where A, A', B, B' are the boundaries of the fundamental domain depicted in Figure 5.1.

For example, translating α along A by g should return α up to some integer covector. In other words, we want

$$g^*\alpha = \alpha + \bar{m}$$

and

$$h^*\alpha = \alpha + \bar{n}.$$

Denoting by $\bar{m} = (k, r)$ and $\bar{n} = (l, s)$, the integer covectors obtained from transporting along A and B respectively, we can then translate $g^*\alpha + \bar{m}$ by h along B' . This should return the same one-form as first translating along B and then translating along A' . For our first lattice, we see that this always holds true and so there is no restrictions on the covectors $\bar{m}$ and $\bar{n}$.

Now we calculate the required integral. We note that on A and A' , the one form α differs by $\bar{m}$ and so

$$\begin{aligned} \int_A \alpha - \int_{A'} \alpha &= \int_A \alpha - \int_A \alpha + \bar{m} \\ &= - \int_A \bar{m}. \end{aligned}$$

Similarly, we have $\int_B \alpha - \int_{B'} \alpha = - \int_B \bar{n}$. Putting this together yields

$$\begin{aligned} \int_U d\alpha &= \int_A \alpha - \int_{A'} \alpha + \int_B \alpha - \int_{B'} \alpha \\ &= - \left(\int_A \bar{m} + \int_B \bar{n} \right) \\ &= - \left(\int_{(0,0)}^{(a,c)} kdx + rdy + \int_{(0,0)}^{(b,d)} ldx + sdy \right) \\ &= -(ka + rc + lb + sd) \end{aligned}$$

with $k, l, r, s \in \mathbb{Z}$ which gives the required invariants.

Case 2 $(\mathcal{B}_{n,a,b})$. Recall that for this family our fundamental domain is generated by

$$g = \left(1, \begin{pmatrix} a \\ 0 \end{pmatrix} \right) \text{ and } h = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ b \end{pmatrix} \right).$$

In this case, our integer covectors $\bar{m}$ and $\bar{n}$ are not arbitrary. We must have that

$$h^*(g^*\alpha + \bar{n}) + \bar{m} = g^*(h^*\alpha + \bar{m}) + \bar{n}$$

and this implies that $h^*n = n$ or more algebraically

$$\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}^T \begin{pmatrix} l \\ s \end{pmatrix} = \begin{pmatrix} l \\ s \end{pmatrix}.$$

This implies that $\bar{n} = \begin{pmatrix} 0 \\ s \end{pmatrix}$ and this is the only restriction on our integer co-vectors $\bar{m}$ and $\bar{n}$. We now calculate the integral over our fundamental domain. We let $\alpha = \alpha_1(x, y)dx + \alpha_2(x, y)dy$. Again, applying Stokes theorem, we have that

$$\begin{aligned} \int_{\partial U} \alpha &= \int_A \alpha - \int_{A'} \alpha + \int_B \alpha - \int_{B'} \alpha \\ &= \int_A \alpha - \int_A (h^*\alpha + \bar{m}) + \int_B \alpha - \int_B (g^*\alpha + \bar{n}) \\ &= \int_A (\alpha - h^*\alpha - \bar{m}) - \int_B \bar{n}. \end{aligned}$$

CHAPTER 5. CLASSIFICATION OF LAGRANGIAN FIBRATIONS OVER CLOSED SURFACES

Now, recalling that $h^* = \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix}$ we have that

$$\begin{aligned}\alpha - h^*\alpha &= (\alpha_1(x, y) - \alpha_1(x, y))dx + n\alpha_1(x, y)dy \\ &= n\alpha_1(x, y)dy.\end{aligned}$$

Now, we have that

$$\begin{aligned}\int_A n\alpha_1(x, y)dy &= \int_{(0,0)}^{(a,0)} n\alpha_1(x, y)dy \\ &= 0\end{aligned}$$

as there is no dx contribution. Hence, our invariant becomes

$$\begin{aligned}\int_{\partial U} \alpha &= - \left(\int_A \bar{m} + \int_B \bar{n} \right) \\ &= - \left(\int_{(0,0)}^{(a,0)} \bar{m} + \int_{(0,0)}^{(0,b)} \bar{n} \right) \\ &= -(ka + sb)\end{aligned}$$

where $k, s \in \mathbb{Z}$ and this completes the calculation of the cohomological invariants of shifts.

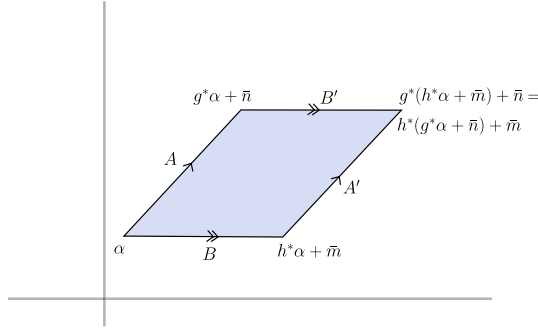


Figure 5.2: Fundamental domain for the second lattice.

□

5.3 Luttinger Surgery

We will use Luttinger surgery in order to construct a Lagrangian fibration over the torus or the Klein bottle with a given Chern class. The main idea is that given a Lagrangian fibration with a fixed Chern class over one of these surfaces, we can perform Luttinger surgery on it in order to obtain a bundle with a global section. This corresponds to a Lagrangian fibration with the trivial Chern class. By performing the surgery in the "opposite" direction i.e. starting with a bundle with a global section and performing Luttinger surgery on this, we will be able to produce a fibration with any given Chern class.

Luttinger surgery was first introduced in [Lut95] and provides a way to modify a symplectic four-manifold (M, ω) along an embedded Lagrangian tori $L \subset M$ such that the symplectic structure is preserved and the geometry is altered only in a neighborhood of the torus. Topologically it is equivalent to Dehn surgery with the framing chosen to be the Weinstein framing. A framing is, roughly speaking, a parameterization of the tubular neighborhood of the torus $L \subset M$. For a Lagrangian submanifold there is a canonical choice of such a framing given by the Weinstein-Lagrangian tubular neighborhood theorem.

Theorem 5.10 ([Wei71]). Let (M, ω) be a symplectic manifold and $L \subset M$ be a Lagrangian submanifold. Then there exist tubular neighborhoods $U \subset M$ and $V \subset T^*L$ of L together with a symplectomorphism:

$$f : (U, \omega|_U) \rightarrow (V, \omega_{\text{can}}|_V)$$

with $f|_L = Id_L$.

The surgery construction works for general Lagrangian tori in a symplectic four-manifold but we will focus on the case where our Lagrangian torus is given by a fibre of a Lagrangian fibration.

Let B be the base space of such a Lagrangian fibration and let $T^*B/\Lambda \rightarrow B$ be the symplectic reference bundle. We view B as the quotient $\mathbb{R}^2/\langle A, B \rangle$ where A, B are the generators of the affine holonomy group. Let $p \in \mathbb{R}^2$ be in the interior of a fundamental domain for the quotient and pick a small disc D of radius 2ϵ centered at this point and a neighborhood U of D . Over this neighborhood, our fibration looks like $\pi : U \times T^2 \rightarrow U$. By Theorem 5.10, the symplectic form on the tubular neighborhood $\pi^{-1}(D)$ of our Lagrangian torus is inherited from the symplectic form on T^*T^2 .

Now, if we take $\partial\pi^{-1}(D) = \partial(T^2 \times D)$ this is a three torus $T^2 \times S^1$. If we remove the interior of this region from the total space of the fibration, we are left with a manifold with boundary T^3 . The surgery now amounts to specifying an attaching map which glues together the boundary of the removed $T^2 \times D$ back into the boundary created in the total space of the fibration. This attaching map is given by a diffeomorphism $F : T^3 \rightarrow T^3$.

Of course, so far this attaching map has not taken into account the symplectic structure of the total space of the fibration. Indeed, the three torus has odd dimension and hence it does not make sense to ask that the attaching map F itself be a symplectomorphism. However, we can ask that F is the restriction of a symplectomorphism of the bundle $\pi^{-1}(N)$ where N is a small annulus around the boundary of the disc ∂D .

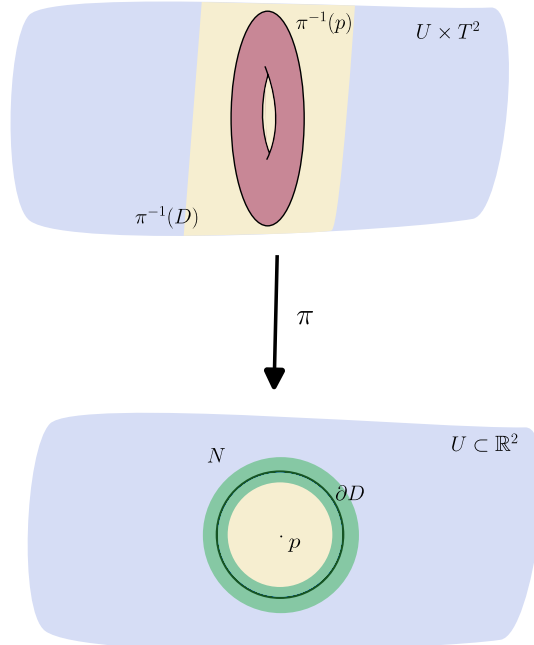


Figure 5.3: Performing Luttinger surgery on the Lagrangian torus $\pi^{-1}(p)$

CHAPTER 5. CLASSIFICATION OF LAGRANGIAN FIBRATIONS OVER CLOSED SURFACES

Of course, one way to define such a map is to pick a Lagrangian section of $\pi^{-1}(N)$ which is given by a closed one-form α , and perform a shift by this one-form. Then, by Proposition 2.18 this will be a symplectomorphism and since it preserves the fibres, its restriction to ∂D will induce a map $F_\alpha : T^3 \rightarrow T^3$. A section of our bundle over the boundary of the disc is a map $f : S^1 \rightarrow T^2$ and is determined up to homotopy by an element of $\pi_1(T^2) = \mathbb{Z}^2$ as such a map is a loop in T^2 . The pair of integers (m, n) represent the winding number of such a map. Let $(x, y) = (r \cos \varphi, r \sin \varphi)$ be coordinates on N centered at p . Then we can define a representative section for any (m, n) to be

$$s = \frac{1}{2\pi}(m\varphi dx + n\varphi dy).$$

One checks that after winding once around the circle i.e. $\varphi \mapsto \varphi + 2\pi$, we have that our section changes by $m dx + n dy$ and hence corresponds to the winding number given by (m, n) . This indeed defines a section of the torus bundle since it is locally given by the symplectic reference bundle and hence over U looks like $T^*U/\langle dx, dy \rangle_{\mathbb{Z}}$.

In principle, one could perform the surgery by performing a shift using this section. The issue is, this section is not Lagrangian. Indeed one calculates

$$ds = \frac{1}{2\pi}(m d\varphi \wedge dx + n d\varphi \wedge dy).$$

In order to make this closed, one can add the correction term

$$\beta = \frac{1}{2\pi}(mx + ny)d\varphi$$

to s . It is easily checked that this does not alter the homotopy class of the section s as incrementing φ by 2π does not alter β . Hence, the section $F_{m,n} = s + \beta$ determines a symplectomorphism on a neighborhood of the bundle over ∂D given by performing a shift by the closed one-form $F_{m,n}$ as in Definition 2.17. This completes the surgery construction.

5.4 Chern Class and Lagrangian Class

The strategy in classifying the Lagrangian bundles over B will be to start with the topological reference bundle T^*B/Λ and using this, produce new bundles by gluing these reference bundles together. We will then show that we can essentially obtain any Lagrangian bundle over B in this way. In [Dui80], Duistermaat defined the *Chern class* of a Lagrangian fibration. It measures the obstruction for a Lagrangian fibration to be topologically equivalent to T^*B/Λ . We now illustrate how this class is defined.

We have seen previously that given a regular Lagrangian fibration with compact, connected fibres, there is a canonical period lattice $\Lambda \subset T^*B$ and a fibrewise transitive action of $T^*B/\Lambda \rightarrow B$ on $E \rightarrow B$. This endowed $E \rightarrow B$ with the structure of an *affine* torus bundle, however we could not yet identify it with the torus bundle T^*B/Λ . The obstruction to topologically identifying these two fibrations is the obstruction to $E \rightarrow B$ admitting a global section. We now answer the question of when $E \rightarrow B$ admits a global section.

If we cover the base space B with open sets $\{U_i\}$ such that each U_i is contractible, then, restricting the fibration to some U_i , all the obstructions to this fibration admitting global action angle coordinates vanish or in other words, locally this can be identified with T^*B/Λ . Hence, if we pick our U_i small enough we can choose sections $s_i : U_i \rightarrow E$. Since the action of T^*B/Λ on $E \rightarrow B$ is transitive, on the intersections $U_i \cap U_j$, we know that $s_i = \mu_{ij} + s_j$ whereby μ_{ij} is a section of T^*B/Λ and $\mu_{ij} + s_j$ denotes the abelian vector space action of this section on s_j . We now make the following claim.

Proposition 5.11. The sections μ_{ij} satisfy the cocycle condition and hence define a cohomology class $[\mu] \in H^1(T^*B/\Lambda)$. Moreover, the cohomology class does not depend on the choice of sections s_i .

Proof. We show that $\mu_{jk} + \mu_{ij} = \mu_{ik}$. We have that

$$\begin{aligned}\mu_{jk} + \mu_{ij} &= s_j - s_k + s_i - s_j \\ &= s_i - s_k\end{aligned}$$

and so the condition is immediate. Suppose we choose different sections s'_i, s'_j . Since the T^*B/Λ action is transitive we have

$$s'_i = s_i + \sigma_i$$

We now compute

$$\begin{aligned}\mu'_{ij} - \mu_{ij} &= s'_i - s'_j - s_i + s_j \\ &= s_i + \sigma_i - s_j - \sigma_j - s_i + s_j \\ &= \sigma_i - \sigma_j\end{aligned}$$

and hence the difference is a Čech co-boundary and so is zero in cohomology as required. $\square$

It is clear that if μ itself equals to zero, then we have a global section of $M \rightarrow B$. Indeed if this is the case, then our sections are equal on the overlaps and hence there is no problem gluing them together to obtain a global section. However, this condition turns out to be in some sense too strong. Indeed, if we loosen the condition to $[\mu] = 0 \in H^1(T^*B/\Lambda)$, it turns out that we will be able to *redefine* our sections so that they equal on the overlap and hence glue together. The following Lemma proves this.

Lemma 5.12. Suppose that $[\mu] = 0 \in H^1(T^*B/\Lambda)$. Then there exists a global section of $M \rightarrow B$.

Proof. If $[\mu] = 0$, then it follows that $[\mu]$ is a coboundary and so $\mu_{ij} = \sigma_i - \sigma_j$ where $\sigma_i : U_i \rightarrow T^*B/\Lambda$. We redefine our sections as

$$s'_i = s_i - \sigma_i.$$

Now, on intersections we have that

$$\begin{aligned}s'_i - s'_j &= s_i - \sigma_i - s_j + \sigma_j \\ &= s_i - s_j - (\sigma_i - \sigma_j) \\ &= s_i - s_j - \mu_{ij} \\ &= 0\end{aligned}$$

and hence we have no problem gluing our new local sections into a global section of $M \rightarrow B$. $\square$

Hence, the obstruction to the existence of a global section of $M \rightarrow B$ lies in the sheaf cohomology group $H^1(T^*B/\Lambda)$. In practice, this group may not be so easy to calculate. We would hence like to relate this sheaf cohomology group to a nicer cohomology group. Namely, we will be able to show that there is an isomorphism $H^1(T^*B/\Lambda) \cong H^2(\Lambda)$. The sheaf of

sections of the period lattice Λ is a locally constant sheaf as there are local trivialisations of the period lattice such that it looks like $U \times \mathbb{Z}^n$. Locally constant sheaves are equivalent to objects called *local systems*.

Calculating the cohomology of these local systems turns out to be much nicer than calculating sheaf cohomology more generally. For example, it is possible to compute this cohomology using cellular methods. Moreover, in the case that the base manifold in question is an Eilenberg-MacLane $K(G, 1)$ -space, we can in fact use group cohomology to calculate this cohomology group which we shall later see. For now, we shall look at how we can obtain this isomorphism.

5.4.1 The Sheaf Theory Approach

From our discussion in the previous section, the obstruction to $E \rightarrow B$ topologically identifying with T^*B/Λ lived in $H^1(T^*B/\Lambda)$. However, we noted that this may not be easy to calculate in general and so we wished to relate it to a nicer cohomology group. We will see now how we can make use of some standard sheaf theory to do this. A reference for some of the standard sheaf cohomology results used here is [Bre97].

Lemma 5.13. The following diagram of sheaves is exact in all rows and columns.

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & \Lambda & \xrightarrow{i} & \mathcal{Z}(T^*B) & \xrightarrow{\text{pr}} & \mathcal{Z}(T^*B/\Lambda) \longrightarrow 0 \\
 & & \parallel & & \downarrow & & \downarrow j \\
 0 & \longrightarrow & \Lambda & \longrightarrow & T^*B & \xrightarrow{\text{pr}} & T^*B/\Lambda \longrightarrow 0 \\
 & & & & \downarrow d & & \downarrow d \\
 & & & & \Lambda^2 B & \xlongequal{\quad} & \Lambda^2 B \\
 & & & & \downarrow & & \downarrow \\
 & & & & 0 & & 0
 \end{array}$$

Proof. It is clear that the two horizontal sequences given by the inclusion and projection maps are exact. We show that

$$0 \rightarrow \mathcal{Z}(T^*B) \rightarrow T^*B \xrightarrow{d} \Lambda^2 B \rightarrow 0$$

is exact. The map $\mathcal{Z}(T^*B) \rightarrow T^*B$ is the inclusion map and so is injective. It is also clear that the differential map d is surjective onto the space of closed 2-forms by definition. The kernel of the differential d is the closed one-forms on T^*B but these are given by Lagrangian sections. The other vertical sequence follows by the same argument and Remark 5.3. $\square$

Lemma 5.14. The following two sheaf cohomological facts hold;

1. T^*B is a fine sheaf so that $H^i(T^*B) = 0$ for $i > 0$.
2. $H^i(\mathcal{Z}(T^*B)) \cong H^{i+1}(B, \mathbb{R})$ for $i > 0$

Proof. That T^*B is a fine sheaf amounts to the fact that smooth manifolds admit partitions of unity. Denoting by $C^\infty(B)$ the sheaf of smooth functions on B this is a fine sheaf. We have a

short exact sequence

$$0 \rightarrow \mathbb{R} \rightarrow C^\infty(B) \rightarrow \mathcal{Z}(T^*B) \rightarrow 0$$

where $\mathbb{R} \rightarrow C^\infty(B)$ is the inclusion and $C^\infty(B) \rightarrow \mathcal{Z}(T^*B)$ is the given by the differential. Since $df = 0$ if and only if f is a constant function, this sequence is exact. Taking the associated long exact sequence in sheaf cohomology we obtain

$$\dots \rightarrow H^i(B, \mathbb{R}) \rightarrow H^i(C^\infty(B)) \rightarrow H^i(\mathcal{Z}(T^*B)) \rightarrow H^{i+1}(B, \mathbb{R}) \rightarrow \dots$$

Now since $C^\infty(B)$ is a fine sheaf, for $i > 0$ we obtain $0 \rightarrow H^i(\mathcal{Z}(T^*B)) \rightarrow H^{i+1}(B, \mathbb{R}) \rightarrow 0$ giving the required isomorphism. $\square$

5.4.2 The Chern Class

We can now show the isomorphism $H^1(T^*B/\Lambda) \cong H^2(\Lambda)$ which will lead to the definition of the Chern class.

Theorem 5.15. There is an isomorphism $\delta : H^1(T^*B/\Lambda) \rightarrow H^2(\Lambda)$.

Proof. We had the short exact sequence

$$0 \rightarrow \Lambda \rightarrow T^*B \rightarrow T^*B/\Lambda \rightarrow 0$$

from Lemma 5.13. Taking the associated long exact sequence in cohomology yields

$$\dots \rightarrow H^i(\Lambda) \rightarrow H^i(T^*B) \rightarrow H^i(T^*B/\Lambda) \rightarrow H^{i+1}(\Lambda) \rightarrow \dots$$

Again, by Lemma 5.13, we have that $H^i(T^*B) = 0$ for $i > 0$ which gives an isomorphism $H^i(T^*B/\Lambda) \cong H^{i+1}(\Lambda)$ for $i > 0$. $\square$

Definition 5.16 (Chern Class). The image of $[\mu] \in H^1(T^*B/\Lambda)$ in $H^2(\Lambda)$ under δ is called the *Chern class* and we shall denote it by $[\mu_C]$.

The Chern class was defined by Duistermaat in [Dui80]. The vanishing of this class implies that the Lagrangian fibration $E \rightarrow B$ is topologically equivalent to $T^*B/\Lambda \rightarrow B$. We note that this already depends on the integral affine structure on B as this determines our period lattice Λ . It is a feature of the regularity of these Lagrangian torus fibrations that such a structure exists. Indeed, if we allow our fibration to have singular fibres, there is no longer a canonical symplectic reference bundle to compare our fibration $E \rightarrow B$ with. Indeed, in this case one must fix a so called *reference system* for which to compare our Lagrangian fibration to and then in a similar way one may define the Chern class of such a singular fibration. More details on the singular case can be found in [Zun01].

The above discussion shows that given a regular Lagrangian torus fibration, we can associate to it the invariant $[\mu_C] \in H^2(\Lambda)$. However, we have not yet answered whether the converse of this statement holds true - that is given a class $[\eta] \in H^2(\Lambda)$ can this be realised as the Chern class of some Lagrangian torus fibration?

To answer this question, we consider the short exact sequence

$$0 \rightarrow \Lambda \rightarrow \mathcal{Z}(T^*B) \rightarrow \mathcal{Z}(T^*B/\Lambda) \rightarrow 0$$

CHAPTER 5. CLASSIFICATION OF LAGRANGIAN FIBRATIONS OVER CLOSED SURFACES

and take the long exact sequence in sheaf cohomology. This leads to the exact sequence

$$\cdots \rightarrow H^1(\mathcal{Z}(T^*B/\Lambda)) \xrightarrow{\Delta} H^2(\Lambda) \xrightarrow{\mathcal{D}_{(B,\Lambda)}} H^2(\mathcal{Z}(T^*B)) \cong H^3(B, \mathbb{R}) \rightarrow \cdots$$

The map $\mathcal{D}_{(B,\Lambda)}$ was considered in [DD87].

They showed that a class $[\eta] \in H^2(\Lambda)$ is realisable as the Chern class of a Lagrangian torus fibration if and only if $\mathcal{D}_{(B,\Lambda)}(\eta) = 0$. They proved this using methods from Poisson geometry. One can show that Lagrangian torus fibrations are the same thing as complete symplectic realisations of the Poisson zero structure on a manifold. In their paper, they considered more generally symplectic realisations of regular Poisson structures of which the zero Poisson structure is a special case. Indeed there is a cohomology class determined by the characteristic foliation $\mathcal{F}_\pi$ and they showed that under the map $\mathcal{D}_{(B,\Lambda)}$, the Chern class must map to this cohomology class. In the case of the Poisson zero structure, the characteristic foliation of $(B, \pi \equiv 0)$ consists of just the points of B and this cohomology class will vanish. For more information on Poisson geometry see the following sources [CFM21].

5.4.3 The Lagrangian Class

This answers the question of which $[\eta] \in H^2(\Lambda)$ can arise as the Chern class of some Lagrangian fibration. However, this invariant is only topologically sharp. Once we have fixed an integral affine manifold B and a Chern class $[\mu_C] \in H^2(\Lambda)$, we can have Lagrangian fibrations over B with the same Chern class $[\mu_C]$ but which are not Lagrangian equivalent to each other. If we recall how the Chern class was constructed this fact should not be surprising. Indeed, we did not take into account the symplectic structure of our Lagrangian fibration in the construction of our Chern class since we allowed the sections of T^*B/Λ to be arbitrary so it would be too much to hope for that we would achieve a symplectic classification. We can fix this problem by choosing our sections to be sections of $\mathcal{Z}(T^*B/\Lambda)$ as we shall now see.

Definition 5.17 ([Zun01]). Let $M \rightarrow B$ be a Lagrangian fibration with period lattice Λ . Let $\{U_i\}_{i \in \mathbb{N}}$ be a trivializing cover and let $s_i : U_i \rightarrow \mathcal{Z}(T^*B/\Lambda)$. Then $\mu_{ij} = s_i - s_j$ defines a one-cocycle independent of the choice of section and hence a cohomology class $[\mu] \in H^1(\mathcal{Z}(T^*B/\Lambda))$. The class $[\mu]$ is called the *Lagrangian class* of $M \rightarrow B$.

Remark 5.18. The fact that the class $[\mu] \in H^1(\mathcal{Z}(T^*B/\Lambda))$ is well-defined and independent of the choice of sections follows from the same arguments as in Proposition 5.11.

If the class $[\mu] = 0$ then we can obtain a global *Lagrangian* section of $M \rightarrow B$ and hence we obtain that $M \rightarrow B$ is Lagrangian equivalent to $T^*B/\Lambda \rightarrow B$. In fact, the following proposition shows that every element of this cohomology group will be the Lagrangian class of some Lagrangian torus fibration.

Proposition 5.19. Every class $[\mu] \in H^1(\mathcal{Z}(T^*B/\Lambda))$ can be realised as the Lagrangian class of some Lagrangian torus fibration.

Proof. Suppose we are given some $[\mu] \in H^1(\mathcal{Z}(T^*B/\Lambda))$. Then picking a good cover $\{U_i\}_{i \in \mathbb{N}}$ of B we may assume $[\mu]$ can be represented by a one-cocycle $\alpha_{ij} : U_i \cap U_j \rightarrow \mathcal{Z}(T^*B/\Lambda)$. Now, each α_{ij} determines a fibrewise affine

$$s_{\alpha_{ij}} : T^*(U_i \cap U_j)/\Lambda \rightarrow T^*(U_i \cap U_j)/\Lambda$$

$$(b, m) \mapsto (b, m + \alpha_{ij}(b))$$

where the additive action of α_{ij} comes from the action of T^*B on $M \rightarrow B$. On triples $U_i \cap U_j \cap U_k$ we require that transitioning a point in the fibre by $s_{\alpha_{ij}} \circ s_{\alpha_{jk}} = s_{\alpha_{ik}}$ but this holds precisely because α_{ij} satisfies the cocycle condition. By Proposition 2.18, the $s_{\alpha_{ij}}$ is a symplectomorphism of the Lagrangian fibration $T^*(U_i \cap U_j)/\Lambda$ as α is closed. Hence, the transition maps are symplectomorphisms and so our local forms glue together to form a global symplectic form on M and this becomes a Lagrangian fibration as locally in each chart U_i it is by construction a Lagrangian fibration. $\square$

Remark 5.20. In fact, taking the long exact sequence in cohomology of the first row of Lemma 5.13, we obtain exactness at $H^1(\mathcal{Z}(T^*B/\Lambda)) \xrightarrow{\Delta} H^2(\Lambda)$ and hence every class $\Delta([\mu]) \in H^2(\Lambda)$ lies in the kernel of $\mathcal{D}_{(B, \mathcal{A})}$ the Dazord-Delzant homomorphism and hence can be realised as a Lagrangian torus fibration. However, this alone tells us only that $\Delta([\mu])$ arises as the Chern class of a Lagrangian torus fibration.

The following theorem summarizes the discussion from this section so far.

Theorem 5.21. Let B be a smooth manifold. The regular Lagrangian torus fibrations over B are determined by

1. The integral affine structure on B . This determines a symplectic reference bundle T^*B/Λ .
2. Fixing the integral affine structure on B , the Lagrangian fibrations over B are topologically determined by a Chern class $[\mu_C] \in H^2(\Lambda)$. The Chern class determines the topological type of the Lagrangian torus fibration and its vanishing implies the fibration is topologically equivalent to T^*B/Λ . An element $[\mu_C] \in H^2(\Lambda)$ is the Chern class of a Lagrangian torus fibration if and only if $\mathcal{D}_{(B, \mathcal{A})}([\mu_C]) = 0$ where $\mathcal{D}_{(B, \mathcal{A})}$ is the Delzant-Dazord homomorphism.
3. The obstruction to our bundle being symplectically equivalent to T^*B/Λ is given by the Lagrangian class $[\mu] \in H^1(\mathcal{Z}(T^*B/\Lambda))$. Every Lagrangian class arises as the Lagrangian class of some Lagrangian torus fibration.

In the case of an integral affine surface, we can make some of this discussion a bit more explicit. We have seen earlier that an element $[\mu] \in H^2(\Lambda)$ is the Chern class of a Lagrangian torus fibration if and only if it vanished under the Delzant-Dazord homomorphism. In the case of integral affine surfaces, it turns out that every class $[\mu] \in H^2(\Lambda)$ can be realised as the Chern class of a Lagrangian torus fibration.

There is a short exact sequence

$$0 \rightarrow \Lambda \rightarrow \mathcal{Z}(T^*B) \rightarrow \mathcal{Z}(T^*B/\Lambda) \rightarrow 0$$

from which we obtain the long exact sequence in cohomology

$$\begin{aligned} \cdots \rightarrow H^1(\Lambda) \rightarrow H^1(\mathcal{Z}(T^*B)) \cong H^2(B, \mathbb{R}) \xrightarrow{d} H^1(\mathcal{Z}(T^*B/\Lambda)) \xrightarrow{\Delta} H^2(\Lambda) \\ \rightarrow H^1(\mathcal{Z}(T^*B)) \cong H^3(B, \mathbb{R}) = 0 \end{aligned} \quad (5.1)$$

where the isomorphisms $H^i(\mathcal{Z}(T^*B)) \cong H^{i+1}(B, \mathbb{R})$ are from Lemma 5.13. By exactness this yields that $H^2(\Lambda) \cong \frac{H^1(\mathcal{Z}(T^*B/\Lambda))}{\ker \Delta}$. By Proposition 5.19, every element of $\frac{H^1(\mathcal{Z}(T^*B/\Lambda))}{\ker \Delta}$ arises as the Lagrangian class of a Lagrangian torus fibration and hence, as this group is

CHAPTER 5. CLASSIFICATION OF LAGRANGIAN FIBRATIONS OVER CLOSED SURFACES

isomorphic to $H^2(\Lambda)$ every class $[\mu] \in H^2(\Lambda)$ can be realised as the Chern class of a Lagrangian torus fibration.

Remark 5.22. Once we have fixed a Chern class we can ask which Lagrangian classes map to this Chern class. The long exact sequence gives us a map $j : H^1(\Lambda) \rightarrow H^2(B, \mathbb{R})$. The group $\ker \Delta = \frac{H^2(B, \mathbb{R})}{\text{Im } j}$ parametrizes the Lagrangian classes $[\mu]$ which map to a fixed Chern class $[\mu_C]$. These correspond to Lagrangian fibrations that are topologically equivalent but not symplectically equivalent.

Remark 5.23. Exactness of the long exact sequence 5.1 shows that the group $\ker \Delta = \frac{H^2(B, \mathbb{R})}{\text{Im } j}$.

Proposition 5.24. The group $\text{Im } H^1(\Lambda) \subset H^2(B, \mathbb{R})$ is generated by the cohomological invariants of shift from Definition 5.5.

Proof. Applying the long exact sequence in cohomology to the second row of Lemma 5.13, we obtain that

$$H^0(T^*B/\Lambda) \xrightarrow{\delta} H^1(\Lambda) \rightarrow H^1(T^*B) = 0$$

and so δ is surjective. We therefore have that $j(H^1(\Lambda)) \subset H^2(B, \mathbb{R})$ is the same as $j \circ \delta(H^0(T^*B/\Lambda)) \subset H^2(\Lambda)$ where j is the map $j : H^1(\Lambda) \rightarrow H^2(B, \mathbb{R})$ from Remark 5.22.

It remains to show that $j \circ \delta$ coincides with the differential $\tilde{d}_\Lambda : H^0(T^*B/\Lambda) \rightarrow H^2(B, \mathbb{R})$ since by definition $H^0(T^*B/\Lambda)$ is the space of latticed one-forms. This follows from another version of $H^1(\mathcal{Z}(T^*B/\Lambda)) \cong H^2(B, \mathbb{R})$. Again, applying the long exact sequence in cohomology to the short exact sequence

$$0 \rightarrow \mathcal{Z}(T^*B) \rightarrow T^*B \rightarrow \Lambda^2 B \rightarrow 0$$

yields the long exact sequence

$$0 \rightarrow H^0(\mathcal{Z}(T^*B)) \rightarrow H^0(T^*B) \xrightarrow{d} H^0(\Lambda^2 B) \rightarrow H^1(\mathcal{Z}(T^*B)) \rightarrow 0$$

since T^*B is a fine sheaf. This gives that

$$H^1(\mathcal{Z}(T^*B)) \cong \frac{H^0(\Lambda^2 B)}{dH^0(T^*B)}$$

and this is the de Rham cohomology group $H^2(B, \mathbb{R})$ by definition.

Now, we have the following short exact sequence

$$0 \longrightarrow \mathcal{Z}(T^*B/\Lambda) \longrightarrow T^*B/\Lambda \longrightarrow \Lambda^2 B \longrightarrow 0$$

and from the long exact sequence in cohomology, there is a map $d_\Lambda : H^0(T^*B/\Lambda) \rightarrow H^0(\Lambda^2 B)$. The required result now follows from commutativity of the following square.

$$\begin{array}{ccc} H^0(T^*B/\Lambda) & \xrightarrow{d_\Lambda} & H^0(\Lambda^2 B) \\ \downarrow \delta & & \downarrow \pi \\ H^1(\Lambda) & \xrightarrow{j} & H^2(B, \mathbb{R}) \end{array}$$

where $\pi \circ d_\Lambda = \tilde{d}_\Lambda$ is the required differential. □

5.5 Computing the Chern Class

After the discussion in the previous section, our strategy will be to compute all of the possible Chern classes for our integral affine closed surfaces. Once we have fixed a Chern class, we can then look at the possible Lagrangian classes that map to it. A lot of the complexity in this process will be in the calculation of the sheaf cohomology groups. Fortunately, since the sheaf of sections of the period lattice Λ is a locally constant sheaf, its sheaf cohomology will be equal to the cohomology of the so called *local system* it determines.

We shall now recall some basic facts and results around the theory of local systems. Some references on the theory of local systems include [ES09], [Whi78] and [DK01].

5.5.1 Local Systems

Recall that a constant sheaf $\mathcal{F}$ over a topological space X , is a sheaf such that for each open set $U \subset X$ we have that $\mathcal{F}(U) = R$ for some fixed group/ring/module R .

Definition 5.25 (Local system). A local system on a topological space X , is a locally constant sheaf $\mathcal{F}$ - e.g. for each $x \in X$, there is a neighborhood $U \subset X$ such that for all $V \subseteq U$ we have that $\mathcal{F}|_V$ is a constant sheaf for some fixed group/ring/module R .

Remark 5.26. For the remainder of this discussion we will take R to be an abelian group although most of this holds in greater generality.

A key feature of these local systems is that if our topological space is nice enough, for example a manifold M , then the local systems over M are in bijection with representations of the fundamental group. In fact, an even stronger condition than this holds. It turns out that this bijection can be upgraded to an equivalence of categories and hence the study of local systems over these topological spaces is equivalent to studying the representations of its fundamental group.

Indeed, given a local system $\mathcal{F}$ over M with fixed module a finite dimensional vector space V , one can associate an "algebraic" monodromy to it. Namely, given $[\gamma] \in \pi_1(M, x)$ we can associate an automorphism of the stalk $\mathcal{F}_x \cong R$. In this way we obtain a monodromy map

$$\rho : \pi_1(M, x) \rightarrow \text{Aut}(\mathcal{F}_x)$$

and this is how we associate a representation of the fundamental group to the local system.

Conversely, given a representation $\rho : \pi_1(M, x) \rightarrow \text{Aut}(R)$ we can associate to it a locally constant sheaf as follows. Let $\pi : \tilde{M} \rightarrow M$ be the universal cover of M and denote by $V_{\tilde{M}}$ the constant sheaf on $\tilde{M}$. Then we define the sections $\mathcal{F}(U)$ to be sections of $V_{\tilde{M}}$ over $\pi^{-1}(U)$ that are equivariant with respect to the action of the fundamental group $\pi_1(M, x)$ which acts on $\tilde{M}$ by deck transformations. This determines a locally constant sheaf as if we pick a small enough neighborhood $U \subset M$ that is evenly covered, then the sheaf is constant on U as $\pi|_{U_\gamma}$ is a homeomorphism and $\pi \circ \gamma = \pi$ where γ is a deck transformation. For more details see [ES09].

5.5.2 Cohomology of Local Systems

We could calculate the cohomology of a local system by calculating its sheaf cohomology directly. However, there turns out to be a nicer way to calculate the cohomology in this case which is amenable to cellular methods.

CHAPTER 5. CLASSIFICATION OF LAGRANGIAN FIBRATIONS OVER CLOSED SURFACES

In fact, the equivalence between local systems and representations of the fundamental group leads to two different ways of calculating the cohomology of the local system. We now briefly illustrate how to calculate through the two different perspectives.

The singular cohomology approach

The first approach we take is the singular cohomology approach. Given a representation $\rho : \pi_1(M) \rightarrow \text{Aut}(R)$ our construction of a local system in fact give a fibre bundle E with fibre R by

$$E := (\tilde{M} \times R) / \pi_1(M)$$

where $\gamma \cdot (\tilde{m}, v) = (\gamma \cdot \tilde{m}, \rho(\gamma)v)$ see [DK01]. Let $\sigma : \Delta^n \rightarrow M$ be a singular k -simplex in M and denote by $e_0 = (1, 0, \dots, 0)$ the base point of σ and $E_{\sigma(e_0)}$ the fibre above $\sigma(e_0) \in X$. We let $S^k(M; E)$ the set of formal sums of functions f that assign to each singular simplex $\sigma : \Delta^k \rightarrow M$, an element $f(\sigma) \in E_{\sigma(e_0)}$. Then $S^k(M; E)$ is an abelian group and we can associate a coboundary operator to it by

$$\delta : S^k(M; E) \rightarrow S^{k+1}(M; E)$$

given by

$$\delta(f)(\sigma) = (-1)^k \left(\gamma_\sigma^{-1}(f(\partial_0 \sigma)) + \sum_{i=1}^{k+1} (-1)^i c(\partial_i \sigma) \right)$$

where the ∂_i are the usual boundary maps for singular simplices and

$$\gamma_\sigma : E_{\sigma(0,1,0,\dots,0)} \rightarrow E_{\sigma(1,0,0,\dots,0)}$$

is an isomorphism induced by the path

$$\tilde{\gamma}_\sigma : [0, 1] \rightarrow M$$

$$t \mapsto \sigma(t, 1-t, 0, \dots, 0)$$

via path lifting as $\rho : E \rightarrow M$ is a covering space.

A computation shows that $\delta^2 = 0$ and we will denote the corresponding cohomology by $H_{\text{sing}}^*(M, R)$ where it is understood that R is viewed as a $\pi_1(M)$ -module through the action of the representation.

Definition 5.27. Let $\rho : \pi_1(M) \rightarrow \text{Aut}(R)$ determine a local system. The *singular cohomology of the local system* is the cohomology of the complex $(S^k(M; E), \delta)$

The equivariant or Borel cohomology approach

Equivariant or Borel cohomology was introduced by Borel in [Bor60] and encodes information about group actions on a topological space X . It has connections to Lie groupoid cohomology and has found applications throughout algebraic geometry and differential geometry. The description that we now give of the cohomology of a local system is a special case of this equivariant cohomology.

We have that there is a right action of the fundamental group by deck transformations on the universal cover $\tilde{M}$ of our manifold M . Denoting by $C^*(\tilde{M})$ the free abelian group of simplices $\sigma : \Delta^n \rightarrow \tilde{M}$, this action turns $C^*(\tilde{M})$ into a right $\pi_1(M)$ -module. Namely, given a simplex and $\gamma \in \pi_1(M)$ we can define this right action as $\sigma \cdot \gamma = \gamma \star \sigma$ where $\gamma \star \sigma$ is the action of γ by deck

transformations and extend to the group ring by linearity. Now, we have that the fixed group of our local system R is a right $\pi_1(M)$ -module via the representation $\rho : \pi_1(M) \rightarrow \text{Aut}(R)$. We would like to apply $\text{Hom}_{\mathbb{Z}[\pi_1(M)]}(-, R)$ to our chain complex of right $\mathbb{Z}[\pi_1(M)]$ -modules $C^*(\tilde{M})$ but we cannot do so as $\text{Hom}_{\mathbb{Z}[\pi_1(M)]}(-, -)$ must take in either two right modules or two left modules. However, we can solve this problem by turning $C^*(\tilde{M})$ into a left $\mathbb{Z}[\pi_1(M)]$ -module by defining the action of $g \in \mathbb{Z}[\pi_1(M)]$ to be $g \cdot z = z \cdot g^{-1}$. Applying $\text{Hom}_{\mathbb{Z}[\pi_1(M)]}(-, R)$ yields a co-complex.

Definition 5.28. The cohomology of this complex is called the *cohomology of M with twisted coefficients* and is the same as the equivariant cohomology of the universal cover $\tilde{M}$. Fixing a representation $\rho : \pi_1(M) \rightarrow \text{Aut}(R)$ we shall denote this cohomology as $H^i(M, R_\rho)$.

The two approaches described can be seen to be equivalent. We shall not give the proof of this but it can be found in Theorem 3.5 of [Whi78].

5.5.3 Relation to group cohomology

There is yet another approach to be taken in this case that our manifold is a "nice" space by which we mean it is an Eilenberg Mac-Lane space of type $K(G, 1)$ as of Definition 1.52.

A local system is determined by a representation of the fundamental group on an abelian group R and hence, one might speculate that the group cohomology of $H^i(\pi_1(M), R)$ where we view R as a $\pi_1(M)$ -module is the same as the cohomology of the local system determined by it. However, this is not the case in general although there are relationships between the two. The following lemma is a special case of when the two coincide.

Lemma 5.29. Let M be an Eilenberg-MacLane space of type $K(G, 1)$ and let $\pi_1(M) \rightarrow \text{Aut}(R)$ be a representation. Then the group cohomology $H^i(\pi_1(M), R)$ and the cohomology of the local system coincide.

Proof. We will give the proof in the case of M being a CW-complex although this is not strictly needed [Whi78]. For $n \geq 2$ we have that S^n is simply connected and so given a map $f : S^n \rightarrow M$ there is a unique lift to the universal cover.

$$\begin{array}{ccc} & & \tilde{M} \\ & \nearrow \tilde{f} & \downarrow \pi \\ S^n & \xrightarrow{f} & M \end{array}$$

and hence we get a map

$$\begin{aligned} \varphi : \pi_n(M) &\rightarrow \pi_n(\tilde{M}) \\ [f] &\mapsto [\tilde{f}] \end{aligned}$$

The covering map also gives a map in the other direction and these maps are easily seen to be inverses of one another. This leads to an isomorphism $\pi_n(M) \cong \pi_n(\tilde{M})$ for $n \geq 2$.

Hence, if M is an Eilenberg-MacLane space of type $K(G, 1)$ then its universal cover has $\pi_n(\tilde{M}) = 0$ for all $n \in \mathbb{N}$. Now if we consider the chain complex

$$\cdots \rightarrow C^2(\tilde{M}) \rightarrow C^1(\tilde{M}) \rightarrow C^0(\tilde{M}) \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$$

CHAPTER 5. CLASSIFICATION OF LAGRANGIAN FIBRATIONS OVER CLOSED SURFACES

where ϵ is the augmentation map. By Whiteheads lemma (Corollary 7.34 [DK01]), $\tilde{M}$ is homotopy equivalent to a point and so $H^i(C^*(M)) = 0$ for $i > 0$ and $H^0(C^*(M)) = 0$. Hence the above sequence is exact and provides a free resolution of $\mathbb{Z}$ by $\pi_1(M)$ -modules and this computes the group cohomology after applying the $\text{Hom}_{\mathbb{Z}[\pi_1(M)]}(-, R)$ functor and taking the cohomology of the resulting complex. $\square$

Remark 5.30. If M is a $K(G, 1)$ space and path connected, then M is often called *aspherical*.

We now make the following observation which we shall use when computing the Chern class of a Lagrangian torus fibration over a complete integral affine manifold.

Lemma 5.31. Let M be a complete affine manifold. Then M is aspherical and hence the cohomology of the local system determined by its linear holonomy representation, is equal to the group cohomology of $\pi_1(M)$ with respect to the linear holonomy representation.

Proof. Since M is complete, the developing map $\text{dev} : \tilde{M} \rightarrow \mathbb{R}^n$ is a diffeomorphism. Since $\mathbb{R}^n$ is contractible it follows that $\tilde{M}$ is also and hence by Lemma 5.29, the group cohomology and the cohomology of the local system coincide. $\square$

In the case of the Chern class, this lived in $H^2(\Lambda)$ where Λ is the locally constant sheaf of sections of the period lattice. Our previous discussion shows that in the case of a complete integral affine manifold with linear holonomy representation $\text{Lin} \circ \text{hol} : \pi_1(M) \rightarrow \text{Aut}(\mathbb{Z}^n)$, the sheaf cohomology of Λ coincides with the group cohomology of $\pi_1(M)$ with respect to this linear holonomy representation once one verifies that these local systems do indeed coincide. In group cohomology, the group H^2 classifies objects called *group extensions*. We quickly recap these now.

Definition 5.32. A *group extension* of G by A is a short exact sequence of groups

$$0 \rightarrow A \rightarrow E \xrightarrow{\pi} G \rightarrow 1$$

where A is abelian.

There is a notion of equivalence for group extensions.

Definition 5.33. An equivalence of extensions is a group isomorphism $\varphi : E \rightarrow E'$ such that the following diagram commutes:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & E & \xrightarrow{\pi} & G \longrightarrow 1 \\ & & \parallel & & \downarrow \varphi & & \parallel \\ 0 & \longrightarrow & A & \longrightarrow & E' & \xrightarrow{\pi'} & G \longrightarrow 1. \end{array}$$

Given a group extension of G by A then there is a unique G -module structure induced on A . Equivalent group extensions induce the same G -module structure on A . Fixing a G -module A , there may be multiple extensions that induce the given G -module structure. We have the following classification result.

Theorem 5.34 ([Bro12]). Fix a G -module A . Let $\epsilon(G, A)$ denote the set of extensions which induce the given G -module structure on A . Then we have that there is a bijection

$$H^2(G, A) \leftrightarrow \epsilon(G, A).$$

This establishes the following perspective on the Chern classes of a Lagrangian fibration over a closed integral affine surface.

Proposition 5.35. Let M be a closed integral affine surface. Then there is a one-to-one correspondence between Lagrangian fibrations over M with a given Chern class and isomorphism classes of group extensions

$$0 \rightarrow \mathbb{Z}^n \rightarrow E \rightarrow \pi_1(M) \rightarrow 1$$

Proof. This follows from the fact that every Chern class of an integral affine closed surface is realisable as the Chern class of a Lagrangian fibration and the bijection established in the previous theorem. $\square$

Remark 5.36. Under this correspondence, the symplectic reference fibration corresponds to the group extension class of

$$0 \rightarrow \mathbb{Z}^n \rightarrow \mathbb{Z}^n \rtimes \pi_1(M) \rightarrow \pi_1(M) \rightarrow 1$$

What remains to show in this section, is that calculating the cohomology of the local system in the sense of the two different definitions given, is equivalent to calculating the sheaf cohomology of the locally constant sheaf. We will not give the proof but we will outline where one can find the required equivalence.

Theorem 5.37. Let M a manifold and let $\mathcal{F}$ be a local system over M with fixed abelian group R . Then the following cohomology theories are equivalent.

1. The sheaf cohomology of the locally constant sheaf $\mathcal{F}$ over M .
2. The singular cohomology of the local system as in Definition 5.27.
3. The cohomology of M with twisted coefficients as in Definition 5.28.
4. The Čech cohomology of the locally constant sheaf $\mathcal{F}$.

Proof. For 1) $\Leftrightarrow$ 2) see Theorem 1.1 in Chapter 3 of [Bre97]. For 2) $\Leftrightarrow$ 3) see Theorem 3.4 of [Whi78]. For 4) $\Leftrightarrow$ 1) see Theorem 4.13 of [Bre97]. $\square$

We also note the following important fact about the homotopy invariance of the cohomology of locally constant sheaves.

Theorem 5.38 (Theorem 21.5. [Dah20]). Let $f : M \rightarrow N$ be a homotopy equivalence. Then for every locally constant sheaf $\mathcal{F}$ on Y , we have that

$$H^*(Y, \mathcal{F}) \cong H^*(X, f^{-1}\mathcal{F}).$$

5.5.4 Classifying Lagrangian fibrations over S^1

Before we give the classification of regular Lagrangian torus fibrations over compact surfaces, we first give the classification over S^1 for completeness.

Lemma 5.39. The Lagrangian fibrations over S^1 are determined by a positive number $y > 0$ and are of the form $T^*S^1/\Lambda_y \rightarrow S^1$ where Λ_y is the period lattice determined by the integral affine structure corresponding to y .

Proof. Since S^1 has dimension one $H^2(\Lambda) = 0$ for any period lattice $\Lambda \subset T^*S^1$. One way to see this is by noting that $\pi_1(S^1) = \mathbb{Z}$ and the group cohomology of $\mathbb{Z}$ vanishes for $i > 1$. Hence, every Lagrangian fibration admits a global section, and since the de Rham cohomology $H^2(S^1) = 0$ there are no other considerations. Hence, the Lagrangian fibrations over S^1 are all topologically trivial, admit a global section and are of the form $T^*S^1/\Lambda_y \rightarrow S^1$ where Λ_y is the lattice determined by $y > 0$. They are classified by the integral affine structure on S^1 and by Lemma 4.1 are therefore determined by a number $y > 0$. $\square$

Remark 5.40. An integral affine structure on S^1 is equivalent to a full rank lattice in $T^*S^1 = S^1 \times \mathbb{R}$ which is determined by the multiples of some number $y \in \mathbb{R} \setminus 0$. This determines the symplectic volume of the space T^*B/Λ_y . Different choices of y change the symplectic volume of T^*B/Λ_y and hence these bundles cannot be symplectomorphic for different choices of y .

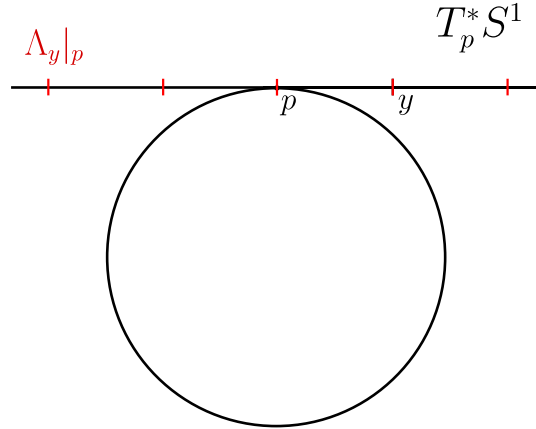


Figure 5.4: A lattice in $T_p^*S^1$ determined by $y > 0$ corresponding to a Lagrangian fibration over S^1 .

5.6 Computing $H^2(\Lambda)$ for the torus

We now use the group cohomology approach in order to calculate $H^2(\Lambda)$ for our compact integral affine surfaces. For $\mathbb{T}^2$, we have that $\pi_1(\mathbb{T}^2) = \mathbb{Z} \oplus \mathbb{Z}$. The group ring for $G = \mathbb{Z} \oplus \mathbb{Z}$ is given by $\mathbb{Z}[G] = \mathbb{Z}[x, x^{-1}, y, y^{-1}]$ where x and y are the generators for $\mathbb{Z} \oplus \mathbb{Z}$.

We will make use of the Koszul resolution which provides us with a nice free resolution of $\mathbb{Z}$ in this case.

Definition 5.41. Let R^n be a free R -module with basis $e_1, \dots, e_n$. The k -th exterior power $\Lambda^k(R^n)$ is the free R -module generated by

$$e_{i_1} \wedge \cdots \wedge e_{i_k}, \quad 1 \leq i_1 < \cdots < i_k \leq n,$$

where the wedge product is alternating:

$$e_i \wedge e_j = -e_j \wedge e_i$$

Lemma 5.42 (Koszul resolution [Wei94]). If $\mathbf{x} = (x_1, \dots, x_n)$ is a regular sequence in R , then the following sequence is exact:

$$0 \longrightarrow \Lambda^n(R^n) \longrightarrow \cdots \longrightarrow \Lambda^2(R^n) \longrightarrow R^n \xrightarrow{x} R \longrightarrow R/I \longrightarrow 0.$$

We have that $(x-1, y-1)$ is a regular sequence in $\mathbb{Z}[G]$ as $x-1$ is a non-zero divisor in $\mathbb{Z}[G]$ and $y-1$ is a non-zero divisor in $\mathbb{Z}[G]/(x-1) = \mathbb{Z}[y, y^{-1}]$. Since $\mathbb{Z}[x, x^{-1}, y, y^{-1}]/(x-1, y-1) \cong \mathbb{Z}$, the Koszul complex yields the following exact sequence

$$0 \mapsto \Lambda^2(\mathbb{Z}[G]^2) \xrightarrow{d_2} \mathbb{Z}[G]^2 \xrightarrow{d_1} \mathbb{Z}[G] \rightarrow \mathbb{Z} \rightarrow 0$$

where $\Lambda^2(\mathbb{Z}[G]^2) = \mathbb{Z}[G]$ and

$$\begin{aligned} d_1(a, b) &= (x-1)a + (y-1)b, \\ d_2(c) &= ((y-1)c, -(x-1)c) \end{aligned}$$

We apply $\text{Hom}_{\mathbb{Z}[G]}(-, \mathbb{Z}^2)$ to the deleted resolution which yields

$$0 \mapsto \mathbb{Z}^2 \xrightarrow{\tilde{d}_1} \mathbb{Z}^2 \oplus \mathbb{Z}^2 \xrightarrow{\tilde{d}_2} \mathbb{Z}^2 \rightarrow 0$$

This immediately yields the following.

Lemma 5.43. Let $G = \mathbb{Z}^2 = \langle x, y \rangle$ and let $\mathbb{Z}^2$ be a G -module. then the second group cohomology group is

$$H^2(G, \mathbb{Z}^2) \cong \frac{\mathbb{Z}^2}{(x-1)\mathbb{Z}^2 + (y-1)\mathbb{Z}^2}$$

Proof. This follows from the above resolution.

$$\text{im}(\tilde{d}_2) = -(x-1)\mathbb{Z}^2 + (y-1)\mathbb{Z}^2$$

and the kernel of $\mathbb{Z}^2 \rightarrow 0$ is all of $\mathbb{Z}^2$. □

For the first family $\mathcal{A}_{a,b,c,d}$, the linear holonomy induces the trivial representation of G on $\mathbb{Z}^2$. We therefore have the following.

Lemma 5.44. For $\mathcal{A}_{a,b,c,d}$, the cohomology group $H^2(G, \mathbb{Z}^2) = \mathbb{Z}^2$.

CHAPTER 5. CLASSIFICATION OF LAGRANGIAN FIBRATIONS OVER CLOSED SURFACES

Proof. Both $(x - 1)\mathbb{Z} = 0$ and $(y - 1)\mathbb{Z} = 0$ as x and y act trivially. Therefore,

$$\begin{aligned} H^2(G, \mathbb{Z}^2) &= \frac{\mathbb{Z}^2}{(x - 1)\mathbb{Z}^2 + (y - 1)\mathbb{Z}^2} \\ &= \mathbb{Z}^2 \end{aligned}$$

□

Lemma 5.45. For $\mathcal{B}_{n,a,b}$, the cohomology group $H^2(G, \mathbb{Z}^2) = \frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}$.

Proof. This amounts to computing the action of $(x - 1)$ and $(y - 1)$ on $\mathbb{Z}^2$ induced by the linear holonomy representation. From our classification in Theorem 4.14 we have that x acts trivially and y acts by $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$. We calculate

$$y \cdot (l, m) - (l, m) = (nl, 0)$$

and so $(y - 1)\mathbb{Z}^2 = n\mathbb{Z} \oplus 0$. This gives

$$\begin{aligned} H^2(G, \mathbb{Z}^2) &= \frac{\mathbb{Z}^2}{n\mathbb{Z} \oplus 0} \\ &= \frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z} \end{aligned}$$

□

Since $H^3(\mathbb{T}^2, \mathbb{R}) = 0$, it follows from Proposition 5.19 that every element in $H^2(\Lambda)$ is the Chern class of some Lagrangian fibration. We summarize this data in the following theorem.

Theorem 5.46. The possible Chern classes for Lagrangian fibrations over the torus are:

1. For the integral affine structures corresponding to $\mathcal{A}_{a,b,c,d}$, every element of the cohomology group $H^2(\Lambda) \cong \mathbb{Z}^2$ is the Chern class of some Lagrangian torus fibration.
2. For the integral affine structures corresponding to $\mathcal{B}_{n,a,b}$, every element of the cohomology group $H^2(\Lambda) \cong \frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}$ is the Chern class of some Lagrangian torus fibration.

5.7 Computing $H^2(\Lambda)$ for the Klein bottle

5.7.1 Spectral Sequences

The fundamental group of the Klein bottle is $\pi_1(K^2) = \mathbb{Z} \rtimes \mathbb{Z}$. In this case, there is a nice spectral sequence that reduces computation of this semi-direct product, to the computation of group cohomologies of $\mathbb{Z}$ with respect to a representation on a certain first cohomology group. We will give a quick outline of spectral sequences with a focus on how to use them to calculate this group cohomology. We will closely follow [McC01].

Consider generally, a cohomology theory and set $H^* = \bigoplus_i H^i$. The goal of a spectral sequence is to approximate the graded object H^* or in some nice cases, to directly compute this. We make the assumption that H^* has a filtration

5.7. COMPUTING $H^2(\Lambda)$ FOR THE KLEIN BOTTLE

$$H^* = F^0 H^* \supseteq \dots \supseteq F^n H^* \supseteq F^{n+1} H^* \supseteq \dots \supseteq \{0\}$$

and for now we will assume that H^* is a graded vector space for simplicity. Given a filtration F^* of H^* it can be collapsed into another graded vector space called *the associated graded vector space* defined by

$$E_0^p(H^*) = F^p H^* / F^{p+1} H^*$$

In the case that H^* is a vector space we can recover H^* up to isomorphism by

$$H^* \cong \bigoplus_{p=0}^{\infty} E_0^p(H^*) = \bigoplus_{p=0}^{\infty} F^p H^* / F^{p+1} H^*$$

However if H^* is not a graded vector space there can be extension problems. We will not discuss this extension problem in more detail as in our computations we will have only one graded piece to consider and hence there is no extension problem.

Now, the idea is that H^* may not be easily computed but we can try to approximate it by the associated graded vector space to some filtration F^* of H^* . We now observe that $E_0^*(H^*)$ is bigraded in the following sense. If we define $F^p H^r = F^p H^* \cap H^r$ then we have that

$$E_0^{p,q} = \frac{F^p H^{p+q}}{F^{p+1} H^{p+q}}$$

The index q is called *the complementary degree* and the index p is called the *filtration degree*.

We note that in the case of vector spaces we can recover H^r as the direct sum of the pieces of $E_0^{p,q}$ of total degree r e.g. where $p + q = r$. We can now give the definition of a spectral sequence.

Definition 5.47. A first quadrant spectral sequence is a sequence of differential bigraded vector spaces. For every $r \in \mathbb{N}$ and $p, q > 0$ we have a vector space $E_r^{p,q}$ equipped with a linear map

$$d_r : E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$$

with $d_r^2 = 0$ a differential of bidegree $(r, 1 - r)$. Furthermore, for all $r \geq 1$ we have that $E_{r+1}^{*,*} = H(E_r^{*,*}, d_r)$ where $H(E_r^{*,*}, d_r)$ denotes taking the cohomology at the $E_r^{*,*}$ spot.

Remark 5.48. We call $E_r^{*,*}$ the E_r -page. Turning a page e.g. going from the E_r -page to the E_{r+1} -page corresponds to taking cohomology. The pages E_r may be pictured as a lattice with each point a vector space and arrows between vector spaces corresponding to the differential d_r .

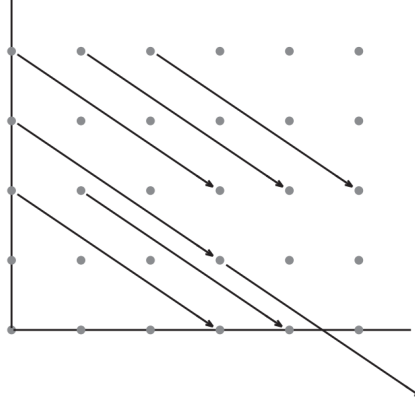


Figure 5.5: A page E_r in a spectral sequence. Arrows mapping of the page correspond to zero mappings. [McC01]

We can now say what it means for a spectral sequence to converge to a graded vector space.

Definition 5.49. A spectral sequence $\{E_r^{*,*}, d_r\}$ of vector spaces is said to converge to a graded vector space H^* if H^* has a filtration F^* and

$$E_\infty^{p,q} \cong F^p H^{p+q} / F^{p+1} H^{p+q} = E_0^{p,q}(H^*)$$

There are still some issues around uniqueness of the convergence that we shall not discuss. We also formulated our definitions in terms of graded vector spaces as oppose to more generally graded R -modules. The theory extends to these in a very similar fashion except there are now more issues with reconstructing our graded module H^* as for example there may be multiple extensions of a R -module A by another R -module e.g. these are not all split as in the vector space case.

We now give the spectral sequence that we shall use in order to compute the group cohomology of $\pi_1(K) = \mathbb{Z} \rtimes \mathbb{Z}$.

Lemma 5.50 (Hochschild-Lydon-Serre Spectral Sequence [HS53]). Let G be a group and $N \subseteq G$ a normal subgroup and let A be a G -module. Then there is a spectral sequence

$$E_2^{p,q} = H^p(G/N, H^q(N, A)) \Rightarrow H^{p+q}(G, A)$$

For the fundamental group of the Klein bottle, we have the short exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rtimes \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0$$

and so $\mathbb{Z}$ sits inside $\mathbb{Z} \rtimes \mathbb{Z}$ and $\frac{\mathbb{Z} \rtimes \mathbb{Z}}{\mathbb{Z} \times 0} \cong \mathbb{Z}$. The differential on the E_2 page is

$$d : E_2^{p,q} \rightarrow E_2^{p+2,q-1}$$

Using the fact that for any $\mathbb{Z}$ -module G , the group cohomology groups $H^i(\mathbb{Z}, G) = 0$ for $i > 1$ it follows that the only entry of the spectral sequence we need to calculate is $H^1(\mathbb{Z}, H^1(\mathbb{Z}, A))$ where $A = \mathbb{Z}^2$ viewed as a $\mathbb{Z}$ -module.

5.7.2 Calculating $H^1(\mathbb{Z}, H^1(\mathbb{Z}, A))$

Let $G = \mathbb{Z}\langle a \rangle$ and let $A = \mathbb{Z}^2$ be a G -module. A free resolution of $\mathbb{Z}[G]$ is given by

$$0 \rightarrow \mathbb{Z}[G] \xrightarrow{a-1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$$

where ϵ is the augmentation map. Applying $\text{Hom}_{\mathbb{Z}[G]}(-, A)$ to the deleted resolution yields

$$0 \rightarrow A \rightarrow A \rightarrow 0$$

and so the cohomology group is given by

$$H^1(G, A) = \frac{A}{(a-1)A}$$

Now in order to calculate the full cohomology group $H^1(\mathbb{Z}, H^1(\mathbb{Z}, A))$, there is the question of what the $\mathbb{Z}$ -module structure on $H^1(\mathbb{Z}, A)$ should be.

Lemma 5.51. The action of $\mathbb{Z}\langle b \rangle$ on $H^1(G, A)$ is given by

$$(b \cdot c)(a) = -(ba^{-1})c(a)$$

where $c \in \text{Der}(G, A)$ and $G = \langle a \rangle$.

Proof. We have that $H^1(G, A) = \frac{\text{Der}(G, A)}{\text{PDer}(G, A)}$. Let $c \in \text{Der}(G, A)$. Then

$$(b \cdot c)(a_1) = b \cdot c(b^{-1}a_1b)$$

where this action is as in [HS53]. Now, using that c is a derivation

$$\begin{aligned} c(a^{-1}a) &= c(1) \\ &= 0 \\ &= c(a^{-1}) + a^{-1}c(a) \end{aligned}$$

and so $c(a^{-1}) = -a^{-1}c(a)$. This gives us that

$$\begin{aligned} b \cdot c(b^{-1}ab) &= b \cdot c(a^{-1}) \\ &= (-ba^{-1}) \cdot c(a) \end{aligned}$$

as required. □

This gives the following expression for the relevant cohomology group keeping in mind the respective actions of $\mathbb{Z}\langle a \rangle$ and $\mathbb{Z}\langle b \rangle$.

Corollary 5.52. The cohomology group $H^1(\mathbb{Z}\langle b \rangle, H^1(\mathbb{Z}\langle a \rangle, A))$ can be expressed as

$$\frac{\frac{A}{(a-1)A}}{(b-1)\frac{A}{(a-1)A}}$$

CHAPTER 5. CLASSIFICATION OF LAGRANGIAN FIBRATIONS OVER CLOSED SURFACES

Now it remains to calculate this group for the holonomy representations of the integral affine structures on the Klein bottle. Firstly, recall that the integral affine structures on the Klein bottle are given by the following.

The possible integral affine structures on the Klein bottle are

1. $a = \left(I, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix} \right), \text{ with } x_2, y_1 > 0.$
2. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ x_2 \end{pmatrix} \right), \text{ with } x_2, y_1 > 0.$
3. $a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix} \right), \text{ with } x_1, y_1 > 0.$
4. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ \frac{n-1}{n}x_2 \end{pmatrix} \right), \text{ with } y_1 > 0 \text{ and } 2y_1 \neq -\frac{n-1}{n}x_2.$

Theorem 5.53. The possible Chern classes for Lagrangian fibrations over the Klein bottle are

$$H^2(\Lambda) \cong \begin{cases} \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}, & \text{for family 1).} \\ \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}, & \text{for family 2) with } n \equiv 0 \pmod{2}. \\ \mathbb{Z}, & \text{otherwise.} \end{cases}$$

Proof.

Case 1 (Family 1). In this case the action of $\mathbb{Z}\langle a \rangle$ on $A = \mathbb{Z}^2$ is trivial and hence $(a - 1)A = 0$. The action of b on $\frac{A}{(a-1)A}$ is given by the matrix action of $-(ba^{-1})$. Since a acts as the identity the action is given by

$$\begin{aligned} b \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} &= \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \\ &= \begin{pmatrix} -x_1 \\ y_1 \end{pmatrix} \end{aligned}$$

and hence the action by $(b - 1)$ is

$$(b - 1) \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} -2x_1 \\ 0 \end{pmatrix}$$

This gives us that $(b - 1) \cdot \frac{A}{(a-1)A} = 2\mathbb{Z} \oplus 0$. Taking the quotient of $A = \mathbb{Z}^2$ by this gives the required result.

Case 2 (Family 2). In this case, a acts non-trivially. The action of $(a - 1)$ is given by

$$(a - 1) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ny \\ 0 \end{pmatrix}$$

It follows that

$$\frac{A}{(a-1)A} = \frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}$$

We calculate $-ba^{-1}$ to be $\begin{pmatrix} -1 & n \\ 0 & 1 \end{pmatrix}$.

and so

$$\begin{pmatrix} -1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} ny_1 - x_1 \\ y_1 \end{pmatrix} \\ = \begin{pmatrix} -x_1 \\ y_1 \end{pmatrix}$$

where in the last equality we are reducing modulo n in the first component. Therefore the action of $(b-1)$ on $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \in \frac{A}{(a-1)A}$ is given by

$$(b-1) \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} -2x_1 \\ 0 \end{pmatrix}$$

and so

$$\begin{aligned} H^2(\Lambda) &= H^1(\mathbb{Z}, H^1(\mathbb{Z}, A)) = \frac{\frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}}{\langle 2 \rangle \oplus 0} \\ &= \frac{\mathbb{Z}}{\gcd(2, n)\mathbb{Z}} \oplus \mathbb{Z} \end{aligned}$$

giving the required result for the second family.

Case 3 (Family 3). In this case, a again acts trivially and so it remains to calculate the action of b . We compute

$$-ba^{-1} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

and so $(b-1) \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y - x \\ -y - x \end{pmatrix}$ and so it follows that

$$\begin{aligned} H^1(\mathbb{Z}, H^1(\mathbb{Z}, A)) &= \frac{\mathbb{Z} \oplus \mathbb{Z}}{\langle (1, 1) \rangle} \\ &= \mathbb{Z} \end{aligned}$$

Case 4 (Family 4). In this case, the $\mathbb{Z}\langle a \rangle$ action on $\mathbb{Z}$ is the same as in Family 2. It remains to calculate the action of b . We calculate

$$-(ba^{-1}) = \begin{pmatrix} -1 & n-1 \\ 0 & 1 \end{pmatrix}$$

and so $(b-1) \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2x + ny - y \\ 0 \end{pmatrix}$. Since this action is on $\frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}$ we reduce mod n in the first component. This now gives that

$$\begin{aligned} H^1(\mathbb{Z}, H^1(\mathbb{Z}, A)) &= \frac{\frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}}{\frac{\mathbb{Z}}{n\mathbb{Z}} \oplus 0} \\ &= \mathbb{Z} \end{aligned}$$

and this completes the calculations. □

5.8 Constructing the fibrations from their invariants.

We now show how to construct the Lagrangian fibrations over the torus and the Klein bottle from their invariants. Let B denote either the torus or the Klein bottle.

From taking the long exact sequence in cohomology associated to the short exact sequence of sheafs

$$0 \rightarrow \Lambda \rightarrow T^*B \rightarrow T^*B/\Lambda \rightarrow 0$$

and using that T^*B is a fine sheaf we obtain that $H^1(T^*B/\Lambda) \cong H^2(\Lambda)$. We will make use of the following interpretation of $H^1(T^*B/\Lambda)$.

We can view the torus or the Klein bottle as a square with the usual identifications which we will denote by Σ in both cases. Take a small disc D in the interior of this square such that the bundle is trivial over this disc. Then a section of this bundle on the boundary of this disc is just a map from $S^1 \rightarrow T^2$ and these are determined up to homotopy by $\pi_1(T^2)$. A section on ∂D can be extended onto all of D if and only if it corresponds to the zero homotopy class.

Now consider the sections on ∂D which are extendable on $\Sigma \setminus D$. This forms some subgroup $F \subset \pi_1(T^2)$. We now have the following lemma.

Lemma 5.54. The group $H^1(T^*B/\Lambda)$ is isomorphic to $\frac{\pi_1(T^2)}{F}$.

Proof. We can cover the Klein bottle or the torus, which we have denoted both by Σ , by two open sets U and V . Let U be a neighborhood of the disc D and let V be a neighborhood of $\Sigma \setminus D$. We consider the Mayer-Vietoris sequence for the sheaf T^*B/Λ .

$$\cdots \rightarrow H^0(U) \oplus H^0(V) \xrightarrow{d} H^0(U \cap V) \rightarrow H^1(\Sigma) \rightarrow H^1(U) \oplus H^1(V) \rightarrow \cdots$$

Now we claim that $H^1(U) \oplus H^1(V) = 0$. Indeed, they both sit into the short exact sequence

$$0 \rightarrow \Lambda|_V \rightarrow T^*B|_V \rightarrow T^*B/\Lambda|_V \rightarrow 0$$

and we note that restriction preserves exactness as exactness can be checked stalk-wise. Taking the corresponding long exact sequence in cohomology and using the fact that $T^*B|_V$ is a fine sheaf, we conclude that $H^1(T^*B|_V) \cong H^2(\Lambda|_V)$. Now, by Theorem 5.37 the cohomology $H^2(\Lambda|_V)$ can be calculated using singular cohomological methods and even via cellular methods. Theorem 5.38 gives us homotopy invariance of this group and since V deformation retracts onto the wedge of two circles this cohomology group vanishes as there are no two cells.

We can apply the exact same arguments to U except now U is contractible and so $H^2(\Lambda|_U) = 0$. It follows that $H^1(U) \oplus H^1(V) = 0$.

We therefore have from the Mayer-Vietoris sequence that $H^1(\Sigma) \cong \frac{H^0(U \cap V)}{\text{Im } d}$. We now show that $\frac{H^0(U \cap V)}{\text{Im } d} \cong \frac{\pi_1(T^2)}{F}$.

Every section of $H^0(U \cap V)$ corresponds to a section on $\partial D = S^1$. We consider the map

$$\begin{aligned} \varphi : H^0(U \cap V) &\rightarrow \frac{\pi_1(T^2)}{F} \\ s &\mapsto [s] \end{aligned}$$

which maps a section to its corresponding homotopy class. It is clear that this map is surjective. Its kernel consists of the maps which are extendable on all of $\Sigma \setminus D$. Recall that the map d is given by

$$d([u] + [v]) = [u|_{U \cap V}] - [v|_{U \cap V}]$$

where $[u] + [v] \in H^0(U) \oplus H^0(V)$. But a section over U corresponded to the zero homotopy class and hence vanishes under φ . Similarly, $[v|_{U \cap V}]$ came from a section on $\Sigma \setminus D$ and hence $\varphi([v|_{U \cap V}]) = 0$. It follows that $\text{Im } d \subseteq \ker \varphi$.

Conversely, suppose that $\varphi([s]) = 0$. Then this implies that $[s]$ is extendable on $\Sigma \setminus D$. But this is exactly the elements of $H^0(V)$ and hence we have $\ker \varphi \subseteq \text{Im } d$ and the result follows. $\square$

5.8.1 The construction

We fix the integral affine structure on the base of our compact surface B . In order to construct a Lagrangian fibration over the base with a given Chern class we first start with the symplectic reference bundle T^*B/Λ . Now, we choose one of the fibres of the fibration - i.e. a Lagrangian torus in the total space of our bundle - and perform Luttinger surgery on this torus with respect to some section s over the boundary of a small disc.

If this disc corresponds to the homotopy class $(m, n) \in \pi_1(T^2)$, then the Luttinger surgery maps this section to the zero section and hence induces addition by $-(m, n)$ in $\pi_1(T^2)$. We have from Lemma 5.54 that the group $H^2(\Lambda)$ which corresponds to the Chern classes is a quotient of $\pi_1(T^2)$. It follows that we can produce any Chern class by picking our section s to lie in the appropriate homotopy class.

In the case of the Klein bottle, we are now done as there are no symplectic considerations. In the case of the torus, we must construct the symplectically different fibrations with the same Chern class.

By Remark 5.22 and Proposition 5.24 the Lagrangian classes mapping to a fixed Chern class is a quotient $\frac{H^2(B, \mathbb{R})}{S}$ where S is the subgroup generated by cohomological invariants of a shift. Since our Lagrangian fibration can always be reduced to one with a global section by means of Luttinger surgery it follows that we can construct the symplectic type by adding an element $[h] \in H^2(B, \mathbb{R})$ considered up to cohomological invariants of shifts. These invariants are calculated in Theorem 5.9 and this completes the construction.

Conclusion

Conclusion

In this thesis we classified the regular Lagrangian torus fibrations over compact surfaces up to Lagrangian equivalence. In order to do this, we showed that there are certain invariants that we can associate to such a fibration, and that these invariants completely determine the nature of the fibration.

The first of these invariants is an integral affine structure on the base space B of the fibration. This completely determines the local nature of any Lagrangian fibration through the symplectic reference fibration. In Chapter 3 we develop the tools required to classify the integral affine structures on compact surfaces. This turns out to depend on group theoretic properties of the fundamental group of the surface. The main result of this chapter is that completeness of an affine manifold is equivalent to parallel volume if the fundamental group is nilpotent.

This result reduces the problem of determining the integral affine structures on a compact surface to a problem in simple group theory and linear algebra. In Chapter 4, we explicitly compute all of the integral affine structures on the two torus and the Klein bottle. The integral affine structures on the two torus are classified in Theorem 4.14. These were found to be the following.

Theorem 4.14. Any integral affine structure on $\mathbb{T}^2$ has its affine holonomy group generated by one of the following subgroups of $\text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$

1. $\mathcal{A}_{a,b,c,d}$ - These integral affine structures are the quotient of $\mathbb{R}^2$ by a lattice of pure translations. The generators in $\text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ are given by

$$\left(1, \begin{pmatrix} a \\ c \end{pmatrix}\right) \text{ and } \left(1, \begin{pmatrix} b \\ d \end{pmatrix}\right)$$

with $a, b, c, d \in \mathbb{R}$ and $ad - bc \neq 0$.

2. $\mathcal{B}_{n,a,b}$ - The generators in $\text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ are given by

$$\left(1, \begin{pmatrix} a \\ 0 \end{pmatrix}\right) \text{ and } \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ b \end{pmatrix}\right)$$

with $a, b > 0$ and $n \in \mathbb{N}$.

Two integral affine structures from distinct families are never equivalent. Integral affine structures in the family $\mathcal{A}_{a,b,c,d}$ are sometimes equivalent as illustrated in Theorem 4.16.

Theorem 4.16. Lattices $\mathcal{A}_{a,b,c,d}$ and $\mathcal{A}_{a',b',c',d'}$ induce equivalent integral affine structures if and only if

$$X = GX'H$$

where $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $X' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ with $G, H \in GL(2, \mathbb{Z})$.

This completes the full classification of the integral affine structures on the two torus.

We also classify the integral affine structures on the Klein bottle. This is the result of Theorem 4.25. We find that there are four distinct families and that integral affine structures from distinct families are never equivalent.

Theorem 4.25. The possible generators for the holonomy representation of the integral affine structures on the Klein bottle are

1. $a = \left(I, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix} \right), \text{ with } x_2, y_1 > 0.$
2. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ x_2 \end{pmatrix} \right), \text{ with } x_2, y_1 > 0.$
3. $a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix} \right), \text{ with } x_1, y_1 > 0.$
4. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ \frac{n-1}{n}x_2 \end{pmatrix} \right), \text{ with } y_1 > 0 \text{ and } 2y_1 \neq -\frac{n-1}{n}x_2.$

The integral affine structure on the base is one invariant of the Lagrangian fibration. However, it is not the only one. The contents of Chapter 2 and Chapter 5 show that there are two more invariants one can associate to such a fibration. These are the *Chern class* and the *Lagrangian class*. The Chern class determines the topological nature of the fibration while the Lagrangian class is a refinement of this and also controls the symplectic structure. The Lagrangian class always maps onto the Chern class in the long exact sequence in cohomology associated to the following short exact sequence in sheaf cohomology;

$$0 \rightarrow \Lambda \rightarrow \mathcal{Z}(T^*B) \rightarrow \mathcal{Z}(T^*B/\Lambda) \rightarrow 0$$

where Λ is the sheaf of sections of the lattice in T^*B inducing the integral affine structure and $\mathcal{Z}(\cdot)$ denotes the Lagrangian or closed sections of the corresponding bundles.

Once we have fixed a Chern class we can ask which Lagrangian classes map onto it. This corresponds to fixing a topological type for the fibration and asking which fibrations are topologically equivalent but not necessarily symplectically. The space of Lagrangian classes which map onto a fix Chern class turns out to be a quotient of the second de Rham cohomology group. For the Klein bottle, the second cohomology class vanishes and hence for a fixed Chern class, all of the Lagrangian fibrations over the Klein bottle which induce this Chern class are symplectically equivalent. For the torus this is not the case; there are Lagrangian fibrations which are topologically equivalent but not symplectically. In both cases we compute the possible Chern classes in Chapter 5. For the torus this is the content of Lemma 5.44 and Lemma 5.45 which we summarize below.

Theorem 5.46. The possible Chern classes for Lagrangian fibrations over the torus are:

1. For the integral affine structures corresponding to $\mathcal{A}_{a,b,c,d}$, every element of the cohomology group $H^2(\Lambda) \cong \mathbb{Z}^2$ is the Chern class of some Lagrangian torus fibration.
2. For the integral affine structures corresponding to $\mathcal{B}_{n,a,b}$, every element of the cohomology group $H^2(\Lambda) \cong \frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}$ is the Chern class of some Lagrangian torus fibration.

Lagrangian fibrations over the torus can be topologically equivalent as bundles but not symplectically equivalent. This is governed by cohomological invariants of shifts which are the differentials of latticed one-forms. These form a subgroup of the second De Rham cohomology group and this subgroup parametrizes Lagrangian classes which map to a fixed Chern class. The calculation of which classes in $H^2(T^2, \mathbb{R})$ correspond to cohomological invariants of shifts is the content of Theorem 5.9.

For the Klein bottle, every Lagrangian torus fibration which is topologically equivalent is also symplectically equivalent. The possible Chern classes for the Klein bottle are found in Theorem 5.53.

Theorem 5.53. The possible Chern classes for Lagrangian fibrations over the Klein bottle are

$$H^2(\Lambda) \cong \begin{cases} \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}, & \text{for family 1).} \\ \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}, & \text{for family 2) with } n \equiv 0 \pmod{2}. \\ \mathbb{Z}, & \text{otherwise.} \end{cases}$$

In order to explicitly construct a Lagrangian fibration with a fixed Chern class, we can perform Luttinger surgery as described in Section 5.3. For the Klein bottle, this completes the construction. For the torus we must add a two form to the base considered up to a cohomological invariant of a shift. This is described in Section 5.8.

Whats next?

There are many directions one can go from here. We will now outline some possible extensions of the work done in this thesis

Integral affine 3-manifolds

In [Koz18], a classification of complete integral affine manifolds on compact three-dimensional manifolds up to a finite-sheeted covering is given. One could attempt to use the methods developed in this thesis in order to compute the possible Chern and Lagrangian classes for Lagrangian fibrations over these manifolds. In this paper they find that there are essentially three different families of such manifold. Up to finite index, the first families has abelian fundamental group, the second family has nilpotent fundamental group and the third class has solvable fundamental group.

For example, the third family which has the most complicated fundamental group, are mapping tori of the torus by some diffeomorphism $A : T^2 \rightarrow T^2$, at least up to a finite covering. The fundamental group of a mapping torus sits into a short exact sequence

$$1 \rightarrow \mathbb{Z}^2 \rightarrow \mathbb{Z}^2 \rtimes_A \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 1$$

and so in principal, one could try to apply the spectral sequence methods we developed in Section 5.7.1 to compute the possible Chern classes for such manifolds.

Integral-integral affine manifolds

One could further try to refine the classification of integral affine manifolds to the classification of integral-integral affine manifolds over compact surfaces. An integral-integral affine manifold is an integral affine manifold whereby the translational components of the transition maps are restricted to also be integers. These structures are also interesting in their own right, for example they have applications to physics [HKY26]. One could attempt to classify Lagrangian fibrations up to Lagrangian equivalence which induce a given integral-integral affine structure on the base.

Investigate further the role of unipotence for integral affine manifolds

A far reaching goal is to show the Markus conjecture holds for the case of integral affine manifolds. As we have seen throughout this thesis, integral affine manifolds have some special geometric features which may be possible to exploit to achieve this. In Chapter 3 we required a lot of technical machinery in order to show that for affine manifolds with nilpotent holonomy, unipotence of the linear holonomy representation is equivalent to completeness of the affine structure.

However, with just the assumption of nilpotency, one can quickly see that at least up to topological type, the Lagrangian fibrations over a fixed integral affine structure depend only on the unipotent part of the representation. Let E_U denote the Fitting submodule, i.e. the unipotent part of the representation. Fix an integral affine manifold B and let Λ denote its corresponding period lattice. From the following short exact sequence of $\pi_1(B)$ -modules

$$0 \rightarrow E_U \rightarrow E \rightarrow E/E_U \rightarrow 0$$

after taking the long exact sequence in cohomology and using that $H^i(\pi_1(M), E/E_U) = 0$ by Lemma 3.10 and Lemma 3.5 as $\pi_1(M)$ is nilpotent, we get that $H^2(\pi_1(M), E_U) \cong H^2(\pi_1(M), E)$. Hence if M has nilpotent holonomy the group corresponding to the Chern classes depends only on the unipotent part of the representation, even without the assumptions of compactness, completeness or parallel volume.

One might wonder if, for compact integral affine manifolds, this can be shown to always hold. This is of course true if the Markus conjecture holds, but this statement in itself is slightly weaker as we are not requiring the holonomy representation itself to be unipotent which would imply completeness. Moreover, inside the group $H^2(\pi_1(M), E)$ there is the subgroup of realisable Chern classes. From the long exact sequence in cohomology of the above short exact sequence, we see that $H^2(\pi_1(M), E_U) \subseteq H^2(\pi_1(M), E)$. It would be interesting to determine whether the group of realisable Chern classes always sits inside this group, which could hold even without the isomorphism $H^2(\pi_1(M), E_U) \cong H^2(\pi_1(M), E)$.

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